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Design of a Telemonitoring System for Patients with Chronic Diseases (Hypertension, Diabetes)

Using Consultation Data, Home Measurements and Possibly Mobile Sensors

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Master of Science Thesis

Data Science and Artificial Intelligence

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List of Abbreviations

ADA	American Diabetes Association
AI	Artificial intelligence
AUC	Area under the receiver operating characteristic curve
BCT	Behaviour change technique
BLE	Bluetooth Low Energy
BP	Blood pressure
CI	Confidence interval
CMS	Centers for Medicare & Medicaid Services (United States)
DBP	Diastolic blood pressure
ECG	Electrocardiogram
eICU	eICU Collaborative Research Database
ENSPD	National Advanced School of Engineering of Douala
ESH	European Society of Hypertension
FK	Foreign key
GDPR	General Data Protection Regulation (European Union)
GPIO	General-purpose input/output
GRU-D	Gated recurrent unit with decay
GSM	Global System for Mobile communications
HbA1c	Glycated haemoglobin
HEARTS	WHO technical package for cardiovascular disease management (healthy-lifestyle counselling, evidence-based protocols, access to medicines, risk-based management, team care, systems for monitoring)
HTTP(S)	Hypertext Transfer Protocol (Secure)
I^2C	Inter-integrated circuit bus
ICD	International Classification of Diseases
IHSG	International Hypoglycaemia Study Group
IoT	Internet of Things
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
LASSO	Least absolute shrinkage and selection operator
LSTM	Long short-term memory
MAD	Median absolute deviation

MAR	Missing at random
MASD	Mean absolute successive difference
MCAR	Missing completely at random
mHealth	Mobile health
MIMIC-IV	Medical Information Mart for Intensive Care, version IV
MNAR	Missing not at random
MQTT	Message Queuing Telemetry Transport
NEWS2	National Early Warning Score 2
PK	Primary key
PPG	Photoplethysmography
PPV	Positive predictive value
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PROBAST+AI	Prediction model Risk Of Bias Assessment Tool, for artificial intelligence
QoS	Quality of service (MQTT delivery guarantee)
RCT	Randomised controlled trial
REST	Representational state transfer
ROC	Receiver operating characteristic
RPS	Ranked probability score
RQ	Research question
SBP	Systolic blood pressure
SHAP	Shapley additive explanations
SMS	Short message service
SNOMED CT	Systematized Nomenclature of Medicine – Clinical Terms
SpO_2	Peripheral oxygen saturation
STEPS	WHO STEPwise approach to noncommunicable disease risk factor surveillance
TLS	Transport Layer Security
TPE	Tree-structured Parzen estimator
TRIPOD+AI	Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis, for artificial intelligence
UML	Unified Modeling Language
UUID	Universally unique identifier
WHO	World Health Organization

List of Symbols

The symbols are grouped by the part of the thesis in which they are defined. Indices: i denotes a patient, t a time, c a channel (systolic and diastolic pressure, pulse, glucose), k a risk level. A few letters are reused in distinct contexts; the column on the right gives the section where each meaning applies.

Symbol	Meaning	Defined in
Measurement and follow-up		
$b_{i,t}$	blood pressure reading of patient i at time t	Section 4.2.2a
μ_i	usual blood pressure of patient i	Section 4.2.2a
$\varepsilon_{i,t}$, σ_ε	measurement error of a reading and its standard deviation	Section 4.2.2a
$\hat{\mu}_i$	estimate of the usual pressure from n readings	Section 4.2.2a
n	number of independent readings	Section 4.2.2a
$\bar{n}$, n_c	required contacts per patient per year; contacts per year once controlled	Section 4.2.2b
Sample size		
p	number of candidate parameters of a model	Section 3.5
S	global shrinkage factor	Section 3.5
R^2_{CS}	anticipated Cox–Snell R^2	Section 3.5
C	C statistic (area under the ROC curve) used to anticipate R^2_{CS}	Section 3.5
$\bar{m}$	size-weighted mean number of samples per patient	Section 3.5
ρ	intraclass correlation of the alert	Section 3.5
$1 + (\bar{m} - 1)\rho$	design effect	Section 3.5
P, u, f	parameters, units and inputs of an LSTM layer	Section 3.5.1
Preprocessing and samples		
$z_{i,t}$	robust (modified) z-score of a reading	Section 3.6.1b
MAD_i, $\mathrm{MAD}_{\mathrm{pooled}}$	median absolute deviation of a patient and of the cohort	Section 3.6.1b
$\widehat{\mathrm{MAD}}_i$, w_i, n_0	shrunk MAD, its weight, and the number of readings at which both estimates weigh equally	Section 3.6.1b
$\mathcal{S}_{i,t}$	training sample anchored at reading t of patient i	Section 3.6.1c

Symbol	Meaning	Defined in
L	length of the input window (hours)	Section 3.6.1c
ε	embargo between the input window and the label	Section 3.6.1c
Δ	forecast horizon	Section 3.6.1c
t'	time of the target reading	Section 3.6.1c
$g(\cdot)$, $h(\cdot)$	labelling rule applied to a reading	Sections 3.6.1c and 3.6.4b
$m_t^{(c)}$	mask: 1 if channel c is observed at t, 0 otherwise	Section 3.6.1d
$n_W^{(c)}$, $\Delta t_{\text{last}}^{(c)}$	number of observations of channel c in the window; time since its last observation	Section 3.6.1d
$r^{(c)}$	adherence ratio, observed over expected readings	Section 3.6.1d
$\tilde{x}_t^{(c)}$	input of the GRU-D model after decay	Section 3.6.1d
$\gamma_t^{(c)}$, κ_c	decay factor and decay rate of channel c	Section 3.6.1d
$\bar{x}^{(c)}$	mean of channel c in the training data	Section 3.6.1d
Labels and features		
$s_{c,t}$	severity level (0 to 3) of channel c at t	Section 3.6.2
y_t^{H}, y_t^{G}	haemodynamic and glycaemic labels	Section 3.6.2
y_t	overall alert label, worst available condition label	Section 3.6.2
s	sample standard deviation within a window	Section 3.6.3a
$\beta^{(c)}$	least-squares slope of channel c (per day)	Section 3.6.3b
θ	fixed threshold of a guideline rule	Section 3.6.3b
$\mathbf{v}_{i,t}$	feature vector of the window ending at t	Section 3.6.4d
Forecasting and evaluation		
λ	rate of routine readings	Section 3.6.4c
$\mathbb{E}[N_\Delta]$	expected number of readings within a horizon Δ	Section 3.6.4c
$\hat{x}_{i,t'}^{(c)}$	value forecast of channel c at the target	Section 3.6.4d
$\hat{\sigma}$	root mean squared error of out-of-fold value forecasts	Section 3.6.5e
Φ	cumulative distribution function of the standard normal law	Section 3.6.5e
p_k, p'_k, w_k	corrected and raw probability of level k; class weight	Section 3.6.5e
π	prevalence of the positive class	Section 3.6.5c
Se, Sp	sensitivity and specificity	Section 3.6.5c
C, w_m, z_m	selection cost, weight and standardised value of measure m	Section 3.6.5c
a, b	calibration intercept and slope	Section 3.6.5c
p_t	decision threshold on the alert probability	Section 3.6.5c
System design		
N, m	number of patients; readings per patient per week	Section 4.1.2c

Symbol	Meaning	Defined in
A	number of alerts per week	Section 4.1.2c
PPV	positive predictive value (precision)	Section 4.1.2c
$t_1, \ldots, t_6$, t'_2	segments of the end-to-end latency	Section 4.3.3
d, T, t_{act}	duty cycle, period between measurements, active window	Section 4.4.2
I_{act}, I_{slp}, I_{avg}	active, sleep and average current	Section 4.4.2
C	battery capacity	Section 4.4.2
$t^\star$	active window at which active and sleep consumption are equal	Section 4.4.2

Chapter 1

General Introduction

1.1 Background and Motivation

Hypertension is the leading modifiable risk factor for premature death worldwide. The first global report of the World Health Organization (WHO) devoted to the condition establishes that the number of adults aged 30–79 years living with hypertension doubled between 1990 and 2019, reaching 1.3 billion, that about 78% of them live in low- and middle-income countries, and that elevated systolic blood pressure is responsible for an estimated 10.8 million deaths each year [1]. Sub-Saharan Africa combines a high and increasing prevalence with persistently poor detection, treatment and control [2]: in the WHO African Region, only 12% of hypertensive adults have their blood pressure controlled.

Diabetes compounds this burden. In Africa, more than one person with diabetes in two is also hypertensive, and the highest sub-regional estimate concerns Central Africa, the region in which the present work is deployed [3]. The two conditions share their risk factors, aggravate each other, and call for the same kind of care: repeated measurement and treatment adjustment sustained over decades.

At the same time, the technological environment of the region is changing. Mobile telephony is now widespread and health technology is among the leading digital investment sectors in sub-Saharan Africa [4], and a controlled study in Cameroonian district hospitals has shown that remote clinician support is feasible in primary care [5]. A care model that fails at follow-up, and an environment in which remote follow-up has become conceivable: these two observations motivate the present work.

1.2 Problem Statement

Care for hypertension and diabetes in Cameroon is episodic. Blood pressure and blood glucose are measured at consultations that occur a few times a year, recorded on paper or in fragmented registers, and rarely assembled into a history that a clinician can read as a

trend. Yet diagnosis and treatment adjustment are longitudinal acts: the WHO protocol requires measurements repeated over several visits and monthly follow-up until the blood pressure target is reached [1, 6]. The consequences are loss to follow-up, which often affects half of patients or more in resource-limited settings [1], and therapeutic inertia, the failure to intensify treatment when blood pressure remains uncontrolled.

Telemonitoring could close this gap, but three obstacles stand in the way. First, the deployment environment imposes technological, organisational, financial, individual, cultural and regulatory barriers [7], and the comparative evidence on telemedicine adoption in the region comes from South Africa, Kenya and Nigeria rather than from Central Africa [8]. Second, the clinical evidence that telemonitoring improves blood pressure and glycaemic control comes almost entirely from high-income settings. Third, a system that processes real patient data through a mobile and cloud pipeline must earn the trust of patients in the confidentiality and security of their data [9].

The research question addressed in this thesis is therefore the following: *how can consultation data, home measurements and mobile sensor data be integrated into a telemonitoring system that detects the deterioration of patients with hypertension and diabetes early enough to support a clinical decision, under the infrastructural, organisational and regulatory constraints of the Cameroonian context?* The specific research questions and hypotheses that follow from the analysis of the literature are stated in Section 2.5.

1.3 Objectives

The general objective is to design a telemonitoring system for patients with hypertension and type 2 diabetes that combines consultation data, home measurements and, where available, mobile sensor data, and that supports the clinician through risk assessment and alert prioritisation. The specific objectives are:

1. to synthesise the evidence on the burden of the two conditions, the effectiveness of telemonitoring, the enabling technologies, the predictive methods and the deployment constraints (Chapter 2), and to derive from it the requirements of the system (Section 4.1);
2. to define a documented data schema, preprocessing pipeline and labelling rule, to develop and compare predictive risk models on public multi-centre data, and to validate them externally on Cameroonian patients (Chapter 3);
3. to design the architecture of the system: acquisition, backend, patient application and clinician dashboard (Chapter 4).

1.4 Scope and Data Actually Used

The system is designed for home follow-up, but the data available to this work were not collected at home. Three sources are used, each for one purpose. The public eICU Collaborative Research Database, which records intensive-care stays in United States hospitals, is used only to train and compare the predictive models. The observation charts of 32 patients hospitalised in Douala, with blood pressure, pulse and capillary glucose recorded morning and evening, are used only to validate those models externally, that is, to measure whether models trained elsewhere transfer to Cameroonian patients. The consultation records of 58 outpatients, most of whom have a single reading, describe the fragmentation of follow-up but cannot support longitudinal modelling. No home self-measurements or sensor data were collected: the home protocol and the acquisition hardware are designed and specified (Chapter 4) but not evaluated on patients. The predictive results therefore concern hospitalised patients, and their transfer to home data remains to be established.

1.5 Organisation of the Thesis

Chapter 2 reviews the literature: it defines the key concepts, analyses previous work in six themes, identifies the gaps and derives the research questions and hypotheses. Chapter 3 describes the methodology: type of research, setting and population, data, sampling, analysis methods, ethics and limitations. Chapter 4 presents the design of the telemonitoring system, the first result of the design approach, together with the home protocol it is built for. The experimental results are presented in the following chapter. The references cited throughout are listed in the bibliography at the end of the document.

Chapter 2

Literature Review

2.1 Introduction to the Review

A literature review situates a piece of research in its scientific field: it establishes what is already known, how it was established, and with what limits, so that the contribution of the work can be stated precisely and defended. For the present thesis this is all the more necessary as the subject lies at the crossing of three fields that rarely meet: the epidemiology of chronic diseases in sub-Saharan Africa, the clinical evaluation of telemonitoring, and the engineering of connected and predictive systems.

The general context is that of hypertension and type 2 diabetes in Cameroon: two frequent, often associated conditions, whose care fails mainly at follow-up, in a health system with few specialists, fragmented records and intermittent connectivity, but where mobile telephony is widespread (Chapter 1).

The review draws on nineteen peer-reviewed sources, each checked against PubMed, PubMed Central, Crossref or the publisher's record, complemented by three institutional documents of the WHO [1, 6, 10], by the TELEMED-CAM study [11], and by the 2023 guidelines of the European Society of Hypertension (ESH) [12], from which the blood pressure thresholds are taken. Two architecture references, Mucchi et al. [13] and Yazdani et al. [14], are used for the system design and are presented where they are used (Section 4.1). It is organised as follows. Section 2.2 defines the key concepts and the models that structure the reasoning. Section 2.3 analyses previous work in six themes, from the problem to the response (Table 2.1); within each theme the studies are presented in chronological order, each study once, and the section ends with a critical comparison of the approaches. Section 2.4 identifies the limits of the existing literature, Section 2.5 links them to the present research and formulates the research questions and hypotheses, and Section 2.6 summarises the review and prepares the transition to the methodology.

Table 2.1: Sources of the review, by theme and in chronological order within each theme.

Theme	Source	Design and setting
1. Burden of disease	Hinneh et al. [3]	systematic review and meta-analysis, Africa
	World Health Organization [1]	institutional global report
2. Care and digital health in sub-Saharan Africa	Kingue et al. [11]	interventional pilot, rural Cameroon
	Bediang et al. [5]	controlled multicentre study, Cameroon
	Dodoo et al. [7]	review of 66 empirical studies, sub-Saharan Africa
	Gafane-Matemane et al. [2]	narrative review, sub-Saharan Africa
	Agbeyangi and Lukose [8]	systematic review, South Africa, Kenya, Nigeria
	Segun-Omosehin et al. [4]	perspective, sub-Saharan Africa
3. Mobile health and clinical evidence	Alessa et al. [15]	review and content analysis of applications, United Kingdom
	Mao et al. [16]	meta-analysis of 51 randomised trials
	Zhou et al. [17]	meta-analysis of randomised trials
	Bandeira et al. [18]	updated meta-analysis of randomised trials
	Mihevc et al. [19]	multicentre randomised trial, Slovenia
4. Sensors and IoT	Alhaddad et al. [20]	review of non-invasive glucose sensing
	Tai et al. [21]	IoT architecture and deep learning technique
	Efendi et al. [22]	IoT and cloud implementation with device validation
	Liu and Wang [23]	review of IoT sensing and machine learning
	Liu et al. [24]	smartphone photoplethysmography study
5. Machine learning	Chatrati et al. [25]	home monitoring system with machine learning
	Mendoza-Cano et al. [26]	cross-sectional predictive study, Mexico
6. Privacy and ethics	Alhammad et al. [9]	systematic review of patient perspectives

2.2 Conceptual Framework

2.2.1 Key concepts

The terms of Table 2.2 are used throughout the thesis with the meaning given here. Several of them are used differently in the literature; fixing them avoids ambiguity.

Table 2.2: Key concepts and the definitions adopted in this thesis.

Concept	Definition adopted
Hypertension	systolic pressure $\geq$ 140 mmHg, diastolic pressure $\geq$ 90 mmHg, or antihypertensive treatment [1]; the ESH grades (grade 1 from 140/90, grade 2 from 160/100, grade 3 from 180/110 mmHg) are used for risk labels [12]; the home threshold is 135/85 mmHg [12]
Type 2 diabetes	chronic hyperglycaemia due to insulin resistance and relative insulin deficiency; its diagnosis and follow-up require a biological measurement (capillary or venous glucose, glycated haemoglobin)
Comorbidity	coexistence of hypertension and type 2 diabetes in the same patient
Care cascade	nested stages of care—hypertensive, diagnosed, treated, controlled—measured on the same population [1]
Loss to follow-up	interruption of the planned follow-up of a patient already in care
Therapeutic inertia	failure to intensify treatment despite blood pressure remaining above target
Telemedicine	delivery of care at a distance through information and communication technologies; it includes teleconsultation, tele-expertise between professionals and telemonitoring
Telemonitoring	transmission of physiological data from the patient's home to a care team, which provides clinical feedback
Mobile health (mHealth)	health practices supported by mobile devices: applications, text messages, connected devices
Self-measurement	measurement of blood pressure or capillary glucose by the patient at home, with a validated device and a standard protocol
Internet of (Medical) Things	connected devices that acquire and transmit data without human intervention; in health, sensors and instruments linked to a platform
Edge and cloud	local processing on a device close to the patient (edge) and remote storage and processing on shared servers (cloud)
Photoplethysmography (PPG)	optical measurement of blood volume variations in the skin, from which heart rate is derived
Clinical decision support system	system that turns patient data into recommendations or alerts for a clinician, who keeps the decision
Supervised learning	estimation of a function mapping input features to a known label; here, a forecast of a future clinical state from a past window of readings
Label leakage	situation in which the label can be recovered from the inputs, so that measured performance reflects a rule rather than a prediction
Risk score and triage	calibrated probability of deterioration, used to order patients by priority for clinical review

2.2.2 Models structuring the work

Four models structure the reasoning of the thesis (Figure 2.1).

The care cascade. The WHO describes hypertension care as a cascade in which patients are lost at each stage [1]. The model locates where an intervention acts: screening acts on diagnosis, whereas longitudinal follow-up acts on treatment and control, where most diagnosed patients in Africa are lost (Section 2.3.1b).

The measurement model. A reading is the patient's usual pressure plus a variable error; repeated readings reduce the error in proportion to their number. This simple model justifies home self-measurement and the use of series rather than isolated values (Section 4.2.2a).

The layered telemonitoring architecture. Remote monitoring systems are described as a chain from acquisition to processing and visualisation [21], distributed over body, environment, edge and cloud layers [13]. The model structures the system design of Chapter 4.

Supervised learning for clinical prediction. A model maps features computed from a past window to a future clinical state. The framework fixes what a valid evaluation requires: a label outside the input window, metrics suited to rare events, and a comparison with the guideline rule (Section 3.6.4b). Behaviour change techniques, coded with the taxonomy used by Zhou et al. [17], complete it on the patient side.

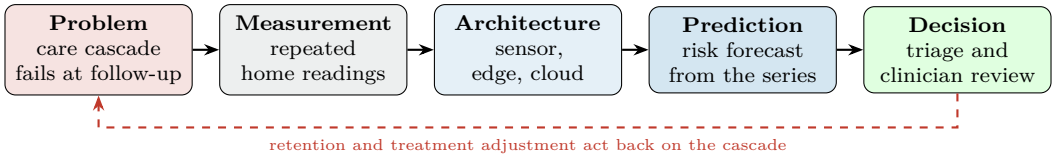


Figure 2.1: Conceptual framework of the thesis.

2.3 Previous Work

This section analyses the published work in six themes, from the burden of the diseases to the protection of data. For each study it reports the method, the main results and the limits, and Section 2.3.7 compares the approaches critically.

2.3.1 Global Burden of Hypertension and Diabetes

2.3.1a The global burden of hypertension

The WHO global report on hypertension [1] is the reference framing of the problem. For population surveillance it defines hypertension as a systolic blood pressure (SBP) $\geq$ 140 mmHg, a diastolic blood pressure (DBP) $\geq$ 90 mmHg, or the use of antihypertensive

medication, in adults aged 30–79 years. This threshold is used throughout this thesis for epidemiological figures. The ESH uses the same 140/90 mmHg threshold and divides hypertension into three grades [12], which are used later to define the risk levels; some publications use the lower 130/80 mmHg threshold of the American guidelines, which explains part of the discrepancies between them.

The report establishes that the number of hypertensive adults rose from 650 million in 1990 to 1.3 billion in 2019, while the age-standardised prevalence barely changed, from 32% to 33%: the increase in absolute numbers is mostly explained by population growth and ageing rather than by a rise in individual risk. Elevated systolic blood pressure is the leading risk factor for premature death among the eighty-seven factors compared, ahead of smoking and hyperglycaemia, with 10.8 million deaths and 235 million disability-adjusted life years each year. Progress towards the global target is insufficient: the prevalence of uncontrolled hypertension fell only from 29% in 2010 to 26% in 2019, and no WHO region is on track. The report notes, however, that the target would be met if about half of the adults already diagnosed were effectively treated. Among the recommended responses, it endorses team-based care, community and home blood pressure measurement, and telemonitoring.

As an institutional synthesis, the report relies on modelled estimates and on references that it does not detail; it does not address patient-level monitoring technology. It is used here for the global framing and for its normative positions, not as primary evidence.

2.3.1b The failure of the care cascade

The WHO structures the care of hypertension as a cascade of nested stages measured on the same denominator, all hypertensive adults (Table 2.3).

Table 2.3: Hypertension care cascade by WHO region, adults aged 30–79 years, 2019 (adapted from [1], Table 2).

Indicator	**African Region**	**World**	**Americas**	**Europe**
Age-standardised prevalence	36%	33%	35%	37%
Diagnosis coverage	43%	54%	70%	66%
Treatment coverage	27%	42%	60%	53%
Effective treatment (control)	12%	21%	36%	26%

Two observations follow. First, the European Region has a higher prevalence than the African Region but a control rate more than twice as high: the African deficit is organisational rather than epidemiological. Second, the conditional losses derived from these figures are spread along the whole chain: in the African Region, 57% of hypertensive adults are undiagnosed, 37% of the diagnosed are untreated, and 56% of the treated are not controlled. Losses at diagnosis and at control are of the same size, but they call for different responses. Among patients already diagnosed, 72% (31 of 43 points) are not

controlled: they are known to the health system and lost during follow-up. A screening device acts only on the first link; a longitudinal follow-up system acts on the second and third.

2.3.1c The hypertension–diabetes comorbidity

The association between hypertension and type 2 diabetes rests on shared risk factors—physical inactivity, unbalanced diet and obesity—and on a direct pathophysiological link: diabetes stiffens small vessels, which raises peripheral resistance and hence blood pressure, while hypertension accelerates the vascular damage of diabetes.

Hinneh et al. [3] produced the first systematic quantification of this comorbidity in Africa. Their meta-analysis, conducted according to PRISMA on PubMed, EMBASE and HINARI up to March 2023, pooled forty-one cross-sectional studies totalling 144 433 participants. The pooled prevalence of hypertension among people with diabetes is 58.1% (95% CI 52.0–63.2), and 60.8% among people with type 2 diabetes alone. Older age, overweight and obesity, longer duration of diabetes, urban residence and male sex are associated with a higher risk. The authors also note that patients with both conditions paradoxically have lower chances of receiving appropriate care before complications occur.

The sub-regional estimates place Central Africa highest (Table 2.4 and Figure 2.2).

Table 2.4: Sub-regional pooled prevalence of hypertension among people with diabetes in Africa [3].

Sub-region	Pooled prevalence	95% CI	Studies
Central Africa	77.6%	53.0–91.4	2
Southern Africa	69.1%	59.8–77.1	10
Eastern Africa	53.0%	45.8–59.1	15
Western Africa	51.5%	41.8–61.1	10

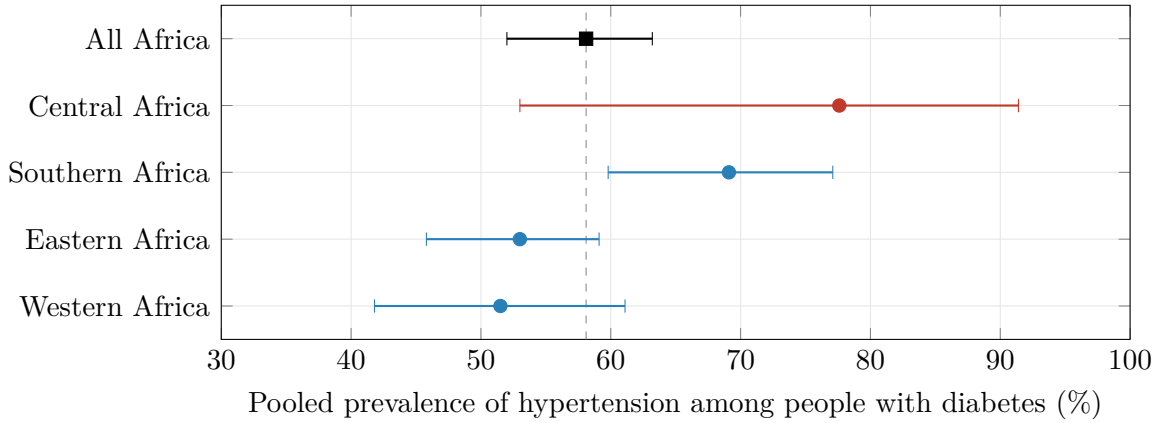


Figure 2.2: Point estimates and 95% confidence intervals of Table 2.4; the dashed line marks the pooled African estimate. The Central African interval spans 38 points.

Reliability of the Central African estimate. The value of 77.6% must be reported with its confidence interval, 53.0–91.4%, and it cannot by itself carry the justification of this work. It rests on two small Cameroonian hospital series (109 and 210 patients), while three other Central African studies listed in the article (1 832 patients, prevalences of 35.0–50.4%) do not appear in this subgroup. Heterogeneity is extreme ($I^2 \approx 98\%$), the included studies are mostly hospital-based and use varying diagnostic thresholds, and the restriction to English-language publications under-represents francophone Central Africa, as the authors themselves acknowledge. The figure also measures hypertension among people with diabetes, not the reverse.

The case for monitoring blood pressure and blood glucose jointly therefore rests on the robust continental estimate of 58.1%, on the pathophysiological mechanism, and on the position of the WHO [1]: uncontrolled hypertension, not hyperglycaemia, is among the risk factors most strongly associated with premature death in people with diabetes; Member States have adopted a target of 80% of people with diagnosed diabetes having good blood pressure control; and the HEARTS package developed for hypertension is explicitly applicable to diabetes and other long-term conditions.

2.3.1d Comparability of estimates

The prevalences quoted in this chapter are not directly comparable. They differ in reference population (general population, people with diabetes or hospital attendees), in diagnostic threshold (140/90 or 130/80 mmHg), in age structure, in the number of measurement visits—a single-visit measurement overestimates prevalence—and in recruitment mode, opportunistic screening and self-selection producing samples systematically different from probability sampling. None of the available sources is nationally representative for Cameroon. The nature of each estimate is therefore stated where it is used.

2.3.2 Hypertension Care and Digital Health in Sub-Saharan Africa

2.3.2a Burden, determinants and care gaps

Gafane-Matemane et al. [2] synthesise the hypertension literature of sub-Saharan Africa published between January 2017 and June 2023, drawn from MEDLINE, PubMed, CINAHL, African Journals Online and grey literature. Most countries of the region reported prevalence data over the period, confirming a high burden with poor detection and control. The authors relate the rise of hypertension to the shift towards a westernised lifestyle—salt, sugar and fat, physical inactivity, obesity, tobacco and alcohol—together with socioeconomic and psychosocial factors, and they identify human immunodeficiency virus infection as a regional specificity; among the mechanisms still under investigation they list

sodium handling, endothelial function and circadian profiles such as nocturnal dipping. They distinguish individual obstacles to care (health literacy, cost, distance) from system obstacles (medicine availability, staffing and training, uneven political will). Community-based task shifting improves linkage to care and medication adherence, but its effect on blood pressure over time is inconsistent.

Their inventory of national surveys is the most recent available for Central Africa (Table 2.5). Dispersion is wide, from 13.5% to 42.1%, and reflects the diversity of instruments as much as real differences; three countries of the sub-region have no data for the period. In Cameroon, the May Measurement Month campaign of 2019 found a prevalence of 20.8%, with only 11.2% of hypertensive participants controlled.

Table 2.5: Most recent hypertension surveys in Central Africa (adapted from [2], Table 1).

Country	**Instrument**	**Year**	n	**Prev.**	**Aware.**	**Treat.**	**Control**
Burundi	hospital screening	2018	353	16.7%	–	–	–
Cameroon	May Measurement Month	2019	30 167	20.8%	29.9%	24.0%	11.2%
Central African Rep.	WHO STEPS	2017	3 265	42.1%	30.2%	–	–
Comoros	national survey	2019	2 602	38.4%	47.9%	45.4%	30.2%
DR Congo	May Measurement Month	2019	29 857	25.5%	33.1%	23.2%	11.9%
Congo (Rep.)	May Measurement Month	2019	3 157	33.5%	42.6%	37.3%	23.3%
Gabon	retrospective study	2013–15	2 104	13.5%	–	–	–
Chad, Eq. Guinea, São Tomé	no data	–	–	–	–	–	–

The review is narrative rather than pooled, the surveys it gathers are heterogeneous, and some figures are inconsistent: the same pair of values (48% and 34%) is presented once as prevalence and once as diagnosis rate. It is used here for the regional profile and the care cascade, not for pooled prevalence.

2.3.2b Why the episodic care model fails

The WHO documents the mechanisms behind the failure of the cascade [1]. *Loss to follow-up* often affects half of patients or more in resource-limited settings; the report therefore holds that screening is useful only if linked to effective follow-up, and it recommends opportunistic detection in health facilities together with retention of identified patients rather than population screening. *Therapeutic inertia*, the tendency not to intensify treatment despite uncontrolled blood pressure, explains why the last link of the cascade is the weakest; a simple standardised protocol helps reduce it. Lifestyle modification

alone lowers systolic pressure by only about 1.3 mmHg in primary-care trials, so almost all diagnosed patients need medication, and its timely adjustment. *Consultation time* is often below five minutes in primary care, which requires digital tools to be fast and to replace paper rather than add to it. Finally, *paper records* are duplicated, lost and unavailable during the visit, whereas a secure longitudinal record is described as essential for hypertension, diabetes and other long-term conditions. The core indicators of the HEARTS programme, two of which are temporal, define what such a record must produce (Section 4.1.1a).

The expected follow-up frequency is set by Recommendation 7 of the WHO guideline on the pharmacological treatment of hypertension: monthly follow-up after initiating or changing treatment until the target is reached, then every three to six months once blood pressure is controlled [6]. The HEARTS treatment-protocol module adds that the return interval should usually be four weeks or less during titration [10]. The gap between this requirement and the one or two contacts a year that an episodic model delivers is quantified in Section 4.2.2b.

The programmes cited by the WHO point in the same direction. In Sri Lanka, 86% of enrolled patients return every three months and 57% are controlled; in the Philippines, India and Cuba, part of the follow-up was moved out of the facility, towards the home or the community. These are associations observed in programmes that changed several components at once, not causal estimates. The Indian programme also shows that retention depends on medicine supply: stock-outs soon after launch discouraged patients from returning, so that a telemonitoring system deployed during stock-outs would probably not improve control, whatever its technical performance.

2.3.2c Telemedicine evidence from Cameroon

Two controlled evaluations of cardiovascular telemedicine have been conducted in Cameroonian primary care.

TELEMED-CAM (2008). Kingue et al. [11] evaluated a hypertension management package in thirty rural health centres supported by a telemedicine centre at Yaoundé General Hospital. The package combined a standardised protocol restricted to generic drugs, lifestyle counselling, training of general practitioners and nurses, and a telecare channel through which providers reported patient data daily by text or voice message and received telephone feedback. Among 268 participants followed for 24 weeks, attendance at scheduled visits was higher in the intervention arm at every grade of hypertension ($p \leq 0.04$), and systolic pressure fell from the second visit. The benefit on control was confined to stage III hypertension (no improvement in 16.7% against 33.3%, $p = 0.04$); for stages I and II the proportions at target were similar (65.2% against 70%, $p = 0.20$). Allocation was not strictly random, clustering was ignored, the arms differed at baseline

(systolic pressure 169 against 161 mmHg, diabetes 9.1% against 43.6%), and telecare was only one component of the package, so its own effect cannot be isolated. The study also quantifies the cost of systematic remote review: about six calls per provider per day, of about five minutes each.

Tele-expertise in district hospitals (2016–2017). Bediang et al. [5] compared two district hospitals equipped with a twelve-lead electrocardiograph and an internet tele-expertise platform (Mbouda and Akonolinga) with two matched control hospitals (Foumbot and Sa'a). Among 171 patients, access to an electrocardiogram with a cardiologist's opinion reached 98.9% against 33.3% (relative risk 2.97), the expert modified the local provider's diagnosis in 46.2% of cases against 17.9%, and patient satisfaction was higher. The study was not randomised and the groups differed at baseline; its primary outcome measures the availability of the service rather than its clinical effect. The reported blood pressure benefit (40.9% against 17.9% of patients with a favourable evolution) counts the 52 control patients who never returned as not improved; restricted to the patients actually reviewed at three months, the proportions become 41.3% against 53.8%. The specialist's opinion also took 10.3 days on average against a protocol maximum of 72 hours, no faster than in the control centres (11.2 days, $p = 0.82$): the bottleneck was expert availability, not transmission.

Table 2.6: The two Cameroonian evaluations compared.

	Kingue et al. [11]	**Bediang et al. [5]**
Period	2008	2016–2017
Function evaluated	telemanagement of treatment	tele-expertise for diagnosis
Design	prospective, partly randomised by centre	controlled, matched centres, not randomised
Participants	268	171
Technology	GSM handset, SMS and voicemail	portable 12-lead ECG, web platform
Principal result	control improved in stage III only	access to ECG with expertise, 98.9% vs 33.3%
Retention result	visit adherence higher at every grade	52 control patients not seen at 3 months
Main weakness	clustering ignored, package effect	non-comparable groups, differential attrition

Read together (Table 2.6), the two studies support three conclusions. Retention is the most robust result, observed by different mechanisms eight years apart. An effect on blood pressure control is not established in Cameroon. And the value added lies in the clinical decision—a revised diagnosis, a supported prescriber—rather than in a tool placed in the patient's hands. Both studies also show that systematic expert review of every case does not scale where specialists are scarce.

2.3.2d Digital health landscape and barriers to adoption

Dodoo et al. [7] reviewed sixty-six empirical studies and government reports on telemedicine in sub-Saharan Africa, drawn from eight databases under adapted PRISMA guidelines. Progress is markedly uneven between countries, and the barriers fall into six dimensions (Table 2.7). Two findings bear directly on design: most projects lost momentum once external funding ended, and health professionals facing a heavy workload perceive telemedicine as an additional task. The authors also report that store-and-forward services suffered from low user satisfaction for lack of real-time interaction. Their inventory lists for Cameroon telecardiology, wound-care and tele-education services, including the Telemed-cam project, and they record forty-one countries with national digital health strategies. Francophone countries, Cameroon included, are under-represented in the literature they review.

Table 2.7: Six dimensions of barriers to telemedicine in sub-Saharan Africa [7].

Dimension	**Definition given by the authors**
Technological	information systems and connectivity
Organisational	stakeholder engagement, sustainability of systems
Individual	users' experience, knowledge and awareness
Financial	financial resources and operating cost
Cultural	social constraints and traditional beliefs
Legal and regulatory	policies, legislation, confidentiality and privacy protection

Agbeyangi and Lukose [8] extend this picture with a PRISMA-guided systematic review of fifty-three studies published between 2014 and 2024, comparing South Africa, Kenya and Nigeria. South Africa leads adoption, including specialist teleconsultation and chronic disease management; Kenya shows strong integration of mobile health; Nigeria is advancing remote diagnosis under infrastructural constraints. Regulatory gaps, digital literacy and infrastructure remain the persistent barriers. Central Africa is absent from the comparison.

Segun-Omosehin et al. [4] assess the digital health landscape of the region from the perspective of cardiovascular care. More than 80% of the population holds a mobile telephone subscription, mobile internet penetration stood at 25% in 2023, and smartphone adoption is projected to reach 88% by 2030; yet about 550 million people lack reliable access to electricity and the internet, and up to 18% of countries had no registered cardiologist in 2020. The collapse of the Rwandan platform Babyl in 2023, after about 2.5 million users had registered, illustrates that adoption is not sustainability. Artificial intelligence remains largely untested in the region, with few validation studies in African populations and a documented risk that externally trained algorithms perform poorly there. As a perspective article without a declared search strategy, this source is used for contextual orders of magnitude, not as pooled evidence.

Taken together, these sources describe a population that is reachable by mobile networks

and a specialist who is not. A system that forwards every reading to a clinician reproduces the scarcity it aims to relieve; a system that sorts readings and orders the clinician's attention addresses it. They also impose offline operation, a simple interface for both patients and providers, and a documented operating cost.

2.3.3 Mobile Health Self-Management and Clinical Evidence

2.3.3a What self-management applications offer

Alessa et al. [15] analysed the 186 hypertension self-management applications available in English on the two main application stores in January 2018, coding thirty of them in depth. Most applications offered a narrow set of functions: 58.1% offered a single function, only 17.7% offered three or more, and none offered all eight identified. Self-monitoring was present in 53.2% of applications, feedback in 28.0% and reminders in 21.0%; most of the applications that tracked blood pressure required the patient to type in the value displayed by the monitor. None of the thirty applications analysed in depth had been scientifically evaluated or certified as a medical device, and none reported the involvement of patients in its design. Three applications claimed to measure blood pressure with the smartphone camera without any evidence of validity. Of the thirty, only twenty had a privacy policy, and fourteen of these declared sharing data with third parties. In terms of behaviour change techniques, coded with the taxonomy of Michie and colleagues, the applications contained a median of nine techniques, with self-monitoring, prompts and cues, and action planning present in all thirty. As a content analysis of a market snapshot, the study describes what applications contain, not whether they work.

2.3.3b Pooled clinical effects

Mao et al. [16] pooled fifty-one randomised trials published between 2008 and 2019, with 13 054 participants with diabetes or hypertension. Mobile health interventions reduced glycated haemoglobin (HbA1c) by 0.39 points (−0.50 to −0.29), fasting glucose by 0.52 mmol/L, systolic pressure by 2.99 mmHg (−4.19 to −1.80) and diastolic pressure by 1.14 mmHg. Two subgroup results are decisive for this work. Interventions combining mobile health with professional management reduced systolic pressure by 6.17 mmHg (−8.83 to −3.50), whereas mobile health alone produced a non-significant 2.16 mmHg. And the effect on systolic pressure was present in developed countries (−5.72 mmHg) but absent in developing countries (+0.25 mmHg, −3.10 to 3.59), whose trials used mobile health without professional management; the glycaemic effect, by contrast, was larger in developing countries (−0.52 against −0.35 points). Only one of the fifty-one trials was conducted in sub-Saharan Africa. Heterogeneity was moderate to high (I^2 of 57–67%) and most trials were short.

Zhou et al. [17] focused on mobile application self-management for adults with hypertension: twenty randomised trials published from 2017 onwards, sixteen of them meta-analysed, with 4 168 participants from nine countries, none in sub-Saharan Africa. Systolic pressure fell by 5.78 mmHg (−7.97 to −3.59; low certainty, $I^2 = 82\%$) and diastolic pressure by 3.28 mmHg (moderate certainty). The subgroup analyses link the effect to the design of the intervention (Table 2.8): interventions with at least eleven behaviour change techniques and those addressing several risk factors were markedly more effective, and three-month interventions tended to outperform longer ones, a difference that is not statistically significant ($P = .09$) and that the authors attribute to the weakening of habit after about twelve weeks. Fifteen of the twenty trials were at high risk of bias, and interventions based mainly on text messages or calls—the type most often evaluated in Africa—were excluded.

Table 2.8: Subgroup analyses on systolic pressure [17]. BCT: behaviour change technique.

Factor	**Modality**	**Effect on SBP (95% CI)**	*P*
Risk factors addressed	multiple	−6.50 mmHg (−9.00 to −3.99)	.007
	single	−1.54 mmHg (−4.15 to 1.06)	
Number of techniques	≥ 11 BCT	−9.68 mmHg (−13.49 to −5.87)	<.001
	< 11 BCT	−2.88 mmHg (−3.90 to −1.86)	
Duration	3 months	−8.87 mmHg (−10.90 to −6.83)	.09
	≥ 6 months	−5.76 mmHg (−8.74 to −2.77)	

Bandeira et al. [18] updated the synthesis of randomised trials of mobile health interventions on blood pressure. They confirm statistically and clinically significant reductions across diverse populations and settings, and identify telemonitoring with clinician feedback as the most effective modality. Heterogeneity between interventions, short follow-up in several trials and the scarcity of data from low- and middle-income countries limit the generalisation of these results.

2.3.3c A randomised trial in the target population

Mihevc et al. [19] conducted the trial most directly relevant to this work: a multicentre, open-label randomised trial in three Slovenian primary-care centres between 2021 and 2023, enrolling 128 patients aged 65 years or more with both hypertension and type 2 diabetes. In addition to usual care, the intervention group measured blood pressure and blood glucose at home with Bluetooth devices that transmitted automatically through a smartphone to a cloud platform; a telemedicine centre coordinated by a general practitioner reviewed the values according to a three-level protocol (green, yellow, red) detailed in Section 4.2.2c. After twelve months, systolic pressure was 6.9 mmHg lower than in the

control group (−11 to −2.7), after a difference of 10.5 mmHg at six months, and HbA1c was 0.5 points lower, as reported by the authors.

Three further results matter. Telemonitoring had no effect on other cardiovascular risk factors, on diabetes-related quality of life or on lifestyle: the authors attribute the clinical effect to improved communication, medication adherence, early detection of exacerbations and treatment adjustment based on accurate data, not to behaviour change. Seven of the 64 patients of the intervention arm (10.9%) withdrew in the first month because they could not use the equipment, despite one hour of individual training. And about a third of the systolic effect was lost between six and twelve months. The trial was conducted in a high-income setting, among older adults, without blinding; its applicability to low-resource settings requires adaptation.

2.3.3d Synthesis of the clinical evidence

Table 2.9 and Figure 2.3 gather the available estimates in chronological order.

Table 2.9: Estimates of the effect of mobile health and telemonitoring on systolic blood pressure and glycaemic control.

Source	Design	Effect on SBP (95% CI)	Glycaemic effect
Mao et al. [16], overall	30 RCTs on SBP	−2.99 mmHg (−4.19 to −1.80)	HbA1c −0.39
with professional management	subgroup	−6.17 mmHg (−8.83 to −3.50)	–
mobile health alone	subgroup	−2.16 mmHg (−5.07 to 0.75), n.s.	–
developing countries	subgroup	+0.25 mmHg (−3.10 to 3.59), n.s.	HbA1c −0.52
Zhou et al. [17]	16 RCTs	−5.78 mmHg (−7.97 to −3.59), low certainty	not assessed
Mihevc et al. [19]	RCT, 128 randomised	−6.9 mmHg (−11 to −2.7)	HbA1c −0.5

Four conclusions emerge. First, mobile health and telemonitoring produce a modest but clinically relevant effect, of the order of −3 to −7 mmHg on systolic pressure and −0.4 to −0.5 points on HbA1c; these are intermediate outcomes, and none of the sources measures morbidity or mortality. Second, the effect depends on professional oversight: Mao et al. [16], Bandeira et al. [18] and Mihevc et al. [19] converge on this point by independent routes, and so do the Cameroonian studies of Section 2.3.2c. Third, the evidence from sub-Saharan Africa is almost absent, and the absence of effect observed in developing countries is plausibly a design gap—no clinician at the end of the chain—rather than a context effect. Fourth, two problems remain open: the erosion of the effect over time, and the inability of a fraction of patients to use the equipment. Telemonitoring must therefore be built as a decision-support system linked to a clinician, not as a self-tracking

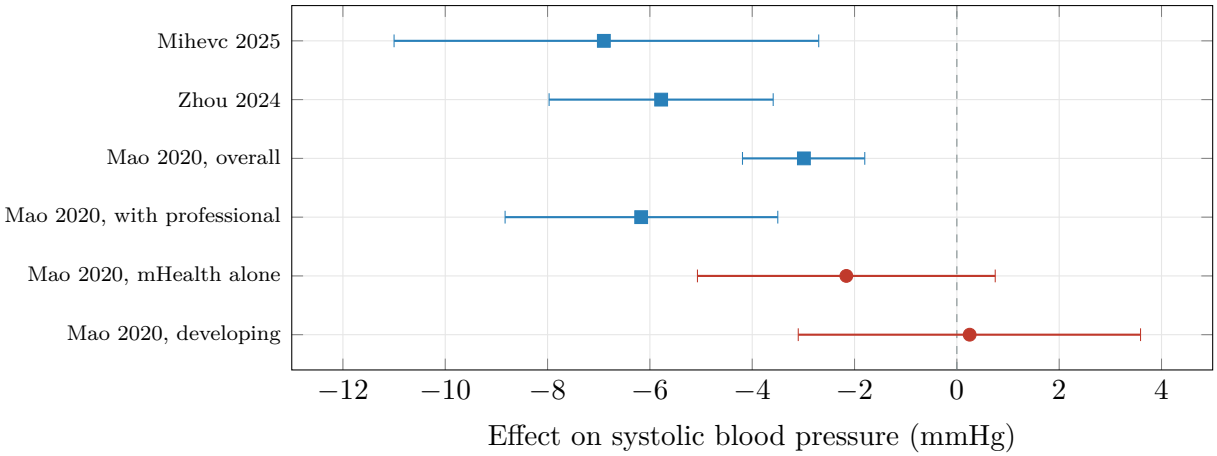


Figure 2.3: Estimates of Table 2.9. Squares mark intervals excluding the null, circles those including it. The two non-significant estimates describe interventions without professional management or conducted in developing countries.

tool handed to the patient.

2.3.4 Sensor and IoT Technologies for Chronic Disease Monitoring

2.3.4a The normative position on measuring devices

The WHO sets the frame within which any sensor must be assessed [1]. Manual devices should be replaced by independently validated automated electronic devices, and automated upper-arm cuff devices are recommended. Cuffless devices, including popular connected wristbands, are not recommended for the management of hypertension, because of unresolved questions about their accuracy and their ability to track blood pressure change; as of 2023, no alternative to the validated upper-arm cuff is accurate enough for diagnosis and management. Only about one automated device in five on the market has evidence of validation.

2.3.4b Non-invasive sensing of glucose and blood pressure

Alhaddad et al. [20] reviewed the non-invasive sensing modalities proposed for blood glucose—electrocardiography, electromagnetic sensing, bioimpedance and photoplethysmography (PPG)—and the machine learning models applied to them. PPG- and ECG-based sensing combined with deep learning shows promise for detecting glucose trends and hypoglycaemia, but not for clinical-grade absolute measurement: many studies have small samples, inter-individual variation limits generalisation, and accuracy remains below the standards recognised by the Food and Drug Administration.

Liu et al. [24] estimated blood pressure from the PPG signal recorded by a smartphone camera alone, without additional hardware. Combining waveform feature engineering with

machine learning, and using SHAP values to trace each prediction back to interpretable physiological features, they report a strong correlation with reference devices and assess agreement by Bland–Altman analysis on 127 participants. The sample is modest and single-site, accuracy remains below that of clinical cuff devices, and generalisation across skin tones, devices and lighting conditions is not established.

These two studies, set against the WHO position and the unvalidated camera applications found by Alessa et al. [15], define a consistent position. Blood glucose must come from a glucometer and blood pressure from a validated upper-arm cuff, because these are the measurements on which clinical decisions rest. Smartphone PPG is a plausible low-cost source of heart rate and an exploratory source of blood pressure trends, to be evaluated rather than relied upon (Section 4.2.3a).

2.3.4c IoT architectures for remote monitoring

Tai et al. [21] address the difficulty of applying conventional machine learning to noisy, heterogeneous and high-dimensional IoT health data. They propose a human-centric deep learning technique for remote chronic disease monitoring, embedded in a standardised three-phase architecture: data acquisition, data processing and data visualisation. The architecture supports accurate and interpretable predictions in their evaluation, which was conducted in controlled conditions and does not address the connectivity constraints of low-resource settings.

Efendi et al. [22] describe and validate a complete IoT system for monitoring elderly people. Its three layers are a physical layer (a MAX30100 optical sensor for heart rate and oxygen saturation and a MAX30205 temperature sensor, read by an ESP32 microcontroller), a network layer (MQTT transmission to a Firebase real-time database) and an application layer (mobile application and website accessed over HTTP). Firebase was chosen for real-time synchronisation, built-in authentication, and local persistence with automatic merging when connectivity returns. Against a digital oximeter and thermometer, six participants measured for ten minutes at rest gave a mean absolute percentage error of 0.90%, and XGBoost was the best of five classifiers of the user's state as stable or not stable. The system measures heart rate, oxygen saturation and temperature, not blood pressure or blood glucose; its validation sample is small, the reference instruments are not identified as validated, and its classifier is examined critically in Section 2.3.5.

Liu and Wang [23] survey the use of IoT mobile sensing and machine learning for cardiovascular disease, cancer, chronic respiratory disease and diabetes. They classify devices into portable, wearable, intelligent clinical, remote, implantable and ingestible categories, describe a seven-layer platform from device access to decision support, and identify the weaknesses of traditional monitoring: manual processes, information silos, lack of real-time feedback and of early-warning mechanisms. They also note that electronic medical records require cleaning of duplicates, unit errors and values carried over identically

between visits. Although presented as a systematic review, the article reports no protocol, flow diagram or number of included studies, so it is treated here as a narrative survey. Controlled settings dominate the studies it covers, and deployment in low- and middle-income countries is poorly represented.

The architectures converge on a common structure: acquisition, transmission and storage, processing, and presentation. None of them has been validated on real patients in routine care in a low-resource setting, and none acquires blood pressure and blood glucose through its own sensors. The detailed implications for acquisition and architecture are developed in Sections 4.1.1 and 4.2.3.

2.3.5 Machine Learning for Clinical Risk Prediction

Chatrati et al. [25] proposed a smart home monitoring system that analyses home blood pressure and glucose readings and notifies the healthcare provider when an abnormality is detected, combining conditional rules for hypertension with a learned classifier for diabetes. It is the closest published precedent of a system combining both measurements with machine learning. As described, however, it classifies single readings rather than series, and it has not been evaluated on patients in routine care.

Liu and Wang [23] report that artificial neural networks, support vector machines, random forests, gradient boosting and deep learning are the dominant algorithm families for chronic disease monitoring. The accuracies they collect cluster between 95% and 100%, whereas the only recurrent network applied to a genuine clinical prediction task, the prediction of readmission in lupus, reaches an area under the curve (AUC) of 0.70. They name the black-box nature of these models and their dependence on large labelled datasets as their main limitations.

The classifier comparison of Efendi et al. [22] illustrates why such figures must be read with care. XGBoost reached an accuracy of 0.973 and the four other models exceeded 0.92. However, the label *stable* is defined by the same normal ranges that are used to discretise the three input features, so the label is a deterministic function of the inputs: the models recover a rule rather than predict a state. Some reported metrics are also mutually inconsistent, and the training configuration described (epochs, sigmoid activation, Adam optimiser) does not apply to tree-based models. The result does not show that XGBoost outperforms other algorithms (Section 3.6.4b).

Mendoza-Cano et al. [26] provide the counter-example of a genuine prediction task. Using 1 308 hypertensive adults from the nationally representative Mexican survey ENSANUT 2022, they predicted uncontrolled blood pressure (130/80 mmHg threshold, prevalence 30.3%) from seven predictors selected by LASSO: sex, age, socio-economic status, years since diagnosis, history of diabetes, source of medical care and adherence to a nutritional plan. Random forest reached an AUC of 0.75 (0.72–0.77), logistic regression

0.61 and XGBoost 0.54 (0.49–0.60), the last being indistinguishable from chance. Variable importance and SHAP values ranked age and years since diagnosis first, and the data and code are public. The best model nevertheless detects only 51% of uncontrolled cases (specificity 0.98); the logistic regression row is computed with a different positive class from the other two, and the discussion attributes to random forest a sensitivity that is in fact its specificity (Section 3.6.5c). The design is cross-sectional and the predictors are demographic and administrative rather than physiological.

Three lessons follow. Reported performance is governed by how the task is constructed—label integrity and informativeness of the features—far more than by the choice of algorithm. Ensemble tree models, random forest and XGBoost, are plausible candidates, and interpretability through feature importance and SHAP is now expected practice, as Mendoza-Cano et al. [26] and Liu et al. [24] both show. And no source forecasts a future physiological state from a patient's own series of blood pressure and blood glucose readings, nor evaluates such a model in a low-resource setting. The methodological consequences are developed in Section 3.6.4.

2.3.6 Data Privacy, Security and Ethics in Digital Health

Alhammad et al. [9] conducted the first systematic review of patients' perspectives on the confidentiality, privacy and security of the data collected by mobile health applications: thirty-three studies published between 2014 and 2022, identified in PubMed, Ovid and ScienceDirect under PRISMA guidance. Patients' concerns about their data are a consistent and significant barrier to adoption. Encryption, secure authentication and regular risk assessment are recommended as baseline safeguards, and transparency about how data are handled improves trust. Most included studies come from high-income countries and measure self-reported perceptions rather than actual incidents.

These findings converge with earlier observations of this review: most commercial applications either lack a privacy policy or declare sharing data with third parties [15], the legal and regulatory dimension is one of the six barriers to telemedicine in the region [7], and the reference implementation of Efendi et al. [22] stores its data in a cloud region outside the country of use, which raises the question of data residency. Since the present work processes real, hospital-sourced patient data through a mobile and cloud pipeline, data protection is treated as a first-class design constraint: anonymisation before modelling, encrypted transmission, role-based authentication, and explicit identification of the applicable Cameroonian legal framework (Sections 3.7.2 and 4.3.5).

2.3.7 Critical comparison of approaches and current trends

The studies reviewed follow five approaches, compared in Table 2.10. None of them combines all the elements the problem requires: clinical validity, professional oversight,

longitudinal data, prediction and suitability for low-resource settings.

Table 2.10: Comparison of the approaches found in the literature.

Approach	Studies	Strengths	Weaknesses
Self-management applications	Alessa et al. [15] and Zhou et al. [17]	low cost, behaviour change techniques	little clinical validation; effect weak without a professional
Rule-based telemonitoring with a clinician	Kingue et al. [11] and Mihevc et al. [19]	validated effect on BP and HbA1c; simple, transparent	fixed thresholds, reacts to level not trend, no learning; costly expert time
Tele-expertise between professionals	Bediang et al. [5]	changes clinical decisions; feasible in Cameroon	no home data; specialist delay unchanged
IoT platforms with machine learning	Tai et al. [21], Efendi et al. [22], and Liu and Wang [23]	complete technical chain; real-time and offline storage	no BP or glucose sensors; tiny or controlled validation; label leakage
Predictive models on clinical data	Chatrati et al. [25] and Mendoza-Cano et al. [26]	interpretable risk predictors	cross-sectional or single readings; modest sensitivity; no deployment

Three trends emerge from the chronology of the studies. Telemonitoring moves from patient self-management towards systems linked to a clinician, which the most recent evidence identifies as the condition of effectiveness [16, 18, 19]. Prediction moves from rules towards machine learning, with interpretability tools such as SHAP becoming standard [24, 26]. And sensing moves towards cuffless and smartphone-based measurement [20, 24], with cloud platforms designed to work offline [22].

Four points remain controversial. Cuffless blood pressure measurement is promising in research [24] but not accepted for clinical decisions [1]. The very high comorbidity reported for Central Africa rests on two small hospital series (Section 2.3.1c). Whether mobile health works in Africa is open: the effect disappears in developing countries when no professional is involved [16]. And the same algorithm, XGBoost, is reported as the best [22] and as the worst [26], a contradiction explained by how the tasks were constructed rather than by the algorithm (Section 3.6.4b).

2.4 Limits of the Existing Literature

The analysis reveals four kinds of limits.

Blind spots. Central Africa is almost absent: one of the 51 trials of Mao et al. [16] was conducted in sub-Saharan Africa, the comparative reviews cover South Africa, Kenya and Nigeria [8], and francophone countries are under-represented [7]. No evaluated system combines consultation data, home measurements and sensor data for patients with both hypertension and diabetes.

Methodological weaknesses. The Cameroonian studies are not randomised and have unbalanced or incomplete groups [5, 11]; most trials are short and measure intermediate outcomes only; the technical validations rely on few volunteers, favourable conditions and unvalidated reference instruments [22]; and the machine learning studies suffer from label leakage, unsuitable metrics or reporting errors (Section 2.3.5).

Absence of local data. No nationally representative estimate exists for Cameroon, and no artificial intelligence tool in circulation has been validated on an African population [4]. Models trained elsewhere cannot be assumed valid here.

What remains to be explored. The only validated protocol for patients with both conditions [19] relies on fixed thresholds identical for all patients, reacts to the level of a reading rather than to its trend, and uses no learned model. Four questions therefore remain open: personalising alert thresholds from each patient's history; detecting drift before a threshold is crossed; forecasting deterioration from home series; and ordering scarce clinician time, with the operating cost and sustainability of the system documented.

2.5 Link with the Present Research

Continuity. The present work builds directly on the existing studies: it adopts the validated three-level home protocol of Mihevc et al. [19], the layered architecture with an edge tier of Mucchi et al. [13], offline-first storage as in Efendi et al. [22], and the modelling and interpretability practice of Mendoza-Cano et al. [26]. It also follows the conclusion of the clinical literature that telemonitoring must be linked to a clinician.

Originality. It differs from them on five points: a system designed to combine consultation records, home measurements and, optionally, sensor data for both conditions; a learned model applied to each patient's own series, to forecast deterioration rather than classify the current reading; a hybrid design in which the guideline rule is a floor that the model can raise but never lower; the explicit sizing of alert volume and clinician time; and models developed on public data and validated externally on Cameroonian patients, a transfer that no study identified in the region has measured.

Research questions. Four questions follow from the gaps identified.

RQ1. Can a telemonitoring system combining consultation, home and sensor data be designed to work under the connectivity, workload and cost constraints of Cameroonian care?

RQ2. Do learned models applied to the patient's series predict deterioration better than the guideline rule, and does a deep temporal model outperform classical models?

RQ3. Can forecasting anticipate deterioration early enough to be clinically useful?

RQ4. Do models developed on data from another health system transfer to Cameroonian patients?

Hypotheses. RQ1 is a design question: it is answered by the system of Chapter 4 and by the traceability of each of its elements to a requirement of the literature (Tables 4.9 and 4.38), not by a statistical test. RQ2 to RQ4 lead to four testable hypotheses, stated with the performance measures of Section 3.6.5c and examined against the evidence in Section 3.6.4c.

H1. At least one learned model predicts the risk level of the next reading with a lower ranked probability score than the guideline rule applied to the last reading (persistence).

H2. This advantage persists when the lead time grows from 12 to 24 and 48 hours.

H3. The deep temporal models do not outperform the tabular models.

H4. Performance is lower on the Cameroonian patients than on the development data, and updating the intercept on the local data recovers part of the loss of calibration.

2.6 Synthesis of the Review

Table 2.11 summarises, for each theme, what the literature establishes, what it leaves open, and the consequence for this work.

The scientific achievements are clear: the burden is high and most diagnosed patients are lost during follow-up; telemonitoring linked to a clinician improves blood pressure and glycaemic control; validated instruments, layered architectures and interpretable models are available. The gap is equally clear: no system combining multi-source data with a learned, personalised and anticipatory model has been designed and evaluated for patients with hypertension and diabetes in Central Africa. Filling it requires a methodology able to collect longitudinal data under local constraints and to evaluate predictive models without the biases identified in this chapter; this is the object of Chapter 3.

Table 2.11: Synthesis of the literature review.

Theme	**Established**	**Not established**	**Consequence for this work**
Burden	high burden; cascade fails at every stage; 72% of diagnosed patients not controlled; 58% of people with diabetes are hypertensive	reliable Central African and Cameroonian estimates	longitudinal follow-up of diagnosed patients, joint BP and glucose monitoring
Sub-Saharan context	loss to follow-up, inertia, paper records; six barrier dimensions; specialist scarcity	BP effect of telemedicine in Cameroon; evidence from Central Africa	offline-first, low workload, triage of clinician attention
Clinical evidence	modest effect on BP and HbA1c, conditional on professional oversight	effectiveness in sub-Saharan Africa; long-term effect	clinician-in-the-loop decision support
Sensors and IoT	validated cuff and glucometer required; layered architectures with offline storage	validated cuffless BP; validation on real patients in low-resource settings	instruments for BP and glucose, smartphone sensing exploratory
Machine learning	ensemble trees plausible; interpretability expected; task construction drives performance	forecasting from individual BP and glucose series	leakage-free forecasting formulation, honest metrics
Privacy	data concerns hinder adoption; baseline safeguards known	evidence from African patients	anonymisation, encryption, declared data residency

Part I

Methodology, System Design and Results

Chapter 3

Research Methodology

3.1 Introduction

This chapter describes how the research is conducted and justifies each methodological choice. It answers the research question of Section 1.2: how to integrate consultation data, home measurements and mobile sensor data into a telemonitoring system that detects deterioration early enough to support a clinical decision, under the constraints of the Cameroonian context. More precisely, it sets out how the research questions of Section 2.5 are investigated and how hypotheses H1 to H4 are tested.

The literature imposes three requirements on the method: use longitudinal data whose instruments and quality can be assessed; evaluate predictive models without the biases found in previous work—label leakage, unsuitable metrics, overfitting on small samples; and respect an ethical and legal framework for real patient data. The chapter presents in turn the type and approach of the research, its setting and population, the data sources, the sampling plan, the analysis methods, the ethical considerations and the limits of the methodology, before a synthesis that prepares the presentation of the results.

3.2 Type and Approach of Research

The research is **applied**: it addresses a practical problem, the follow-up of patients with chronic diseases, by designing an artefact, a telemonitoring system. It follows a design-science logic in three stages: requirements are derived from the literature, the system is designed and built (Chapter 4), and its predictive component is evaluated on retrospective data. Within this frame, the predictive component is **experimental**, since models are compared under a protocol fixed in advance, and the analysis of the cohort is **descriptive** (adherence, data quality, missing data).

The approach is **quantitative**, with a limited qualitative component (Table 3.1). The hypotheses are expressed as measurable comparisons—ranked probability score,

calibration, lead time—which calls for quantitative methods. The labels, however, are clinical judgements: the partner physician validates the labelling rule and relabels a sample of readings.

Table 3.1: Type of research by objective.

Objective	**Type**	**Approach**
Design the telemonitoring system (RQ1)	applied, design science	requirements analysis, prototyping, traceability
Compare predictive models (RQ2, H1, H3)	experimental	quantitative
Validate the models on Cameroonian patients (RQ4, H4)	experimental	quantitative
Evaluate short-horizon forecasting (RQ3, H2)	experimental	quantitative
Describe the cohort, adherence and data quality	descriptive	quantitative
Validate the labels	evaluative	relabelling by the partner physician

3.3 Research Setting and Population

The research is carried out at the Laboratory of Computer Engineering, Data Science and Artificial Intelligence of the National Advanced School of Engineering of Douala (ENSPD), in collaboration with a partner physician practising in a Cameroonian health facility. The physician facilitated access to the anonymised records, validates the labelling rule and relabels a sample of readings. The deployment context is that described in Section 2.3.2: patients followed episodically, paper or fragmented records, few specialists and intermittent connectivity.

3.3.1 Target population and eligibility criteria

The target population consists of adults with diagnosed hypertension, type 2 diabetes, or both, followed at the partner facility. The eligibility criteria of a prospective collection (Table 3.2) are adapted from the trial of Mihevc et al. [19] to the local context; they would be finalised with the partner physician and the ethics committee before any recruitment.

3.3.2 Collection scenarios

The initial specification of the thesis subject foresaw real anonymised data from a partner physician in a hospital, which corresponds to scenario A. Which streams are actually available determines most of what follows, so four scenarios are distinguished (Table 3.3).

The target of that specification—at least thirty patients and three hundred measurements, about ten per patient—fits a prospective cohort followed for about five weeks, or

Table 3.2: Eligibility criteria.

Inclusion	**Exclusion**
Age 18 years or more	type 1 or gestational diabetes; pregnancy
Hypertension and/or type 2 diabetes documented in the facility record	cognitive impairment preventing the use of the devices, without a caregiver
Followed at the partner facility	severe condition requiring hospitalisation at enrolment
Able to measure at home, alone or with a caregiver, with access to a smartphone	refusal or withdrawal of consent
Written informed consent	

Table 3.3: Collection scenarios and their consequences.

	A: retrospective outpatient	**B: prospective**	**C: mixed**	**D: retrospective inpatient**
Source	consultation records	recruited cohort with home devices	records plus a prospective sub-cohort	in-hospital observation charts
Streams	A only	A + B (+ C)	A + B on a subset	A only, daily
Interval between readings	$\approx$ 90 days	3–4 days	both	$\approx$ 1 day
Ethics	secondary use	individual consent	both	secondary use
Feasible horizon	next consultation	$\approx$ 3–7 days; 24–48 h in verification	per stream	24–48 h
Low-cost acquisition evaluated	no	yes	yes	no

an inpatient stay of about ten days with daily observations; it does not fit ninety-day outpatient intervals. The low-cost acquisition contribution can only be evaluated under scenario B or C, which require recruitment, individual consent and devices. The method is written so that it applies to any of these scenarios; only the numerical values of the window length, the horizon and the minimum counts change with the scenario.

Data obtained. Two datasets were obtained, both transcribed from anonymised paper records of a public hospital in Douala. The first corresponds to scenario A: consultation records of 58 patients, 81 readings in all, of whom 49 (84%) have a single reading. It documents the fragmentation of follow-up but cannot support longitudinal modelling, and it is used descriptively. The second corresponds to scenario D: the observation charts of 32 inpatients, 366 readings taken between 8 June and 30 August 2026, morning and evening—two readings on 177 of the 189 patient-days, and no gap longer than one day. Blood pressure and pulse are recorded at almost every reading, capillary glucose at 69% of them, and 8 of the 32 patients have no glucose at all. This second dataset fixes the numerical values of the scenario; it is too small to develop models (Section 3.5) and is used for external validation. Scenario B was not obtained, so the low-cost acquisition contribution is not evaluated in this work. The transcription does not record the admission diagnosis, so the eligibility criteria of Table 3.2 cannot be verified patient by patient; this limit is stated in Section 3.8.

3.4 Data Collection Methods and Instruments

The data of this work are secondary: no patient was recruited and no measurement was taken for it. Three sources are used, each for a single purpose (Table 3.4). The home protocol, the instruments and the record schema of the system, designed for a prospective collection, are described with the system in Chapter 4.

3.4.1 Data sources

The Douala charts were transcribed by hand, one row per reading. The transcription records whether a reading was taken in the morning or the evening but not its hour, and it does not record the admission diagnosis. In the wards, blood pressure and pulse follow the nursing schedule, whereas capillary glucose is ordered according to the clinical picture; the consequences of this difference for missing data are treated in Section 3.6.1d. In eICU, vital signs come from nurse charting and from periodic monitoring, and glucose from bedside measurements; how these dense records are brought to the rhythm of the charts is described in Section 3.5.

Table 3.4: The three data sources and their role.

Source	Content	Collection	Role
eICU Collaborative Research Database [27]	intensive-care stays in 208 United States hospitals, 2014–2015; open demonstration subset of 2 520 stays in 186 hospitals [28]	electronic records of routine care, de-identified	training and comparison of the models only
Douala inpatient charts	32 patients, 366 readings, 8 June to 30 August 2026; blood pressure, pulse and capillary glucose, morning and evening	paper observation charts of a public hospital, transcribed without names	external validation only
Douala consultation records	58 outpatients, 81 readings, 49 patients with a single reading	paper consultation records of the same hospital, transcribed without names	description of the fragmentation of follow-up

Units. Values are stored in one canonical unit per parameter and converted at entry (Table 3.5). Glucose is the riskiest field, because three conventions coexist (1 mmol/L = 18.016 mg/dL = 0.1802 g/L, from the molar mass of glucose, 180.16 g/mol) and Cameroonian practice uses g/L; a value whose magnitude matches the wrong convention is flagged. The exact factor matters: with the rounded value of 18.0, a reading of 1.80 g/L falls just below the 180 mg/dL threshold of Table 3.11.

Table 3.5: Units and plausible ranges.

Parameter	Unit	Plausible range	Role
SBP / DBP	mmHg	60–260 / 30–160	clinical
Blood glucose	mmol/L	1.0–35.0	clinical
Heart rate	bpm	30–220	clinical (label, Section 3.6.2)
SpO_2	%	50–100	covariate
Temperature	°C	30.0–45.0	covariate

3.4.2 Validity and reliability

Validity and reliability are assessed at three levels. For the **instruments**, the devices used in the Douala wards are not named, so their type and the consequences of their tolerance are inferred from the readings themselves (Section 3.4.2a); the eICU records come from routine monitoring devices whose validation is not reported either. For the **data**, controls applied at entry detect implausible, duplicated and carried-forward values (Section 3.4.2b). For the **labels**, a written protocol fixes the labelling rule, and the partner physician relabels a sample, with agreement measured by the weighted kappa (Section 3.6.2).

3.4.2a Instruments of the data obtained

A blood pressure device is validated when it meets the universal standard ISO 81060-2 [29, 30]: against a reference, the mean of the differences must not exceed 5 mmHg and their standard deviation 8 mmHg. A glucose meter meets ISO 15197:2013 when 95% of its results lie within 15 mg/dL of the laboratory value below 100 mg/dL, and within 15% above [31]. The observation charts do not name the devices, so their validation cannot be checked; two analyses of the readings bound what this uncertainty implies.

Type of instrument. A manual aneroid or mercury scale is graduated in steps of 2 mmHg, so manual readings end in even digits, with an excess of zeros; an electronic display shows any integer [32]. In the charts, 50% of systolic and 46% of diastolic values end in an odd digit, zeros account for 10% and 13%, and the terminal digits do not depart from uniformity (χ^2 test, $p = 0.74$ and 0.26). The readings were therefore taken with an electronic device. Pulse values behave differently: 84% end in an even digit ($p < 10^{-50}$), consistent with values read from the graphical pulse curve of the chart, with a resolution of about 2 beats per minute. The pulse thresholds of the label lie far from this resolution: the four readings at or above 131 beats per minute range from 135 to 144.

Label uncertainty. A reading close to a threshold may change level under an error that a validated device is allowed to make. Shifting each reading by the tolerance of its standard—5 mmHg for pressure, 15 mg/dL or 15% for glucose—changes the level of 27% of systolic, 39% of diastolic and 56% of glucose readings. Part of the disagreement between a model and the label is therefore measurement noise that no model can remove. Two consequences follow: an error between adjacent levels must weigh less than an error across several levels (Section 3.6.5c), and the relabelling by the partner physician measures how much of this noise reaches clinical judgement.

3.4.2b Quality controls at entry

Errors are cheapest to catch at entry, when the information needed to correct them is still available. The controls of Table 3.6 learn nothing from the data and can be applied to all records before the train–test split. Flagged records are kept, not deleted: a run of identical values is itself information about how the data were produced. Impossible values (Table 3.8) are the exception: they are removed from features and labels. In the inpatient charts the controls remove one diastolic value of 24 mmHg and one pair of 102/100 mmHg, and flag runs of identical values (6 diastolic, 9 pulse) and seven extreme values, all contradicted by the next reading and sent for clinician review.

Table 3.6: Quality controls applied at entry.

Control	Rule	On violation
Plausibility	value within the range of Table 3.5	flag `out_of_range`
Coherence	SBP > DBP and SBP − DBP ≥ 10 mmHg	flag; pair excluded, since one of its values is wrong and which cannot be known
Duplicate	same patient, parameter and minute	keep one, flag the rest
Carried forward	three or more identical consecutive values	flag `carried_forward`
Unit	magnitude consistent with another convention	flag `unit_suspect`
Device	unvalidated device for BP or glucose	reject; device unknown in retrospective charts: flag
Time	measurement time later than record time (prospective only)	flag
Protocol	reading pair present, 45–180 s apart (prospective only)	informational flag

3.4.2c Label leakage

The main threat to the validity of the labels is the circularity found in the classifier of Efendi et al. [22] (Section 2.3.5). Their label *stable* is defined as all three parameters lying in their normal ranges, and each feature is recoded as low, normal or high using the same ranges, so the label is a deterministic function of the inputs:

$$y = 1 \iff (x_1 = 1) \wedge (x_2 = 1) \wedge (x_3 = 1). \tag{3.1}$$

The classifiers recover a rule instead of predicting a state, and the perfectly balanced classes (2 250 against 2 250) confirm that the dataset was constructed rather than observed. The same circularity arises in this work if the risk label is derived from guideline thresholds and the features include the current values; the way out is to predict a future state from a past window (Section 3.6.4b).

3.5 Sampling Plan

Sample size. The initial specification targets at least thirty patients and three hundred measurements, about ten per patient; the inpatient cohort meets it, with 32 patients and 366 readings. Whether that is enough to develop a prediction model is a separate question, answered with the criteria of Riley et al. [33] and Riley et al. [34]. For a binary outcome, the minimum sample size must keep the optimism of the model small (a global shrinkage factor of at least 0.9), keep the apparent and adjusted Nagelkerke R^2 within 0.05 of each

other, and estimate the overall risk within ±0.05. The first criterion gives

$$n \geq \frac{p}{(S-1)\,\ln\!\left(1 - R^2_{\mathrm{CS}}/S\right)}, \tag{3.2}$$

for p candidate parameters, a shrinkage $S = 0.9$ and an anticipated Cox–Snell R^2_{CS}, derived from an anticipated C statistic [35]. The criteria are computed with a Python port of the reference implementation [36], checked against its published example. The outcome is the alert of each condition (level high or critical at the target). The samples of one patient are correlated, so the effective sample size divides the number of samples by the design effect $1 + (\bar{m} - 1)\rho$ [37], where $\bar{m}$ is the size-weighted mean number of samples per patient and ρ the intraclass correlation of the alert.

Table 3.7: Sample size of the inpatient cohort against the criteria of Riley et al. Parameter budget for an anticipated C statistic of 0.70 to 0.80.

	Haemodynamic alert	**Glycaemic alert**
Samples (patients)	282 (32)	190 (24)
Prevalence of the alert	0.23	0.59
Intraclass correlation ρ; design effect	0.32; 3.9	0.31; 3.6
Effective sample size	72	53
Sample size for the overall risk (±0.05)	270	372
Parameters allowed, samples taken as independent	2–6	2–6
Parameters allowed, effective sample size	0–1	0–1
Patients needed for 8 parameters (C = 0.75)	≈ 215	≈ 170

The inpatient cohort is therefore far below the minimum for developing a model (Table 3.7): even the overall risk cannot be estimated to the required precision, and the parameter budget is at most one. Flexible learners need more still: random forests, support vector machines and neural networks may need several times as many events per variable as logistic regression to reach the same optimism [38].

Two datasets with distinct roles. The modelling study is therefore exploratory, and it separates development from validation. The candidate models are trained and compared on the eICU Collaborative Research Database, a public multi-centre database of 200 000 intensive-care stays in 208 United States hospitals [27, 39], and on it only; its open demonstration subset of 2 520 stays of 1 841 patients from 186 hospitals [28] is used to build and test the pipeline, and the full database, available under a data use agreement, for the reported benchmark. eICU is preferred to the other public intensive-care databases, such as MIMIC-IV, because it covers many hospitals: the outer folds can then be grouped by hospital, which estimates how a model transfers between sites before it meets the Douala charts (Section 3.6.5a); a single-centre database cannot provide that estimate. The Douala cohort is used for external validation only: it measures whether models developed on United States data transfer to Central African inpatients, the question raised

by Segun-Omosehin et al. [4] (RQ4). No study identified in the region has measured such a transfer for a learned model. The closest work concerns early warning scores: in African wards, scores derived elsewhere lose discrimination, which motivated a score derived on African patients [40]; a recent review of these scores in low- and middle-income countries, most of them African, concludes that local validation and recalibration are required [41]; and a Ghanaian study is validating intensive-care scores derived in high-income countries on West African patients [42]. The intercept update of Section 3.6.5e answers the same need. External validation has its own sample size requirements—about 100 events and 100 non-events as a rule of thumb, and formal criteria on calibration and discrimination [43, 44]. The Douala cohort has about 64 haemodynamic and 112 glycaemic alert targets, so its estimates are reported with patient-level confidence intervals and read as exploratory.

Sampling method and harmonisation. In eICU, every adult stay with hypertension or diabetes recorded in the diagnoses or the past history is eligible, which makes eligibility verifiable, unlike in the charts (Section 3.3.2). Intensive-care monitoring is far denser than a ward chart, so each eICU stay is resampled to the rhythm of the charts: two slots a day, at 08:00 and 20:00, each taking the non-invasive cuff pressure, the pulse and the bedside capillary glucose nearest to the slot within two hours. The slot hours are an assumption, since the charts record only morning and evening; the two-hour tolerance balances completeness against the risk of taking a reading that belongs to another slot, and a sensitivity analysis varies it. The choice matters mostly for glucose, which is measured rarely in intensive care: in the demonstration subset, tolerances of one, two and three hours give 645, 1 356 and 1 983 glycaemic samples, against 4 018 to 4 249 haemodynamic samples. Both datasets then share the sample definition of Section 3.6.1c, and the remaining differences—intensive care against the ward, the United States against Cameroon—are the shift that external validation measures. Figure 3.1 illustrates the resampling.

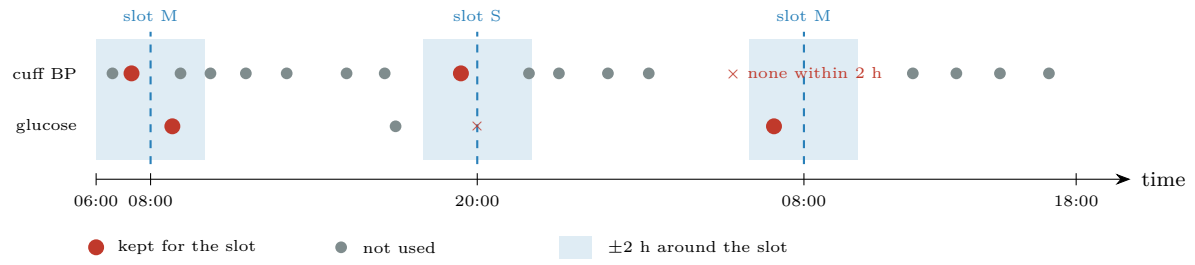


Figure 3.1: Harmonisation of an intensive-care stay to the rhythm of the ward charts. At each 08:00 and 20:00 slot, the reading nearest in time within two hours is kept for each channel; a slot with none is missing for that channel, as an unrecorded value on a chart would be.

In Douala, all the charts made available by the hospital are used; this is a convenience sample at a single site, and its characteristics are reported so that its representativeness can be judged.

3.5.1 Sample size and model capacity

The 282 haemodynamic and 190 glycaemic samples of the Douala cohort have an effective size of 72 and 53 (Table 3.7). This is small against the capacity of any flexible model, and above all of a recurrent one. An LSTM layer has

$$P = 4\Big[u(f+u)+u\Big] \tag{3.3}$$

parameters for u units and f inputs; even a small layer ($u = 16$, $f = 10$) has 1 728 parameters, that is 24 to 33 per independent sample. Trained on the Douala cohort, a recurrent model would fail for lack of data, and the comparison of H3 would measure the sample size rather than the models. A second risk follows: rare levels concentrated in a few patients or hospitals could leave a fold with no positive case, so the folds are stratified on the presence of events.

Four responses were possible: increase the sampling frequency for a sub-cohort; run the model comparison on a public dataset and use the local cohort as external validation; restrict the comparison to a few simple models; or report a recurrent model as a negative result limited by capacity. The second is adopted (Section 3.5): it gives the deep models a sample on which their comparison means something, and it turns the small local cohort into a test of transportability rather than a development set. On the development data the parameter budget is recomputed with the same criteria, and the hyperparameter search is sized to it. The forecast horizon follows the choice of Section 3.6.4c.

3.6 Data Analysis Methods

The analysis has two parts. A **descriptive** part characterises the cohorts, data quality and missing data, with medians and interquartile ranges for continuous variables and proportions for categorical ones. A **predictive** part follows five steps: preprocessing, labelling, feature engineering, model formulation, and training and evaluation. Figure 3.2 shows how they connect and where each dataset enters. The software tools are listed at the end of the section.

3.6.1 Preprocessing

This section specifies what happens between the database and the model. It is where label leakage, invented temporal structure and biased imputation are either prevented or built in. The operation performed is a temporal aggregation over several channels (blood pressure, glucose, heart rate), each sampled on its own schedule; the only grouping above the patient is the hospital, used to split the development data.

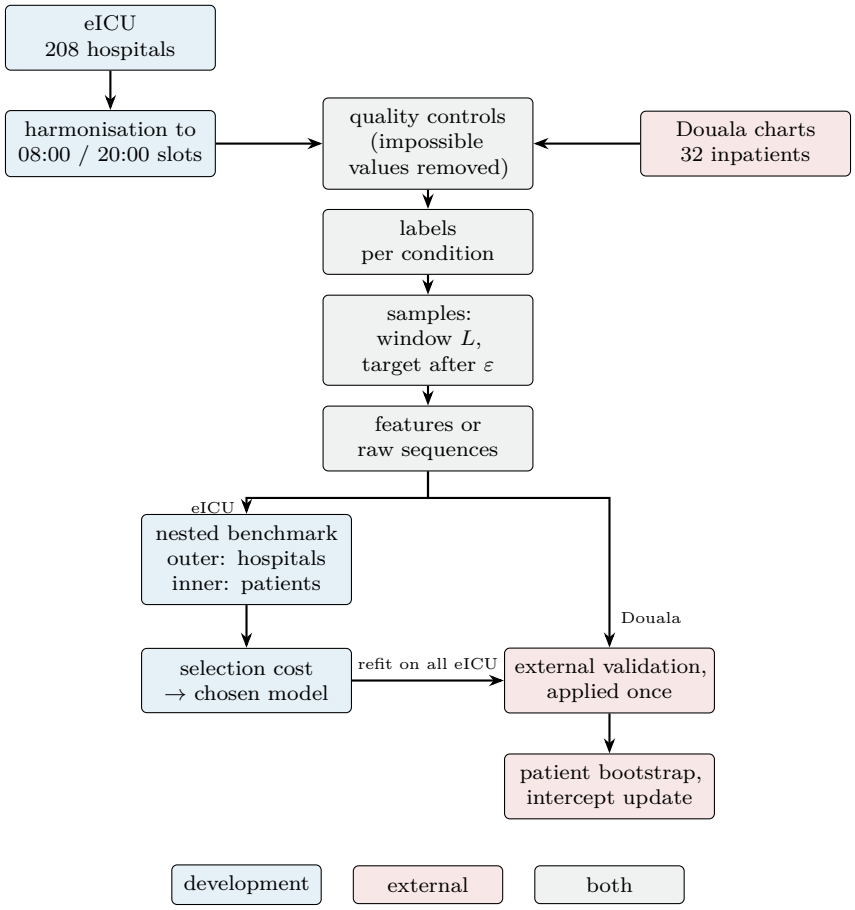


Figure 3.2: Study design. The same preprocessing, labelling and sample definition apply to both datasets; models are developed and compared on eICU only, and the Douala charts are used once, for external validation.

3.6.1a A two-phase pipeline

Every step that learns a parameter from the data must be fitted on the training fold only. The pipeline is therefore split into two phases separated by the train–test split (Figure 3.3), and it is implemented as a single pipeline object rather than as a sequence of notebook cells. A step placed on the wrong side of the split does not cause an error: it silently produces an optimistic estimate that fails in deployment.

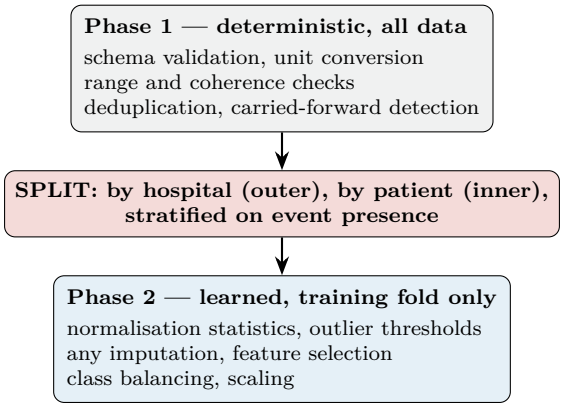


Figure 3.3: The two-phase preprocessing pipeline.

3.6.1b Outliers: artefact or event?

In a risk-detection system the outlier is often the event of interest. A systolic reading of 185 mmHg is a statistical outlier for a patient whose median is 138, and it is also a red alert. An automatic cleaning step would silently delete the positive class, and the model would then perform well on a series in which crises never occur. The dividing line is therefore physiological plausibility, not statistical rarity (Table 3.8).

Table 3.8: Three kinds of extreme value.

Category	Criterion	Action
Impossible	outside the range of Table 3.5 (e.g. 400 mmHg)	excluded from features and labels
Possible but extreme	within range, beyond the statistical threshold (e.g. 195 mmHg)	kept: it may be the event
Contradicted	extreme value contradicted by a reading within hours	flagged for clinician review

When a statistical threshold is used, for flagging only, it relies on the median absolute deviation (MAD), which is not inflated by the outliers it is meant to detect:

$$z_{i,t} = \frac{x_{i,t} - \text{median}(x_i)}{1.4826\,\text{MAD}_i}. \tag{3.4}$$

With about ten readings per patient, a personal MAD is unstable, so it is shrunk towards the cohort value:

$$\widehat{\mathrm{MAD}}_i = w_i \,\mathrm{MAD}_i + (1 - w_i)\,\mathrm{MAD}_{\mathrm{pooled}}, \qquad w_i = \frac{n_i}{n_i + n_0}. \tag{3.5}$$

With $n_0 = 20$, a patient with ten readings takes one third of the estimate from their own data and two thirds from the cohort; as the series grows, the estimate becomes personal. The value of n_0 is the number of readings at which a patient's own estimate and the cohort estimate weigh equally; no source fixes it, so the flags are recomputed with $n_0 = 10$ and 40: in the Douala charts they flag 5, 7 and 9 readings for $n_0 = 10$, 20 and 40, so the choice changes little. Because flags serve clinician review and never enter the features, the patient's whole series is used to compute them. A reading is flagged when $|z_{i,t}| > 3.5$, the cut-off recommended for this modified z-score [45], and it is marked as contradicted when the next reading of the same patient is not extreme. This implements the personalised threshold missing from the fixed protocol, in a way that degrades gracefully when data are scarce.

3.6.1c Irregular sampling and windows

The two datasets used here are aligned on two slots a day, but the home readings for which the system is designed are irregularly spaced, and the method is written for both. Interpolating home readings onto a daily grid would invent five values for every two real ones, and because interpolation smooths the series, it would understate variability, one of the quantities of interest. The raw timestamps are therefore kept, and the interval structure enters the features explicitly, per channel: time since the last observation, number of observations in the window, and their regularity. Blood pressure and glucose are never assumed to share a time axis, since at home they are measured on different days.

A training sample is a window of past observations and a label taken outside it:

$$\mathcal{S}_{i,t} = \Big(\{x_{i,s} : t - L \le s \le t\},\ g\big(x_{i,t'}\big) \Big), \qquad t' \in [\,t + \varepsilon,\ t + \Delta\,], \tag{3.6}$$

where L is the window length and $\varepsilon > 0$ an embargo that guarantees no observation is used both as input and as label (Figure 3.4). Because readings are irregular, the label is the first observation in the search interval; a sample with no observation there is discarded.

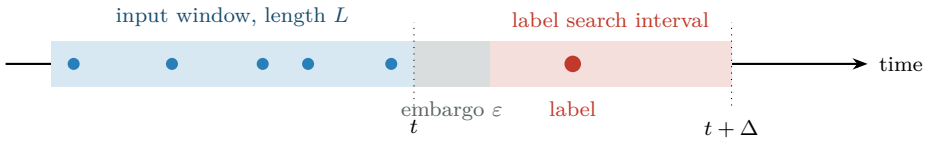


Figure 3.4: Construction of one training sample.

Windows are defined in time (the last L hours) rather than in count (the last k

readings), because every clinical reference is expressed in time—the seven-day profile, the one-to-four-week confirmation, the monthly follow-up—and five readings may span four days for one patient and seven months for another. A feature is computed only if the window contains at least two observations for a level and three for a slope; otherwise it is missing, and the number of such windows is reported as a measure of adherence.

In the inpatient charts, readings are taken morning and evening; the hours are not recorded, so the two slots are placed 12 h apart, and every interval in the cohort is then exactly 12 h. Each reading t of a patient is a candidate anchor, the last reading of an input window. Because the window length is a hyperparameter searched in the inner folds (Section 3.6.5a), the set of anchors must not depend on it, or each length would be evaluated on different targets. An anchor is therefore eligible when the shortest window considered, $L = 24$ h, holds at least two blood pressure readings; longer windows add history to the same anchors. The candidate lengths and the features they support follow from the number of readings they contain (Table 3.9): a window can carry at most as many independent quantities as it contains observations (Section 3.6.3a).

Table 3.9: Window length and the features it can support, at two readings a day.

L (h)	Readings	Features supported
24	2	level only
48	4	level and slope; variability marginal
72	6	level, slope and variability
96	8	level, slope and variability

Overlapping windows multiply the number of rows but not the information: two windows of the same patient that share readings are not independent. The 283 samples of the cohort come from 32 patients; the patient is the independent unit, confidence intervals are computed at the patient level, and all the windows of a patient fall on the same side of every train–test split.

3.6.1d Missing data

Imputation is valid when data are missing completely at random (MCAR) or at random given observed values (MAR), and invalid when missingness depends on the unobserved value itself (MNAR). Consultation data are MNAR, because attendance depends on the patient's state; home data are MNAR by a different route, because discouraging readings may be omitted; only sensor gaps (battery, connection) are approximately MCAR. In hospital, the vital signs follow the chart schedule, but glucose is ordered according to the clinical picture, so whether it is measured depends on the patient's state: its absence is MNAR as well.

The direction of the resulting bias matters. If abnormal values are less likely to be recorded, the observed values are biased towards normal. Mean imputation and forward

filling then teach the model that absence predicts stability, whereas in reality absence predicts deterioration. This is a sign error that more data cannot correct, and it contradicts the system's own rule that silence must raise an alarm (Section 4.1.2b). Imputation, which suits cross-sectional datasets such as the one used by Chatrati et al. [25], is therefore not the default here.

Absence is instead represented as information (Table 3.10); the feature sets of Table 3.13 carry it through the number of glucose readings and the time since the last one. The adherence ratio applies to home deployment: since the home protocol fixes the expected number of readings, the ratio of observed to expected readings measures adherence continuously, and retention is the most robust finding of the Cameroonian studies (Section 2.3.2c). In the charts the schedule is the nurses', not the patient's, so the ratio is not used.

Table 3.10: Missingness features, per channel.

Feature	**Meaning**
$m^{(c)} \in \{0,1\}$	was channel c observed in the window
$n_W^{(c)}$	how many times
$\Delta t_{\text{last}}^{(c)}$	time since the last observation
$r^{(c)} = n_W^{(c)} / n_{\text{expected}}^{(c)}$	adherence ratio against the protocol

Tree ensembles that support missing values learn, at each split, which branch to follow when a value is missing. A recurrent model must be adapted, with the mask as an extra input, no loss computed on padded positions, and a decay of the last observed value towards the mean of the channel in the training data as the gap grows, as in the GRU-D model [46]:

$$\tilde{x}_t^{(c)} = m_t^{(c)} x_t^{(c)} + \left(1 - m_t^{(c)}\right)\left[\gamma_t^{(c)}\, x_{\text{last}}^{(c)} + \left(1 - \gamma_t^{(c)}\right)\bar{x}^{(c)}\right], \qquad \gamma_t^{(c)} = \exp\!\left(-\kappa_c\, \Delta t_{\text{last}}^{(c)}\right). \quad (3.7)$$

The decay rate κ_c differs by channel, because a three-day-old blood pressure is more informative about today than a three-day-old fasting glucose. The decay is always used together with the mask, since on its own it would again assert stability during long gaps. In the sequence models, blood pressure and glucose are never imputed; only covariate sensor channels may be, with the mask kept. The proportional-odds model cannot accept missing inputs, so its missing features are replaced by the training median together with a missingness indicator, which keeps the absence visible to the model (Table 3.14).

3.6.1e Scale of the inputs

The label is defined on absolute thresholds, so the features keep their clinical units. Normalising each patient's values by their own median and dispersion would hide how far a

reading lies from a threshold, which is precisely what the label measures; the patient's own distribution is used only to flag extreme values (Section 3.6.1b). The proportional-odds model and the neural networks are sensitive to the scale of their inputs, so each input is standardised with the mean and standard deviation of the training fold (Table 3.14); the tree ensembles need no scaling.

3.6.2 Labelling

The label states how much clinical attention a reading calls for, on four ordered levels: stable, moderate, high, critical. It must satisfy six requirements: every threshold comes from an official source; low values count as well as high ones; it applies to a single reading; a missing channel does not bias it; thresholds are versioned; and its agreement with clinical judgement is measured. The thresholds below are those of in-hospital monitoring, the setting of the data actually collected (scenario D, Section 3.3.2); a home deployment would substitute the out-of-office thresholds of Table 4.12.

Grades as severity bands. The ESH grades classify a patient's usual blood pressure from repeated measurements [12]; they are not defined for one reading. They are used here as severity bands of a single reading, and the repetition that diagnosis requires is carried by the input window (Section 3.6.1c). The other thresholds keep their own function: 135/85 mmHg governs the home triage of Section 4.2.2c, and 140/90 mmHg the epidemiological comparisons [1].

Table 3.11: Severity of a single reading, per channel. BP in mmHg, pulse in beats per minute, capillary glucose in mmol/L.

Level	Haemodynamic channels	Glucose
3 critical	SBP $\geq$ 180 or DBP $\geq$ 110, grade 3 [12]; SBP $\leq$ 90; pulse $\leq$ 40 or $\geq$ 131 [47]	$<$ 3.0, level 2 hypoglycaemia [48, 49]; $>$ 16.7 (300 mg/dL), severe hyperglycaemia [50]
2 high	SBP 160–179 or DBP 100–109, grade 2	3.0 to $<$ 3.9, level 1 hypoglycaemia; $\geq$ 10.0 (180 mg/dL), treatment threshold [51]
1 moderate	SBP 140–159 or DBP 90–99, grade 1	$>$ 7.8 (140 mg/dL), in-hospital hyperglycaemia [51]
0 stable	otherwise	otherwise

Four choices in Table 3.11 need a justification.

- **Low pressure and pulse.** The National Early Warning Score 2 (NEWS2) adds the scores of six physiological parameters; only a score of 3 in a single parameter, the "red score", calls on its own for urgent review by a clinician [47]. Only that threshold is used for the two NEWS2 parameters recorded here, systolic pressure and pulse; the intermediate bands have meaning only inside the aggregate score. No official source sets a lower threshold for diastolic pressure, so none is used.

- **Pulse.** Tachycardia and hypotension accompany the hypovolaemia of hyperglycaemic crises [52], so the pulse belongs in the label of these patients, at the NEWS2 red-score threshold only.
- **Hypoglycaemia.** Level 1 (< 3.9 mmol/L) is an alert value and level 2 (< 3.0 mmol/L) is clinically significant [48]; they map to high and critical. Level 3 is defined by a clinical event, not a glucose value, and cannot be observed in the records.
- **Severe hyperglycaemia.** The diagnostic criteria of ketoacidosis and of the hyperosmolar state require ketones, pH or osmolality, which the charts do not record [52]. No clinically accepted cut-off for severe hyperglycaemia exists; the 300 mg/dL threshold is the one adopted by the official hospital-harm measure [50], and it is used as a convention. A sensitivity analysis caps hyperglycaemia at the high level; a conclusion that changes between the two versions depends on this convention and is reported as such.

Boundaries are stored in the unit of their source and converted at comparison, so that no rounding moves a reading across a level (Table 3.5).

One label per condition. Within a condition, the worst channel governs: an additive score is validated only on the complete NEWS2 parameter set, not on two or three of its parameters. Across conditions, one label is computed per condition, following the dual-task formulation (Section 3.6.4d):

$$y_t^{\mathrm{H}} = \max\Big(s_{\mathrm{SBP},t},\, s_{\mathrm{DBP},t},\, s_{\mathrm{pulse},t}\Big), \qquad y_t^{\mathrm{G}} = s_{\mathrm{glu},t} \text{ if glucose is measured at } t, \tag{3.8}$$

$$y_t = \max\Big(y_t^{\mathrm{H}}, y_t^{\mathrm{G}}\Big), \tag{3.9}$$

where s is the level of a channel in Table 3.11 and the maximum runs over the labels available at t. A single label would mistake a missing glucose value for a normal one; since glucose is ordered for some patients and not for others, the absence of a measurement is itself informative (Section 3.6.1d), and the error would be systematic. The glucose label is therefore defined only where glucose is measured. Figure 3.5 summarises the construction.

Four rules prevent the label from leaking into the features:

1. the label is computed on the first reading of the label interval $[t+\varepsilon, t+\Delta]$ and the features on readings of the input window, with no overlap;
2. no feature measuring the distance to a threshold is computed at the label instant;
3. the guideline rule applied to the last observed value is a mandatory baseline, and the contribution of the model is its margin over that rule;
4. performance is also reported against the physician's relabelling, which is not derived from the thresholds.

The labelling rule and its threshold table are versioned so that someone else can reproduce

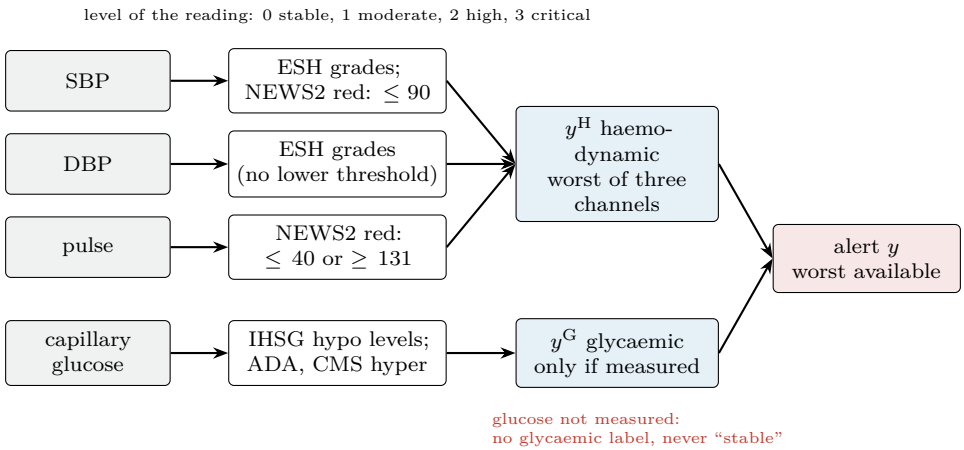


Figure 3.5: Construction of the label at a target reading. Each channel is graded by its sourced thresholds (Table 3.11); the worst channel governs within a condition; the alert is the worst of the condition labels that exist.

them. The partner physician relabels a random sample of readings without seeing the rule's output, and agreement is reported with the quadratically weighted kappa, which suits ordered levels. The sample is sized so that the 95% confidence interval of the kappa has a half-width of at most 0.10: a simulation on the observed distribution of levels gives about 100 readings for a kappa near 0.75 and about 150 for a kappa near 0.6, that is about a third of the 366 readings of the charts.

3.6.3 Feature Engineering

The tabular models receive descriptors computed on the input window; the sequence models receive the readings themselves. This subsection fixes how many descriptors a window can support, which ones are computed, and how many enter the models.

3.6.3a How much a short window can say

Every descriptor computed from a window of n readings is a function of the same n numbers, so no set of descriptors can carry more than n independent quantities. The case $n = 2$ shows this exactly. The sample standard deviation is $s = |x_2 - x_1|/\sqrt{2}$, the mean absolute successive difference is $\mathrm{MASD} = |x_2 - x_1|$, so that

$$s = \frac{\mathrm{MASD}}{\sqrt{2}}, \tag{3.10}$$

the slope is $\pm\mathrm{MASD}/\Delta t$, and the median equals the mean. Level, dispersion, slope and last value are then only two quantities, a level and a signed difference. Supplying all four adds columns without adding information, and columns are the scarce resource here (Section 3.6.3c). Robust statistics such as the median and MAD need observations to

be robust with, so they are used at the level of the patient's history, not within a short window.

Learned representations, such as the convolutional feature learning described by Liu and Wang [23], are designed for densely sampled signals and large samples; with two readings a day over stays of a few days they are not applicable to the tabular models, and hand-constructed descriptors are used instead. The sequence models receive the raw readings of the window, with a mask of what was measured.

3.6.3b The feature families

Table 3.12 lists the candidate features, computed per channel on the readings of the input window only. The slope is the ordinary least-squares coefficient on the real reading times, which handles gaps without resampling:

$$\beta^{(c)} = \frac{\sum_k (t_k - \bar{t})(x_k - \bar{x})}{\sum_k (t_k - \bar{t})^2}, \tag{3.11}$$

expressed per day. Its variance explodes when the readings are bunched in time, so it is computed only from at least three readings spanning at least half the window; a dispersion also needs three readings. A feature the window cannot support is missing, not zero: zero would assert a stability that was not observed.

Table 3.12: Candidate features, per channel $c \in \{\text{SBP, DBP, pulse, glucose}\}$.

Family	**Feature**	**Definition or purpose**
Level	last value, window mean	current and typical level
Trend	slope $\beta^{(c)}$	change per day
Variability	standard deviation	dispersion within the window
Cross-channel	pulse pressure	SBP − DBP of the last complete reading, a marker of arterial stiffness
Measurement	count, time since the last value	how much was measured; for glucose, whether it was ordered at all
Static	age, sex	the only covariates recorded in both datasets

Four remarks explain what is not in the table.

- **No signed margin to a threshold.** For a fixed threshold θ, the margin $x_{\text{last}} - \theta$ is an affine function of the last value. A linear model absorbs it into its intercept, and a tree, which splits on order, finds exactly the same partitions; the margin adds a column and no information. The same holds for the grade band, a coarsening of the last value.
- **No disease duration or regimen.** Years since diagnosis and treatment are recorded neither in the charts nor in eICU; a feature that exists in only one dataset cannot be validated externally.
- **Pulse pressure helps trees.** A tree splits on one variable at a time and would

need many splits to approximate a condition on SBP − DBP; supplying it explicitly turns that staircase into one split.

- **The glucose count is informative.** Glucose is ordered for some patients and not for others (Section 3.6.1d); the count of glucose readings lets the model use that pattern openly instead of through imputed values.

3.6.3c The dimensionality budget

A feature set built without restraint grows quickly: four channels with six descriptors each, plus cross-channel and static features, already give about thirty columns. The number of parameters a dataset can support is set by the criteria of Section 3.5, recomputed on the development data. Three feature sets are defined in advance (Table 3.13); the primary set gives the main result, the others are secondary analyses, and every result is reported with its number of features.

Table 3.13: Feature sets, fixed before any result.

Set	p	**Features**
Primary	8	last SBP, slope of SBP, last DBP, last pulse, last glucose, slope of glucose, number of glucose readings, age
Extended	16	primary, plus the standard deviations of SBP and glucose, the slopes of DBP and pulse, pulse pressure, time since the last glucose, mean SBP, sex
Descriptors	16	mean, standard deviation, slope and last value of the four channels, as in the preliminary exploratory model of this project, kept for comparison

The primary set spends its columns on the channels that carry the decision: systolic pressure and glucose get a level and a trend, diastolic pressure and pulse a level, and the glucose count summarises how the patient is monitored. Dispersion is left out of the primary set because a short window estimates it poorly. The window length is not a feature choice but a hyperparameter, searched in the inner folds over 24, 48, 72 and 96 hours (Table 3.9).

3.6.3d Feature selection and scaling

The primary feature set is fixed in advance on clinical grounds, which removes a source of variance and makes the result reproducible. A data-driven selection, such as the LASSO used by Mendoza-Cano et al. [26], is run only as a secondary analysis, inside the cross-validation loop (Section 3.6.5a); selecting on the full dataset would let the test data choose the features. How often each feature is selected across folds is reported, as a measure of stability.

Model families differ in their preparation needs (Table 3.14). Every preparation step that learns from the data is fitted on the training fold only.

Table 3.14: Preparation required by model family.

Model	Scaling required	Missing values
Proportional-odds logistic regression	yes	median imputation with missing indicators
Random forest, CatBoost	no	handled natively
GRU-D, masked LSTM	yes	mask and decay (Section 3.6.1d)

3.6.3e Baselines

Physiological series are strongly autocorrelated, so carrying the current level forward is a strong predictor. When the label is the guideline rule applied to a reading, persistence and the guideline rule applied to the last reading are the same baseline; it is turned into probabilities by the matrix of transitions between levels observed in the training data. Every candidate is compared with two baselines (Table 3.15). The contribution of the modelling work is the gap between persistence and the candidates. If that gap is small, the honest conclusion is that, at this sampling density and sample size, the models add little to the guideline rule—a result in itself, and more useful to a clinician than an inflated accuracy.

Table 3.15: The baselines.

Baseline	Description
Prior	the frequency of each level in the training data, the same for every sample
Persistence	the guideline level of the last reading of each channel, with the training transition probabilities to each level at the target

3.6.4 Learning problem and candidate models

This subsection states which models are compared, how the prediction task is constructed, and which horizon is predicted.

3.6.4a Candidate algorithm families

No model is chosen in advance: the candidates are the families that published benchmarks of comparable problems put forward, and the benchmark of this work selects among them (Table 3.16). The problem is a small tabular one, derived from short, irregular and incomplete clinical series. On clinical prediction tasks, machine learning has shown no systematic benefit over logistic regression [53]. On tabular data, tree ensembles generally match or beat deep networks [54], and in the most recent living benchmark CatBoost is the best gradient-boosting model when tuned without ensembling [55]. Only the deep temporal models read the data as a sequence, which is what hypotheses H1 and H3 compare; among

them, GRU-D was designed for multivariate series with missing values [46], and the masked LSTM [56], the most widely used recurrent architecture, serves as its reference.

Table 3.16: Candidate families and their justification.

Family	**Candidate**	**Why it is in the pool**
Linear	proportional-odds logistic regression [57] with an L2 penalty; the same on principal components	no systematic loss against machine learning in clinical prediction [53]; the order of the levels is part of the model; principal components address the redundancy of the descriptors
Bagging	random forest [58]	the reference of the reviewed literature [26]
Boosting	CatBoost [59]	best tuned gradient-boosting model of the living benchmark [55]
Value forecast	ridge or CatBoost regression per channel, levels from the threshold table	the second formulation of Section 3.6.4d
Deep temporal	GRU-D [46]; masked LSTM [56]	hypothesis H3

Four families are left out, each for a stated reason. A single decision tree is dominated by its ensembles. Support vector machines produce poorly calibrated probabilities, and the outputs here are probabilities shown to clinicians. XGBoost ranks below CatBoost and LightGBM in the same benchmark, so one gradient-boosting model is enough. The tabular foundation model TabPFN leads on small datasets, but only up to 10 000 training samples [55, 60], which the development data exceed; it is a candidate for future work on smaller development sets.

Hyperparameter search. Each candidate has a search space declared before any run, split into structural hyperparameters, which change the capacity or the input of the model (window length, depth, number of units, number of components), and training hyperparameters, which change how it is fitted (learning rate, penalties, dropout, class weights, weight of the regression head). The search uses the tree-structured Parzen estimator of Optuna [61, 62], inside the inner folds only: each proposed configuration is scored by its ranked probability score on pooled inner-validation predictions, and the configuration kept is the one with the lowest selection cost among those tried (Section 3.6.5c). The number of configurations per candidate is fixed in advance and reported, since the optimism of a selection grows with the number of configurations tried [63]. The spaces and budgets are listed with the code. Figure 3.6 shows the procedure in one outer fold.

The benchmarks support an expectation, stated before the results so that the comparison is not read as a surprise: on tabular inputs, the deep temporal models should not outperform the tree ensembles and the penalised linear model, which is hypothesis H3.

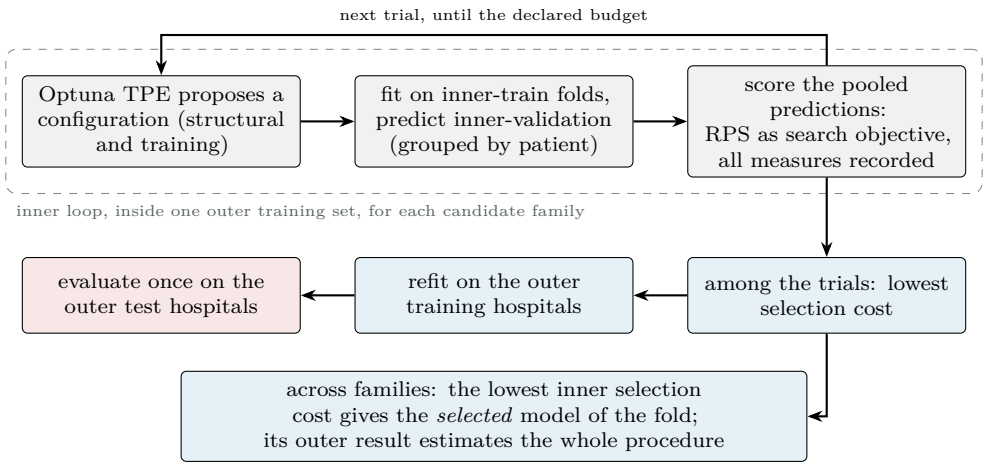


Figure 3.6: Hyperparameter search and selection in one outer fold. The search objective is the ranked probability score; the configuration kept is the one with the lowest selection cost; the outer test hospitals are used once.

3.6.4b Task construction determines reported performance

The same algorithm, XGBoost, is reported as the best model by Efendi et al. [22] and as no better than chance by Mendoza-Cano et al. [26] (Section 2.3.5). The difference does not come from the algorithm: the first study measures the recovery of a rule already encoded in the features (Section 3.4.2c), the second a genuine prediction from weak predictors. Reported performance depends first on how the task is built—the integrity of the label and the informativeness of the features—and only last on the algorithm.

For this work, the consequence is direct. If the label y_t is a threshold rule applied to the current reading x_t and the model sees x_t, the task is label recovery:

$$\hat{y}_t = g(x_t) \quad \text{with} \quad y_t = h(x_t). \tag{3.12}$$

The legitimate task predicts a future state from a past window,

$$\hat{y}_{t+\Delta} = g(x_{t-k}, \ldots, x_t), \tag{3.13}$$

in which the label lies outside the input by construction (Figure 3.7). Its performance will be much lower, and that lower figure is the honest one. The forecasting formulation is therefore not an extra ambition but the condition for the result to mean anything.

3.6.4c Hypotheses, forecast horizon and sampling frequency

The literature gives each hypothesis of Section 2.5 a different standing. For **H1**, persistence is a strong baseline for autocorrelated physiological series (Section 3.6.3e), and the only comparable result, a recurrent network predicting lupus readmission with an AUC of 0.70, suggests a modest margin at best. **H2** has never been tested: no source forecasts

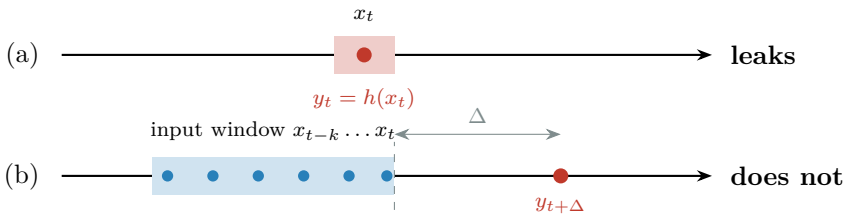


Figure 3.7: (a) Classifying the current state when the label is a rule applied to the same reading. (b) Forecasting a state outside the input window.

physiological state from series of blood pressure and glucose readings, which is the strongest claim to novelty of this work and also its main risk. **H3** is the best supported, since tree ensembles generally match or beat deep networks on tabular data [54]. **H4** follows from the loss of performance of scores transferred between settings [40, 41].

H2 meets a constraint that follows from the protocol. With routine readings at a rate λ, the expected number of readings inside a horizon of length Δ is

$$\mathbb{E}[N_\Delta] = \lambda\Delta, \qquad \lambda = \frac{2}{168\,\mathrm{h}}, \tag{3.14}$$

which gives 0.29 readings within 24 hours and 0.57 within 48 hours (Figure 3.8). Most prediction points would have nothing to predict. The horizon is therefore matched to the regime of the patient. In the *routine* regime, the model predicts the next scheduled reading or the next seven-day mean, a horizon of three to seven days that is both estimable and clinically meaningful. In the *verification* regime, where the protocol prescribes two readings a day, a 24 to 48 hour horizon contains about four readings; this is also the moment when a clinical decision is pending, so a short-horizon forecast would change an action.

The inpatient charts follow the verification regime: a reading every 12 h. The lead time is then set by the embargo ε of the sample definition. With $\varepsilon = 0$ the target is the next reading, 12 h ahead; with $\varepsilon = 12$ h it is the reading 24 h ahead; with $\varepsilon = 36$ h, the one 48 h ahead. The primary task uses the next reading, and H2 is tested by how performance decays with lead time, on 283, 251 and 187 samples respectively. A forecast that only matches the guideline rule applied to the last reading anticipates nothing, so the comparison is always made against that rule at the same lead time.

Finally, the data are split by hospital and by patient (Section 3.6.5a), and every preprocessing step that learns from the data (imputation, scaling, feature selection) is fitted on the training fold only; within a patient, the embargo of Section 3.6.1c prevents any look-ahead.

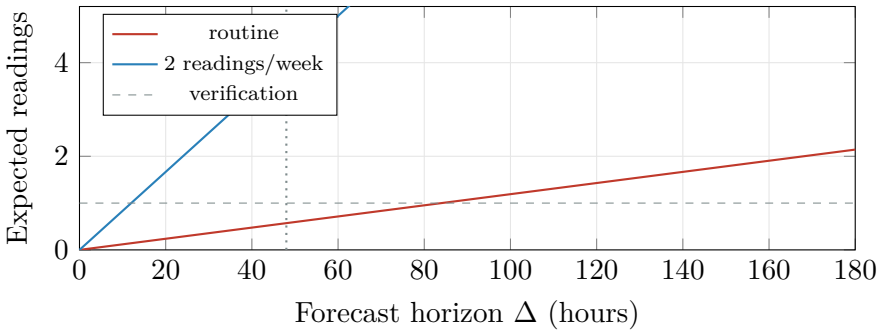


Figure 3.8: Expected number of readings inside the forecast horizon, by sampling regime.

3.6.4d Formulation

Two formulations reach the same clinical output—the probability of each level at the target—and neither is assumed better; both enter the benchmark (Section 3.6.4).

- **Value forecast.** The model forecasts the value of each channel at the target, and the level follows from the threshold table of Table 3.11:

$$\hat{x}_{i,t'}^{(c)} = f\Big(\mathbf{v}_{i,t}\Big), \qquad c \in \{\text{SBP}, \text{DBP}, \text{pulse}, \text{glucose}\}, \tag{3.15}$$

  where $\mathbf{v}_{i,t}$ is the feature vector of the window ending at t and t' the target time. The forecast error distribution turns the value into level probabilities (Section 3.6.5e). Every sample gives a training signal in clinical units, a 5 mmHg error near a threshold is distinguished from one far from it, and a guideline revision changes the decision layer without retraining. The price is an assumption on how the errors of correlated channels combine within a condition.
- **Direct ordinal classification.** The model predicts the level of each condition directly. It needs no assumption on the channels, but it sees only the level, not the distance to a boundary, and suffers more from the imbalance between levels.

The prediction is "dual-task" in two complementary senses, both adopted: one target per condition, haemodynamic and glycaemic, as defined in Section 3.6.2; and, where the model family allows it, a regression head and a classification head trained together, the regression loss providing a training signal on every sample when high levels are rare. Each condition is evaluated on the targets where its label exists, and the overall alert on the targets where both exist. The horizon is the next reading, 12 h ahead, with longer lead times for H2 (Section 3.6.4c).

3.6.4e The label

The four risk levels—stable, moderate, high, critical—are ordered. A standard multi-class loss would treat predicting "critical" for a stable patient as no worse than predicting "moderate"; the value forecast carries the order naturally, and the classification formulation

must use an ordinal loss. Two label sources are reported separately: the rule-derived label, available for every sample and used as the primary endpoint, and the physician's relabelling of a sample of Douala readings (Section 3.6.2), which does not derive from the thresholds and is used as a secondary endpoint. Once the system is deployed, the clinician verdicts of the event record (Section 4.2.4) will provide a third, larger source. The prevalence of each level at the targets is measured before any balancing and reported with the results. Among inpatients the alert (high or critical) is frequent: 17% of haemodynamic targets in the eICU demonstration subset and 23% in Douala, 29% and 59% of glycaemic targets. No source gives the corresponding prevalence among outpatients followed at home, so the sizing of a home deployment (Section 4.1.2c) uses $\pi = 0.02$ for critical readings as an explicit planning assumption, and shows how the conclusions change with π (Figure 4.4).

3.6.5 Training and evaluation

Models are trained and evaluated under a protocol fixed in advance, with metrics suited to rare events.

3.6.5a Training protocol

All constraints combine into a nested cross-validation on the development data (Figure 3.9). Five outer folds estimate performance; they are grouped by *hospital*, so that every test fold contains hospitals the model has never seen. This internal–external validation estimates how a model transfers across sites before it meets the Douala charts [64]. Inside each outer training set, four inner folds, grouped by patient, select the window length and the hyperparameters; every step that learns from the data is fitted there and nowhere else. Five outer folds leave about forty hospitals of the full database, and about 37 of the demonstration subset, in each test fold; four inner folds keep three quarters of the outer training set for fitting while bounding the cost of the search. Selecting a model on the same folds that estimate its performance biases the estimate upwards, and the bias grows with the number of configurations tried [63]; the outer test fold is therefore used once, and the size of the search is reported. eICU does not release calendar dates, so folds cannot be ordered in time across patients; within a patient, the embargo of Section 3.6.1c already prevents any look-ahead.

The whole procedure is repeated three times with different fold assignments, a number set by the cost of the search, and the mean and spread across repetitions are reported. Two models evaluated on the same folds are compared with the corrected resampled t test, which accounts for the overlap between training sets; a plain test across repetitions treats them as independent and overstates significance [65].

After the cross-validation, each retained configuration is refitted on all the development data, with the window length and hyperparameters chosen by the same inner procedure,

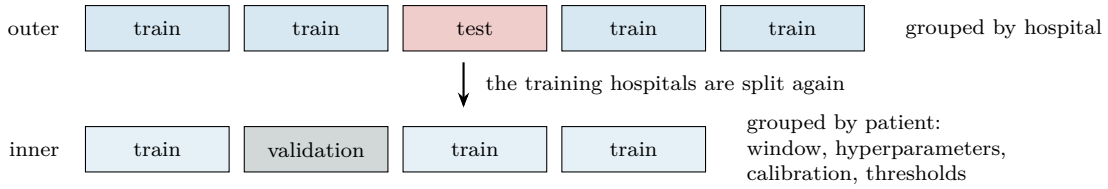


Figure 3.9: Nested cross-validation on the development data, repeated with different fold assignments.

and applied once to the Douala charts. Nothing is tuned on them. Their confidence intervals come from a bootstrap that resamples patients rather than samples, so that the correlation between the samples of a patient is kept [43].

3.6.5b Class imbalance

Three mechanisms handle imbalance (Table 3.17). The first two are the most common; the third is often the best, because it leaves the probabilities intact and only moves the operating point, which is a clinical choice anyway (Section 4.1.2c). Any resampling is applied inside the training fold: oversampling before the split places copies of the same window in training and test, and the resulting estimate is meaningless. A model trained with class weights has its probabilities corrected before use (Section 3.6.5e), and per-level measures are always reported, since an aggregate hides exactly the failure that matters.

Table 3.17: Mechanisms for handling imbalance.

Mechanism	**Implementation**	**Effect on probabilities**
Class weighting	a weight w on positive cases in the loss	distorted
Oversampling	duplicate or synthesise positive cases	distorted
Threshold adjustment	move the decision point only	none

3.6.5c Metrics

With prevalence π of the positive class, accuracy decomposes as

$$\text{Accuracy} = \pi\,\text{Sensitivity} + (1-\pi)\,\text{Specificity}. \tag{3.16}$$

When positives are rare, a model that never predicts them reaches an accuracy of $1-\pi$. The logistic regression of Mendoza-Cano et al. [26] illustrates this: its accuracy of 0.70 is close to the proportion of controlled patients, while its specificity is 0.03. The same study shows two reporting errors to avoid: the positive class is not the same in all rows of its results table, and the discussion attributes to random forest a sensitivity of 0.98 that is in fact its specificity; its true sensitivity is 0.51.

The measures are therefore not chosen by habit but by explicit criteria [66]. A measure is admissible when it has a clear focus—discrimination, calibration, overall accuracy of the probabilities, or clinical utility—and, for overall accuracy, when it is a proper scoring rule, one that a model cannot improve by reporting probabilities other than its own [67]. Accuracy, the F1 score and recall at a fixed threshold fail these criteria: they are improper, depend on an arbitrary cut-point and mix statistical with decision performance. Table 3.18 lists the admissible measures; the positive class is the alert (level high or critical), and every measure carries a patient-level confidence interval.

Table 3.18: Admissible performance measures.

Focus	**Measure**	**Why**
Overall, four levels	ranked probability score [68]	proper for ordered levels: an error across several levels costs more than an adjacent one, which matters when a large share of labels lies near a threshold (Section 3.4.2a)
Overall, alert	Brier score, log loss	proper
Discrimination	area under the ROC curve of the alert	independent of any threshold; read against 0.5
Calibration	calibration intercept and slope [69]	a probability shown to a clinician must mean what it says
Clinical utility	net benefit and decision curve [70]	weighs true alerts against false alarms at the clinical threshold

Selection cost. In the inner folds, configurations are compared on one number. The four foci are standardised across the configurations of the same comparison—minus the median, divided by the scaled median absolute deviation, oriented so that lower is better—and combined:

$$C = \sum_m w_m \, z_m, \qquad \sum_m w_m = 1, \tag{3.17}$$

with one term per focus: the ranked probability score, the area under the curve, the calibration error $|a| + |\ln b|$ for intercept a and slope b, and the net benefit averaged over the decision thresholds. The weights express how much each focus matters clinically, so they are a value judgement, not a parameter: tuning them on the data that judge the models would let any weighting favour a chosen model. They are fixed before the search, equal in the provisional analysis, and replaced by weights derived from the costs of errors stated by the partner physician. The decision threshold p_t follows from the same statement: alerting at p_t means accepting $(1 - p_t)/p_t$ false alarms per true alert, so the range $0.1 \le p_t \le 0.3$ used provisionally covers from nine to about two. A sensitivity analysis draws the weights at random, uniformly over all weightings, and reports how often the selected configuration remains first; a choice that changes with the weights is reported as such. The outer folds report every admissible measure separately, never the composite alone. Performance is also reported by sex and by age band (below and above 65 years), so that a model that serves one group worse than another is detected, and across the hospitals of the outer folds,

whose spread measures the heterogeneity between sites; the Douala validation extends that comparison to a population the development data do not contain.

3.6.5d Interpretability

Mendoza-Cano et al. [26] explain their models with two complementary tools: permutation importance for random forest and SHAP values for XGBoost [71]. Both ranked age and years since diagnosis first, but the rankings differ in detail. Two lessons follow. First, an explanation describes a model, not a phenomenon: explaining a model that does not discriminate, such as an XGBoost at an AUC of 0.54, has little value, so interpretability is reported only for a model whose discrimination has first been shown to be better than that of persistence. Second, importance measures are unreliable when features are correlated: permuting one of two correlated features lets the model fall back on the other, and gain-based importance credits whichever happens to be chosen at a split [72, 73]. Systolic and diastolic pressure are strongly correlated here, so the features of one channel are permuted together, and the loss in ranked probability score measures what the channel contributes (*grouped permutation importance*). The measure applies to every candidate, the deep models included, so all are explained on the same scale; for CatBoost, exact tree SHAP values are reported in addition, and the direction of their effects is checked against physiology.

3.6.5e From model output to risk score

Every candidate returns the probability of each of the four levels at the target; the alert probability is the sum of the probabilities of the high and critical levels. Under the value formulation, the forecast $\hat{x}$ of a channel becomes level probabilities through the spread of its forecast errors: the normal distribution $\mathcal{N}(\hat{x}, \hat{\sigma}^2)$ is integrated over the intervals of the threshold table (Table 3.11),

$$P\big(s_c = k\big) = \sum_{[a,b)\,:\,\text{level } k} \Big[\Phi\Big(\frac{b-\hat{x}}{\hat{\sigma}}\Big) - \Phi\Big(\frac{a-\hat{x}}{\hat{\sigma}}\Big)\Big], \tag{3.18}$$

where $\hat{\sigma}$ is the root mean squared error of out-of-fold forecasts on the training data, so that it measures generalisation rather than fit. Within a condition the worst channel governs, so $P(y \le k) = \prod_c P(s_c \le k)$, under the stated assumption that the errors of the channels are independent given the inputs. Figure 3.10 illustrates the computation for one channel.

When a classifier is trained with class weights w_k, its outputs overstate each level in proportion to its weight. The correction is exact by Bayes' rule:

$$p_k = \frac{p'_k / w_k}{\sum_j p'_j / w_j}, \tag{3.19}$$

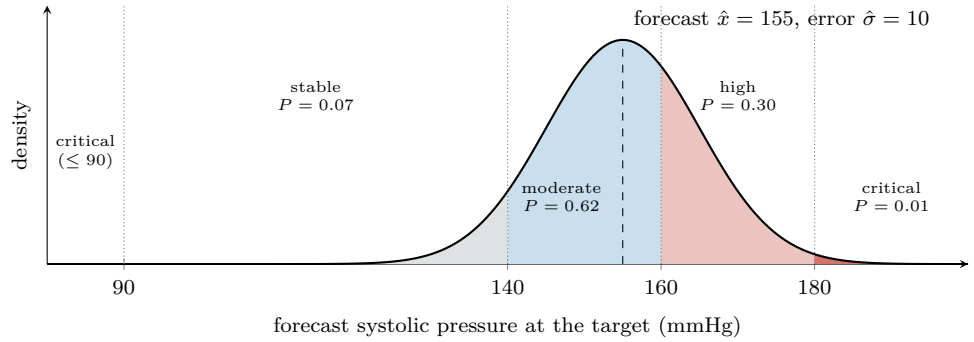


Figure 3.10: From a value forecast to level probabilities, for one channel. The forecast error distribution is cut by the thresholds of Table 3.11; each area is the probability of a level. Within a condition, the channels are then combined by their worst level.

where p' is the raw output. In the two-class case with the conventional weight $w = (1-\pi)/\pi = 49$ at $\pi = 0.02$, a raw output of 0.5 corresponds to a true probability of exactly 0.02, the base rate: a clinician shown 0.5 would read a risk twenty-five times too high. The correction is applied to every weighted model, and whether weighting helps at all is a training hyperparameter of the search.

All candidates are fitted with proper losses—the log-likelihood of the ordered levels, or squared error for the value forecasts—so their probabilities are not recalibrated during development; calibration is assessed instead, with the intercept and slope of Table 3.18 and a calibration curve of four or five bins with confidence intervals [69]. The Brier score is always reported next to the score of a model predicting the base rate for everyone, $\pi(1-\pi)$. In external validation the prevalence of the alert differs between United States intensive care and Douala wards, which shifts calibration-in-the-large even for a model that transfers well; performance is therefore reported as is, and again after updating the intercept alone on the Douala data, the lightest of the updating methods [74]. Only the first is a test of transportability; the second shows what a local recalibration would recover.

3.6.5f Thresholds and the three levels

The three levels need two cut-points on the calibrated score. The yellow level costs the patient a verification profile and costs the clinician nothing, so many false alarms are acceptable and its threshold is low; the red level costs clinician time, so its threshold is higher. Each cut-point is a decision threshold in the sense of Section 3.6.5c: it follows from the number of false alarms accepted per true alert at that level, stated by the partner physician, and the net benefit at that threshold is reported with the decision curve. Provisionally, 0.1 and 0.3 are used: they lie within the range of thresholds over which decision curves are usually read [70], and they correspond to accepting nine and about two false alarms per true alert. A threshold applied to calibrated probabilities needs no tuning on the data, so it is applied unchanged to every test fold. The alert volume at the chosen

point is reported; what it implies for a home deployment is computed in Section 4.1.2c.

The model may raise a level set by the guideline rule, but never lower it:

$$\text{level}_{\text{final}} = \max\Big(\text{level}_{\text{rule}}(x_{\text{last}}),\ \text{level}_{\text{model}}\big(s(\mathbf{v})\big)\Big), \tag{3.20}$$

with green $\prec$ yellow $\prec$ red. A patient at 185/110 mmHg is red whatever the model says; the best model in the literature misses half of the positive cases; and if the model fails entirely, the system falls back to the standard of care rather than to nothing.

3.6.5g Tracking the evolution of the patient's state

The level answers whether a reading calls for attention; the clinician also needs to know whether the patient has changed, which is the direction of the trend shown on each queue row (Section 4.6.4). The state of a patient is the vector $\mathbf{s} = (\bar{x}_{\text{SBP}}, \bar{x}_{\text{DBP}}, \bar{x}_{\text{HR}}, \bar{x}_{\text{glucose}})$ of the mean of each channel over a window. The current state $\mathbf{c}$ covers $(t - 24\,\text{h}, t]$ and the baseline $\mathbf{b}$ the three days before, $(t - 96\,\text{h}, t - 24\,\text{h}]$, the same four days as the rule floor; evolution is a distance $d(\mathbf{c}, \mathbf{b})$ computed on the channels present in both.

Cosine similarity is the first measure that comes to mind, but it compares directions, not sizes: if every channel rises by half, $\mathbf{c} = 1.5\,\mathbf{b}$ and the cosine distance is zero, so a patient whose pressures and pulse rise together appears unchanged. On raw values it is moreover dominated by the channels with the largest units (systolic pressure near 130 mmHg against glucose near 6 mmol/L), and its behaviour depends on how the vectors are scaled [75]. Patient-similarity research uses many measures without a common benchmark [76, 77], so no measure is assumed. The candidates come from the largest independent comparison of similarity measures, nine measures on 38 data sets [78], which separates lock-step measures, comparing aligned elements, from elastic ones such as dynamic time warping. Elastic measures align sequences of points; a state is a vector of four means, so only lock-step measures apply. Two families are added for monitoring an individual: scaling each change by the patient's own variability, the idea of the reference change value of laboratory medicine [79], and the Mahalanobis distance [80], which is the statistic of multivariate control charts for individual observations [81]. Control charts for individual patients, used in hypertension and diabetes, are promising but little evaluated [82]. Table 3.19 lists the eight candidates.

The measures are compared on the forecast samples of Section 3.6.1c, with every candidate scored on the same samples (at least two channels in both states, since a cosine on one channel is always 0 or 2). A measure is useful if its distance ranks higher the samples in which the patient's level changes: the primary outcome is a level at the next reading that differs by at least one from the level of the baseline state, and the secondary outcome is a new alert, a baseline below high followed by a high or critical reading. Each measure is evaluated by the area under the ROC curve, with an interval

Table 3.19: Candidate measures of the change between the current state and the baseline, from the simplest to the most elaborate. σ_j: between-patient SD of channel j; σ_{ij}: SD of patient i, shrunk towards the pooled within-patient SD; Σ: pooled within-patient covariance of changes. Scales are estimated on the development data only.

Measure	**Definition**	**Property**
Cosine	$1 - \mathbf{c} \cdot \mathbf{b}/(\|\mathbf{c}\|\|\mathbf{b}\|)$	blind to a common change of size
Euclidean, raw	$\|\mathbf{c} - \mathbf{b}\|$	dominated by the largest unit
Euclidean, cohort scale	$\|(\mathbf{c} - \mathbf{b})/\boldsymbol{\sigma}\|$	units removed; one scale for all patients
Cosine, centred	cosine of the deviations from the cohort mean, in σ units	direction of the deviation, not its size
Euclidean, patient scale	$\|(\mathbf{c} - \mathbf{b})/\boldsymbol{\sigma}_i\|$	change relative to the patient's own variability
Manhattan, patient scale	$\sum_j \|c_j - b_j\|/\sigma_{ij}$	less weight to one large change
Chebyshev, patient scale	$\max_j \|c_j - b_j\|/\sigma_{ij}$	the largest single change
Mahalanobis	$\sqrt{(\mathbf{c} - \mathbf{b})^\top \Sigma^{-1} (\mathbf{c} - \mathbf{b})}$	correlated channels (systolic, diastolic) not counted twice

from the patient-level bootstrap of Section 3.6.5a. Scales and covariance are estimated on the eICU development data, and the Douala charts serve as an external check, never to choose. The measure retained is the one with the highest development AUROC on the primary outcome, or, when a simpler candidate of Table 3.19 has an interval containing that value, the simplest such candidate. Protocol and candidates are declared before the run (`configs/state_compare.yaml`, `scripts/12_state_comparison.py`). The selected distance is shown to the clinician next to the change of each channel; the patient sees the change of each channel only. No alert is raised on the distance: a threshold would need control limits validated on home data, which this work does not have.

3.6.5h Reporting and pre-registration

Table 3.20 lists what the experimental part reports. Five choices are also fixed now, before the test data are seen: the admissible measures of Table 3.18, with the ranked probability score as the primary overall measure; the selection cost and its weights; the primary comparison, every candidate against the guideline rule and persistence at the same lead time, no model being favoured in advance; the threshold rule, decision thresholds from the stated costs of errors; and the eight-feature primary set. Everything else is labelled secondary. This is the simplest answer to the question of whether a result was found or constructed.

Table 3.20: What the experimental part reports.

Item	Reason
Patients, stays and hospitals per dataset; eligibility	representativeness
Readings per patient; intervals per channel	sample size and window length
Missingness per channel; flag counts	data quality
Samples, patients, effective sample size; p	Sections 3.5 and 3.6.3c
Prevalence of each level before balancing	Section 3.6.4e
The full baseline ladder, including persistence	Section 3.6.3e
Every admissible measure, per condition and per lead time	Section 3.6.5c
Spread over the repetitions; corrected tests between models	Section 3.6.5a
Brier score with base rate; calibration curve with intervals	Section 3.6.5e
Selection cost, its weights and the stability of the ranking	Section 3.6.5c
Thresholds, how they were chosen, and the alert volume	Section 3.6.5f
Seeds, library versions, search space and number of configurations	reproducibility
Douala external validation, with patient-level intervals	Section 3.6.5a
Secondary endpoint against the physician's relabelling	Section 3.6.4e

3.6.6 Software tools

The analysis is carried out in Python: pandas and NumPy for data handling; SciPy for the proportional-odds model and the statistical tests; scikit-learn for the random forest, principal components and the folds; CatBoost for gradient boosting; PyTorch for GRU-D and the LSTM; Optuna for the hyperparameter search; and SHAP for interpretability. The labelling rule, the quality controls, the sample-size criteria and the performance measures are implemented in the same package and checked by automated tests, including parity with the reference implementations where one exists. Configurations—thresholds, selection cost, search spaces—are kept in versioned files, separate from the code. The application stack is described in Chapter 4. Code, configurations and the labelling rule are versioned; the Douala data are not published, for reasons of confidentiality, and the eICU data are available from PhysioNet under its own licence.

3.7 Ethical Considerations

The research involves real patients and their health data. Three principles apply: informed consent, anonymity and confidentiality, and compliance with the applicable ethical and legal standards.

The data actually used. The two datasets of this work raise different questions. The eICU data are de-identified and distributed by PhysioNet: the demonstration subset under an open database licence, the full database to credentialed researchers who have completed training in human research protection and signed its data use agreement [27, 39]; they are used within those terms and never combined with identifiable data. The Douala charts

are a secondary use of routine records, transcribed without names; secondary use does not require individual consent where an ethics committee waives it, but it does require that committee's opinion and the written authorisation of the hospital. The measures below apply to the Douala data and to any prospective collection.

3.7.1 Anonymity and confidentiality

Three commitments apply: the data are real and anonymised, direct identifiers are removed before processing, and the data are used with the agreement of the partner physician and the institution, for research only. Before collection, the following must also be documented: the ethics committee reference, the written institutional authorisation, the Cameroonian legal instrument governing health data (Section 3.7.2), individual consent if collection is prospective, the retention period and destruction date, and the region where the data are stored.

Removing names is not enough to anonymise a small cohort. In thirty patients from one facility, the combination of exact age, sex, consultation dates and site identifies almost everyone. Four inexpensive measures reduce this risk before any analysis file is shared: age in five-year bands; calendar dates replaced by the time elapsed since each patient's first reading, which is all the analysis uses; exclusion of the site identifier; and banding of rare values. Calendar events shared by all patients, such as a medicine stock-out, matter for retention (Section 2.3.2b), so they are kept in a separate file containing no patient data. Figures plot elapsed time, never real dates. The mapping between pseudonyms and identities stays with the partner physician and is never transferred.

3.7.2 Legal framework and informed consent

Mihevc et al. [19] complied with the European General Data Protection Regulation (GDPR) because their trial took place in the European Union. A system operated in Cameroon for Cameroonian patients falls outside its scope, and claiming GDPR compliance would hide the need to identify the law that does apply. The design therefore adopts the GDPR principles—lawfulness, purpose limitation, data minimisation, storage limitation, integrity and confidentiality—as a voluntary standard, while the applicable framework is Cameroonian. Personal data are governed by Law No. 2024/017 of 23 December 2024 on the protection of personal data [83], in force since June 2026, which creates an independent supervisory authority and makes any transfer of data abroad conditional on an equivalent level of protection in the destination country and on the prior authorisation of that authority; the security of electronic communications falls under Law No. 2010/012 on cybersecurity and cybercrime [84]. Before deployment, these are completed by the institutional policy of the facility and the opinion of an ethics committee, a procedure already followed by Bediang et al. [5] in Cameroon.

Where the data are stored is also a legal choice, since the storage region determines which jurisdiction governs them. The database region is fixed at creation and cannot be changed without migrating the data, so it is chosen deliberately before the first write; under Law No. 2024/017, a region outside Cameroon is a transfer abroad that requires authorisation. The reference implementation of Efendi et al. [22] stored its data in South-East Asia without discussion.

Under a prospective collection, individual consent is required. The consent document covers eight points: what is collected; who sees it; where it is stored and under which law; how long it is kept; that participation does not replace clinical care; that withdrawal is possible; what happens to collected data on withdrawal; and that the link between pseudonym and identity stays with the partner physician. The order of work follows: the legal framework determines the storage region and the consent document, which must both precede collection.

3.8 Limitations of the Methodology

Eight limitations are known in advance; for each, its likely effect on the results is stated.

1. **Small local sample, foreign development data.** The 32 Douala patients are far below the minimum for developing a model (Table 3.7), so the models are developed on United States intensive-care data and only validated locally. Intensive-care patients are monitored and treated differently from ward patients, so performance in Douala may be lower than in development; that gap is a result, measured with wide confidence intervals.
2. **Inpatients, not home users.** The charts come from hospitalised patients, sicker than the outpatients a telemonitoring system serves, so high and critical levels are more frequent than at home; the models cannot be transferred to home data without revalidation.
3. **Data missing not at random.** Glucose is ordered for some patients and not for others, so its absence carries information. One label per condition prevents a missing value from being read as a normal one, but the features still see the pattern of measurement.
4. **Labels derived from thresholds.** The primary labels come from guideline rules, so the model may partly learn the rule; the physician's relabelling, immune to this, covers only a third of the Douala readings, and no clinician verdict exists in retrospective data.
5. **Unknown instruments and hand transcription.** The devices are not named, pulse is read from a graphical curve, and every value was copied by hand from paper. The controls of Table 3.6 catch impossible values, not plausible errors, and the label of a large share of readings near a threshold is uncertain (Section 3.4.2a).

6. **Diagnoses not recorded.** The admission diagnosis is not transcribed, so eligibility cannot be checked and the cohort may include patients with neither condition.
7. **Slot hours assumed.** The charts record morning and evening, not the hour; the eICU slots at 08:00 and 20:00 and their tolerance are assumptions, and the number of glycaemic samples depends strongly on the tolerance (Section 3.5).
8. **Short follow-up and intermediate outcomes.** A hospital stay of a few days cannot measure clinical outcomes such as cardiovascular events; the evaluation concerns prediction, not clinical effectiveness.

The quantitative assumptions on which the analysis rests, and what changes if they do not hold, are stated below.

3.8.1 Working assumptions

A training protocol must name the quantities it rests on; Table 3.21 lists them, with what changes if an assumption does not hold. The risk labels use the thresholds of Table 3.11, each stored with its source and edition (Section 3.6.3b).

Table 3.21: Working assumptions.

Quantity	Value	If overturned
Local data	scenario D, inpatient charts, two readings a day	home data (scenario B) would require revalidation and a longer horizon
Horizon	next reading, 12 h; 24 and 48 h for H2	a sparser home protocol moves the horizon to days (Section 3.6.4c)
Window L	24, 48, 72 or 96 h, searched	longer windows need longer stays
Slots in eICU	08:00 and 20:00, ± 2 h	glycaemic samples vary from 645 to 1 983 (Section 3.5)
Effective sample size	Douala 72 and 53; eICU demonstration subset about 1 290 and 250	the parameter budget of Section 3.6.3c
Features p	8 primary, 16 extended	Table 3.13
Home prevalence π	0.02 for critical readings, planning assumption	alert volume and precision (Section 4.1.2c)

3.9 Synthesis and Transition

Table 3.22 summarises the methodological choices. Together they are designed to answer the research questions while avoiding the weaknesses identified in the literature: instruments whose type and tolerance are assessed, a forecasting task instead of label recovery, metrics and baselines suited to rare events, and an evaluation fixed before the data are seen.

Following the design-science approach, the results are presented in two stages. Chapter 4 presents the first result, the telemonitoring system designed from the requirements of the literature, with the home protocol it is built for. The experimental results—description of

Table 3.22: Summary of the methodology.

Element	**Choice**
Type and approach	applied design-science research; experimental and descriptive; quantitative with clinical input
Setting	ENSPD laboratory with a partner physician in a public hospital in Douala
Population	adults with hypertension and/or type 2 diabetes; inpatients in the data used
Data	eICU (training and comparison only); Douala inpatient charts (external validation only); Douala consultation records (descriptive)
Instruments	unnamed electronic devices in the Douala wards, inferred from terminal digits; routine monitoring in eICU
Sample	32 Douala patients, 282 haemodynamic and 190 glycaemic samples; eICU harmonised to two slots a day
Analysis	two-phase preprocessing; labels per condition from sourced thresholds; eight-feature primary set; proportional-odds logistic regression, random forest, CatBoost, value forecasts, GRU-D and masked LSTM
Evaluation	nested cross-validation, outer folds by hospital and inner folds by patient, repeated three times; ranked probability score, discrimination, calibration and net benefit; comparison with persistence; external validation with intercept update
Ethics	secondary use of de-identified data; ethics committee opinion and hospital authorisation; Cameroonian Law No. 2024/017

the cohorts and evaluation of the predictive models against hypotheses H1 to H4—are presented in the following chapter.

Chapter 4

Design and Architecture of the Telemonitoring System

This chapter presents the telemonitoring system, the first result of the design approach of Section 3.2. Section 4.1 derives the requirements of the system from the literature, and Section 4.2 the home protocol it is built for. Section 4.3 fixes the layers, the data flow, the latency budget, the failure modes and the security measures; Section 4.4 describes the hardware prototype and its transport protocol, Section 4.5 the backend as built, and Section 4.6 the patient application, the health assistant and the clinician dashboard. Throughout, what is built is distinguished from what is specified (Table 4.22). Section 4.7 summarises what is original and what remains open.

4.1 Requirements Derived from the Literature

The requirements on measurement, acquisition and modelling were implemented in the methodology (Chapter 3). This section states the requirements on the architecture and on decision support, before Table 4.9 gathers all of them.

4.1.1 Architectural requirements

The architecture must turn isolated readings into series, place a first filter close to the patient, survive intermittent connectivity and keep the data under a known jurisdiction.

4.1.1a From isolated readings to a series

A paper register records that a consultation took place; it does not record a series. Yet the same reading means different things in different series (Figure 4.1): 152 mmHg after a stable series around 135 mmHg calls for a check, whereas 152 mmHg at the end of a three-month drift calls for treatment intensification. The system must therefore store and display trajectories, not isolated values.

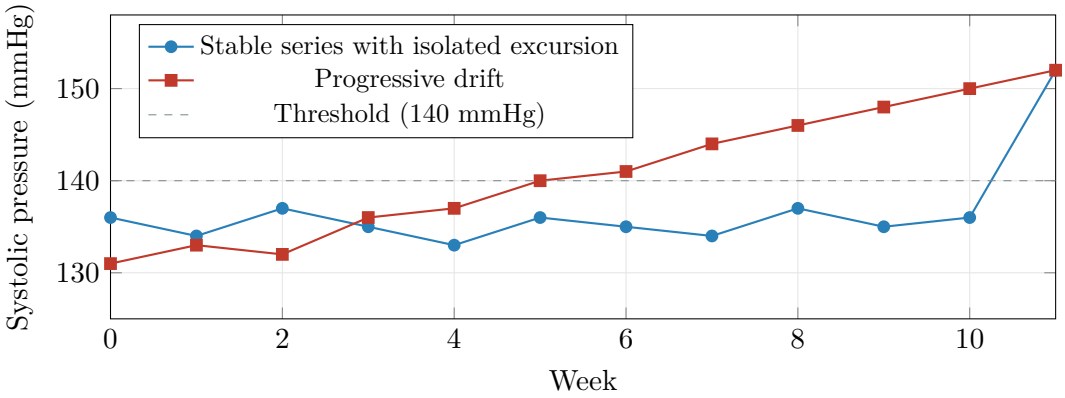


Figure 4.1: Same last value, different meaning (synthetic illustration). A single reading of 152 mmHg is a check-up case in one series and an intensification case in the other.

Two further requirements follow. First, information silos re-form in a digital environment as soon as each device produces its own format [23]; the system must use one documented data schema. Second, electronic records contain duplicates, unit errors and values carried over identically from one visit to the next; the last is the most dangerous for a trend model, because it simulates a stability that did not occur, and it must be detected explicitly.

Finally, the WHO defines what a hypertension information system must produce: the core HEARTS indicators (Table 4.1). Two of the four are temporal and can only be computed from a dated history; the system must compute them automatically.

Table 4.1: Core HEARTS programme indicators [1].

Indicator	**Definition**	**Temporal**
Programme coverage	hypertensive patients registered in the catchment area	no
Blood pressure control	registered patients with BP $< 140/90$ mmHg over the last three months	yes
Retention in care	registered patients with a visit in the last three months, among those registered at least twelve months ago	yes
Availability	essential medicines and measurement devices available	no

4.1.1b A layered architecture with an edge tier

Mucchi et al. [13] define a reference architecture in four layers: a BODY layer of sensors worn by the patient; an ENVIRONMENT layer of ambient sensors; an EDGE layer, typically a smartphone, that performs short-term analysis, detects anomalous values and raises a first warning; and a CLOUD layer that stores the data, performs long-term analysis and manages alerts. The other architectures map onto it (Table 4.2).

The architectures agree on sensing, transport, storage and presentation; they disagree on the edge. This disagreement matters. Without an edge, every reading travels to the

Table 4.2: Mapping of the reviewed architectures onto the layers of Mucchi et al. [13].

Mucchi et al. [13]	**Efendi et al. [22]**	**Liu and Wang [23]**	**Yazdani et al. [14]**
BODY	physical layer	access layer	—
ENVIRONMENT	—	access layer	—
EDGE	*absent*	data transfer only	*absent*
CLOUD	network layer	storage, processing, analysis	database layer
Services and alerts	application layer	application service, decision support	user and function layers

cloud and every decision is taken there; with an edge, readings can be checked, aggregated and given a first warning locally, without a round trip. Where some 550 million people lack reliable electricity and internet access, the edge is a precondition rather than an optimisation.

The edge–cloud split is also exactly the triage the clinical analysis requires (Figure 4.2): the edge resolves the abundant routine readings, the cloud examines the series, and only a few cases reach the scarce clinician. At the clinician's end, Yazdani et al. [14] add two useful elements: patients are listed by priority rather than chronologically, and the checkup period of each patient is a configurable parameter, which turns the WHO follow-up intervals (Section 4.2.2b) into a setting.

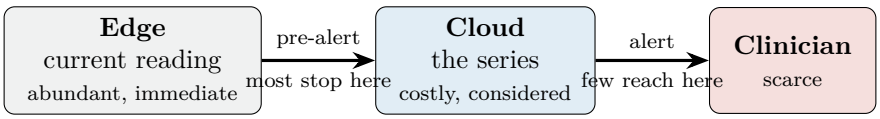


Figure 4.2: Triage as three filters in series, each costlier and scarcer than the previous one.

4.1.1c Communication protocols

Upstream traffic consists of small, frequent messages from constrained devices over unreliable links; downstream traffic is request-driven and must serve web clients. Each link therefore uses a different protocol (Table 4.3). MQTT is a publish–subscribe protocol with a single long-lived connection, minimal overhead and delivery guarantees, which suits the upstream profile of a constrained device; no reviewed source, however, measures its advantage over HTTP. For a smartphone, the client library of the database already provides a local queue, retries and delivery once connectivity returns over HTTPS, so MQTT is kept for the sensor prototype only. These transport and storage choices are engineering choices, made by criteria (offline operation, authentication, cost, effort) rather than by a benchmark: the benchmark of this work concerns the predictive models. JSON is preferred to the XML used by Yazdani et al. [14], because it is native to the storage layer and to mobile runtimes; their idea of routing messages by metadata is kept.

Table 4.3: Protocol selected for each link.

Link	Protocol	Reason
Device → phone	Bluetooth Low Energy	validated devices transmit without manual entry
Sensor → cloud	MQTT over TLS	constrained device, unreliable link, small payloads
Phone → cloud	database client over HTTPS	local queue and retry built in
Cloud → client	HTTP/REST	interoperability, request-driven access
Cloud → client, live	WebSocket	immediate delivery of alerts
Payload format	JSON	native to storage and runtime

4.1.1d Storage and synchronisation

Four requirements follow from the storage technology and the deployment context.

- **Store series, not current values.** The database of Efendi et al. [22] holds one current value per parameter; computing a seven-day mean would be impossible. Records must be keyed by measurement time so that a time window can be read with a single range query.
- **Validate at write time.** The record structure is fixed, so server-side validation rules can reject unit errors and impossible values before they enter the record.
- **Work offline.** Local storage with automatic merge on reconnection turns lost readings into delayed readings, which removes one cause of missing data.
- **Keep two timestamps.** Deferred synchronisation separates the time of measurement from the time of arrival, sometimes by days. Every temporal feature must use the measurement time; a model trained on arrival order would learn the connectivity pattern, which follows geography and income, rather than the physiology.

Alerts are raised at two levels: a local pre-alert at the edge on the current reading, and a considered alert in the cloud on the series. This mirrors the two-stage clinical protocol of Section 4.2.2c; the engineering architecture and the clinical protocol agree.

4.1.1e What the sources leave open

None of the reviewed architectures reports its operating cost, discusses dependence on a proprietary cloud provider, or justifies where the data are stored; all three must be addressed here, given the funding failures of Section 2.3.2d. None has been validated on real patients in routine care either (Table 4.4).

The usability test of Mucchi et al. [13] contains a result worth keeping: usability was rated well, but half of the respondents rated the system's efficiency for managing chronic disease as sufficient or below, and two-thirds rated their own digital skills only fair. The nurses asked for faster alert notification and for immediate feedback from the doctor when an alert is raised. Digital competence of front-line staff is a binding constraint, and an

Table 4.4: Implementation and validation status of the reference architectures.

Source	Implementation	Validation
Yazdani et al. [14]	design only (UML)	none
Liu and Wang [23]	survey diagram	none
Efendi et al. [22]	hardware and cloud	six volunteers, ten minutes, seated
Mucchi et al. [13]	mobile application and cloud repository	ten nurses, twenty days, simulated patients

alert is only useful if it reaches a decision-maker.

4.1.2 Decision-support requirements

Decision support must combine guideline rules with a learned model, keep the clinician in charge of every therapeutic decision, and size the number of alerts to the clinician time available.

4.1.2a Hybrid decision support

Yazdani et al. [14] define a decision-support telemonitoring system as the combination of telemonitoring with clinical decision support, and justify it by the workload that large volumes of home data would otherwise impose on physicians. They also observe that most such systems are built for one disease and must be rebuilt for another; their answer is to separate the platform from the knowledge base, so that each disease has its own configurable rule set. For a system covering two conditions with two guideline sources, this separation is necessary.

Three families of decision support exist (Table 4.5). The proposed system is *hybrid*: guideline rules provide a floor, and the learned model refines it. If the model is uncertain or fails, the system falls back to guideline conformity rather than to nothing.

Table 4.5: Families of clinical decision support.

Family	Basis	Strength	Weakness
Rule-based	guideline thresholds	transparent, needs no training data	no personalisation, no trend
Model-based	learned from data	personalisation, trend detection	opaque, needs data, may not transfer
Hybrid	rules as floor, model as refinement	guideline conformity plus personalisation	two components to validate

Liu and Wang [23] add three techniques. Before choosing an algorithm, the series should be characterised (periodicity, drift, stationarity); because home readings are irregularly spaced, methods valid for uneven sampling must be used rather than resampling onto a grid (Section 3.6.1c). Anomalies can be flagged relative to each patient's own distribution,

which gives the personalised threshold missing from the fixed protocol (Section 3.6.1b). And model training (offline, in the cloud) is separated from detection (online, at reception).

4.1.2b The decision flow

The flow of Yazdani et al. [14] contains three decision points. First, did the data arrive on schedule? If not, the patient receives repeated reminders. Second, is the patient normal or in alert? Third, in alert, the system proposes a prescription and the physician approves it, overrides it or requests a face-to-face consultation. The third point is the most important: the system proposes, the physician decides, and no prescription reaches the patient without a clinician. The first point is also essential: since loss to follow-up affects half of patients or more (Section 2.3.2b), a system that reacts only to received data is blind to the patients most at risk. Silence must itself raise an alarm.

Compared with the validated protocol of Mihevc et al. [19], this flow lacks the intermediate (yellow) level: every abnormal reading goes to a clinician. Combining the two, with the model placed on the verification profile, gives the flow adopted here (Figure 4.3).

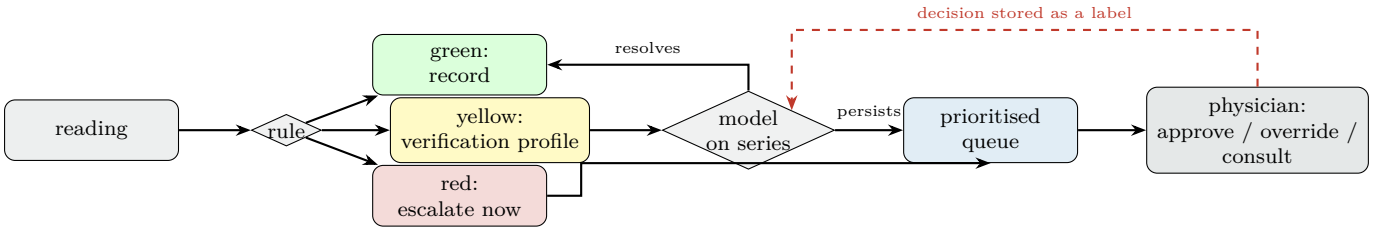


Figure 4.3: The decision flow adopted: rule-based triage, model-based assessment of the verification profile, prioritised escalation and clinician decision. A parallel branch raises reminders when no data arrive.

4.1.2c The arithmetic of alert volume

This subsection applies a standard result, the dependence of precision on prevalence, to the sizing of a telemonitoring service; none of the reviewed telemonitoring sources makes this calculation. Consider N patients sending m readings a week, a proportion π of which genuinely need attention, and a classifier of sensitivity Se and specificity Sp. The number of alerts per week and the proportion of them that are genuine (the positive predictive value) are

$$A = Nm\Big[\pi\,\mathrm{Se} + (1-\pi)(1-\mathrm{Sp})\Big], \qquad \mathrm{PPV} = \frac{\pi\,\mathrm{Se}}{\pi\,\mathrm{Se} + (1-\pi)(1-\mathrm{Sp})}. \tag{4.1}$$

The false-alert term $(1-\pi)(1-\mathrm{Sp})$ does not depend on sensitivity: it is the large volume of normal readings multiplied by the false-positive rate. Applied to the values of Mendoza-Cano et al. [26] at $\pi = 0.303$, these formulas reproduce their reported precisions (0.92

against 0.94 for random forest, 0.40 against 0.42 for XGBoost), which confirms them. In telemonitoring, however, critical readings are rare. At $\pi = 0.02$, the precision falls sharply (Table 4.6 and Figure 4.4): reaching a precision of 80% requires a specificity above 99.5%, whereas the best model in the reviewed literature reaches 0.98 at a prevalence fifteen times higher.

Table 4.6: Precision and false alerts at a prevalence of 2%.

Se	Sp	PPV	False alerts per 1 000 readings
0.51	0.98	0.34	19.6
0.90	0.98	0.48	19.6
0.90	0.99	0.65	9.8
0.90	0.995	0.79	4.9
0.90	0.999	0.95	1.0

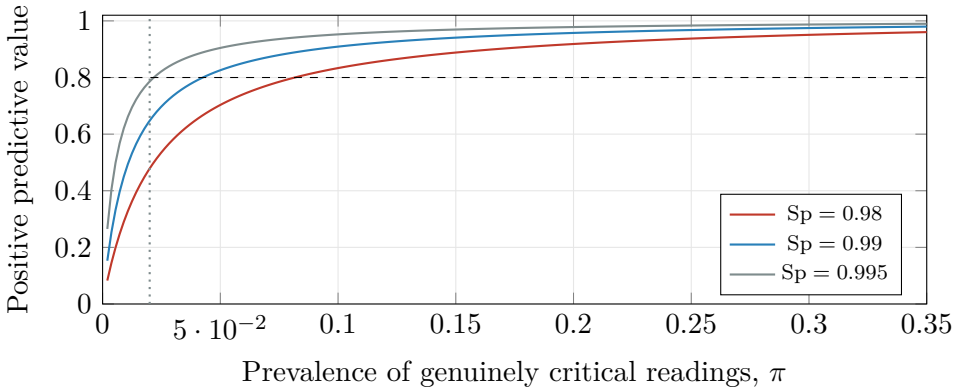


Figure 4.4: Positive predictive value against prevalence at a sensitivity of 0.90. The dashed line marks a precision of 80%; at $\pi = 0.02$ it is reached only beyond a specificity of 0.995.

The multiplier Nm depends on the sampling frequency, which varies by several orders of magnitude between designs (Table 4.7). Thirty patients on the protocol of Mihevc et al. [19] generate about two false alerts a week, which is manageable; the same specificity applied to a 1 Hz sensor stream generates more than eleven thousand false alerts per patient per week, whatever the quality of the model. High-frequency data can be summarised into features, but they cannot be alerted on reading by reading.

Four design responses follow. Alerts are raised on aggregates over a window (a seven-day mean), except at the red level, where one extreme value justifies escalation. The yellow level acts as a volume regulator: averaging the fourteen readings of a seven-day profile makes the separation between normal and abnormal patients 3.7 times larger than for one reading (Section 4.2.2a), so higher specificity becomes reachable. Every performance figure is accompanied by the alert volume it implies. And the decision threshold is a configurable clinical parameter, so that a service with one physician and a service with five can run the same system at different sensitivities.

Table 4.7: False alerts by acquisition design, at Sp = 0.98 and $\pi = 0.02$.

Design	Readings per patient per week	False alerts per patient per week
Protocol of Mihevc et al. [19]: 2 BP + 1 glucose	3	0.06
Manual entry, daily	7	0.14
Continuous sensor at 1 Hz [22]	604 800	11 854

4.1.2d Feedback loops and the limits of automation

Towards the physician. The system presents a proposal that the physician approves or overrides, and every event is recorded [14]. Three findings support this pattern: it addresses therapeutic inertia by presenting the next protocol step when the data show that the target is missed (Section 2.3.2b); remote expertise changed the local diagnosis in 46.2% of cases in Cameroon [5]; and the effect of the only trial in the target population came from treatment adjustment, not behaviour change [19].

Towards the patient. Behaviour change techniques are implemented because they are cheap (reminders, feedback, scheduling) and because their number correlates with the effect [17]; at least eleven are targeted (Section 4.6.2). They are not, however, where the clinical effect comes from.

Closing the loop. Every physician decision is also a label: an override of a red alert is a false positive confirmed by a clinician, an approval a true positive. Recording these decisions systematically builds, over months, a set of clinically adjudicated labels that does not depend on the thresholds the model sees—the way out of the circularity of Section 3.6.4b, and a source of locally valid training data. It costs nothing to design in from the start and cannot be reconstructed afterwards.

Limits of automation. Table 4.8 fixes what the system may do alone. Two reasons justify the boundary: the best model in the literature misses half of the positive cases, and lifestyle advice, the safest thing to automate, is also the least consequential (about 1.3 mmHg of systolic reduction).

4.1.2e Interoperability

Diagnoses are coded in ICD and clinical findings in SNOMED CT, so that the record can later be reconciled with a hospital information system. Every clinical rule is stored with the guideline edition and date it comes from, since the guidelines will be revised and a

Table 4.8: The boundary of automation.

May be automated	**Must pass a clinician**
Measurement reminders	any dose or medication change
Return of the reading and its level	any instruction to stop a treatment
Trend display	any diagnosis
Generic lifestyle guidance	any referral
Scheduling and non-transmission follow-up	any urgency-graded instruction beyond "contact your health centre"

rule base without versions cannot be audited. The United States regulations listed by Liu and Wang [23] do not apply in Cameroon; the applicable framework is discussed in Section 3.7.2.

4.1.3 Summary of Requirements

Table 4.9 gathers the requirements established in this chapter. Those on modelling are implemented in the methodology (Chapter 3); those on measurement and acquisition in the home protocol (Section 4.2); the others in the rest of this chapter.

4.2 The Home Monitoring Protocol

The system is built for patients followed at home. This section fixes what it must collect and how: the data streams, the measurement protocol, the instruments and the record schema. None of it was carried out on patients in this work, whose data are retrospective (Section 1.4); it is the protocol a prospective evaluation of the system would follow.

4.2.1 Data streams

The system combines three streams: the consultation records of the facility (stream A), home self-measurements (stream B) and, optionally, sensor data (stream C). They differ in their statistical properties, not only in their technology (Table 4.10).

Two points deserve emphasis. Consultation data are *missing not at random*: a patient who deteriorates may consult more often, or disappear; a model trained on them partly models attendance. Manual home entry suffers from three distinct problems: transcription errors (digit transposition, systolic–diastolic inversion), a timestamp that records the moment of entry rather than of measurement, and selective omission of discouraging readings. The first two are removed by Bluetooth transmission; the third is a bias, not noise, because it removes precisely the extreme values the model exists to detect, and it must be handled in the analysis (Section 3.6.1d). The risk labels used for modelling are

Table 4.9: Requirements derived from the literature.

Area	Requirement	Basis
Measurement	repeated home readings; reference value is the seven-day mean of retained readings	Sections 4.2.2a and 4.2.2c
	two readings per session, interval enforced by the device, second retained	Mihevc et al. [19]
	three bidirectional levels with a verification profile on the yellow level	Section 4.2.2c
	HEARTS indicators produced automatically	Section 4.1.1a
Acquisition	blood pressure only from a validated upper-arm cuff, glucose only from a glucometer	Section 2.3.4a and Table 4.13
	Bluetooth transmission; measurement time stored apart from entry and arrival time	Sections 4.1.1d and 4.2.1
	camera PPG for heart rate; blood pressure by camera exploratory only	Section 4.2.3a
	device agreement assessed by Bland–Altman against a validated reference	Section 4.4.4
Architecture	four layers with an edge tier performing validation, rule tiering and local storage	Section 4.1.1b
	queued upstream transport, REST downstream, live push, JSON payloads	Section 4.1.1c
	series keyed by measurement time; server-side validation; offline-first	Section 4.1.1d
	encryption, role-based access, de-identification, access log, declared residency	Section 4.3.5
	operating cost, provider dependence and residency documented	Section 4.1.1e
Modelling	forecast a future state; no feature computed at the label instant	Section 3.6.4b
	horizon matched to the sampling regime	Section 3.6.4c
	split by hospital and by patient; learned preprocessing inside the training fold	Section 3.6.5a
	proper scoring rules, calibration and net benefit; per-level measures with intervals	Section 3.6.5c
	guideline rule as mandatory baseline	Section 3.6.3e
Decision support	hybrid: final level = the higher of the rule level and the model level	Section 4.1.2a
	alerts on window aggregates; alert volume reported with every result	Section 4.1.2c
	non-transmission is an alarmed event	Section 4.1.2b
	system proposes, physician decides; decisions stored as labels	Section 4.1.2d
	automated advice limited to Table 4.8	Section 4.1.2d
	ICD and SNOMED CT coding; every rule versioned	Section 4.1.2e

Table 4.10: The three acquisition pipelines.

Property	**1. Consultation**	**2. Home self-measurement**	**3. Sensors**
Frequency	1–2 per year	several per week	up to 1 Hz
Missingness	not at random: attendance depends on the patient's state	not at random: readings may be omitted	at random: technical failure
Dominant error	measurement conditions	transcription, timestamp, selective omission	instrument accuracy, movement
Blood pressure	yes, reference quality	yes, from a validated cuff	no
Blood glucose	yes, laboratory	yes, from a glucometer	no
Role	ground truth, clinical context	principal longitudinal series	contextual covariates

derived from official thresholds for blood pressure, pulse and glucose (Section 3.6.2) and checked against the judgement of the partner physician.

4.2.2 Measurement protocol and procedure

The protocol follows from the clinical requirements of diagnosis and follow-up, which are recalled first.

4.2.2a Why a single measurement is not enough

The WHO diagnostic protocol is longitudinal by construction [1]. Blood pressure is measured twice at the first encounter, under standard conditions (seated, back supported, appropriate cuff), and the decision depends on the level observed (Table 4.11). Hypertension is diagnosed only if the threshold is exceeded at two visits. A care model that cannot guarantee the second visit does not produce imperfect diagnoses: it produces none.

Table 4.11: Recommended action by confirmed pressure at the first encounter [1].

Pressure on both readings of the day	**Recommended action**
SBP $\geq$ 160 or DBP $\geq$ 100 mmHg	start treatment the same day
SBP 140–159 or DBP 90–99 mmHg	confirm at another visit, one to four weeks later

The reason is the variability of blood pressure. For patient i measured at time t, a reading can be written

$$b_{i,t} = \mu_i + \varepsilon_{i,t}, \tag{4.2}$$

where μ_i is the patient's usual pressure, the clinically relevant quantity, and $\varepsilon_{i,t}$ gathers circadian variation, activity, stress, the measurement environment and instrument error. If n readings are spaced far enough apart for their errors to be independent, the usual

pressure is estimated with variance

$$\mathrm{Var}(\hat{\mu}_i) = \frac{\sigma_\varepsilon^2}{n}. \tag{4.3}$$

A consultation provides two to four points a year; home follow-up provides several dozen (Figure 4.5). Repeated home readings therefore estimate μ_i far more precisely, and they also make it possible to estimate σ_ε itself, the patient's own variability. This is the central argument for home measurement.

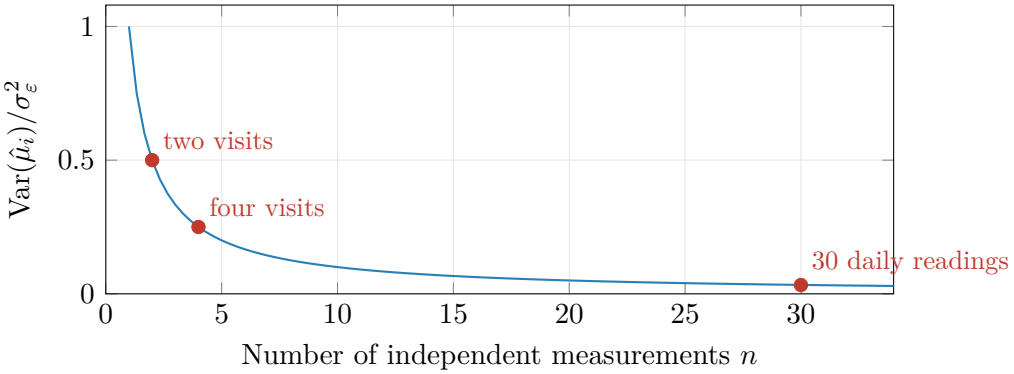


Figure 4.5: Relative variance of the estimate of the usual pressure as a function of the number of independent measurements.

4.2.2b How often patients must be followed

Recommendation 7 of the WHO guideline on the pharmacological treatment of hypertension sets monthly follow-up after any treatment change until the target is reached, then follow-up every three to six months [6]. In the African Region, 56% of treated patients are not controlled (Section 2.3.1b). For an active list with that proportion uncontrolled, the required number of contacts per patient per year is

$$\bar{n} = 0.56 \times 12 + 0.44 \times n_c \approx 7.6 \text{ to } 8.5, \qquad n_c \in [2, 4]. \tag{4.4}$$

An episodic model that delivers one or two contacts a year therefore delivers between an eighth and a quarter of what the guideline requires. This cumulative gap, more than the length of each consultation, is the structural deficit that telemonitoring must fill. It also explains therapeutic inertia: intensifying treatment requires a history showing that the target is persistently missed, and that history does not exist.

4.2.2c The self-measurement protocol and three-level triage

The trial of Mihevc et al. [19] provides the only clinically validated home protocol for patients with both conditions, and it is adopted as the reference.

Blood pressure. Two readings in the morning and two in the evening, after five minutes of seated rest; the device enforces a one-minute interval between the two readings and the second is retained. The reference value is the mean of the retained morning and evening readings over seven days: with fourteen readings, the standard error of the usual pressure is divided by $\sqrt{14} \approx 3.7$ compared with a single reading. The home target is 135/85 mmHg, five points below the office threshold because there is no white-coat effect at home.

Blood glucose. Routinely, one fasting measurement and one 90 minutes after lunch; after an abnormal value, a one-day profile of six measurements (before and 90 minutes after each meal). Targets are a fasting glucose below 7 mmol/L (about 1.26 g/L) and a post-prandial glucose below 10 mmol/L (about 1.80 g/L).

Routine frequency. Blood pressure twice a week and blood glucose once a week, as stated in the text of the article. A low routine frequency keeps the burden on the patient acceptable; a high frequency is triggered only when a signal justifies it.

Each reading is classified into three levels (Table 4.12), and the protocol works in two stages (Figure 4.6).

Table 4.12: The three-level protocol of Mihevc et al. [19]. BP in mmHg, glucose in mmol/L.

Level	Blood pressure	Blood glucose	Action
Green	110–134 / 60–84	4–7 before a meal; 7–10 after	no change
Yellow	135–174 / 85–104	7.1–10 before a meal; 10.1–14 after	verification profile: 7-day BP profile or 6-point glucose profile
Red	≥ 175 / ≥ 105, or < 110 / < 60	< 4; > 10 before a meal; > 14 after	immediate teleconsultation with the general practitioner
Profile mean	$\geq 135/85$ or $< 110/60$	> 7 before or > 10 after a meal	teleconsultation

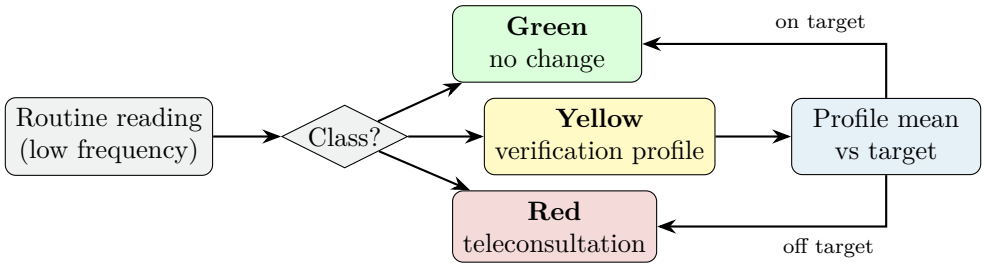


Figure 4.6: The two-stage triage of Mihevc et al. [19]. An isolated abnormal value triggers more measurements, not a clinician; only the mean of the verification profile reaches a clinical decision.

The protocol has three properties that the proposed system adopts. It is *bidirectional*: low values trigger as well as high ones (hypoglycaemia, hypotension in older treated

patients). It is *two-stage*: an isolated abnormal value leads to more data, which separates noise from trend by increasing n rather than by lowering the threshold. And it is *economical in clinician time*: only the red level mobilises a physician, which matters where professionals already perceive telemedicine as extra work (Section 2.3.2d). When systolic and diastolic pressures fall in different levels, the more severe level applies; the source is silent on this point and this is a decision of the present work.

The protocol also has three limits, which define the room for a learned model: its thresholds are fixed and identical for all patients, it reacts to the level of a reading rather than to its trend, and it uses no machine learning (Section 2.4).

4.2.2d Collection procedure

A prospective collection with the system, not carried out in this work, would proceed in four steps. At the **enrolment visit**, at the facility, the eligibility of the patient is checked, informed consent is obtained, the consultation history of the previous twelve months is extracted from the record and pseudonymised, and the patient receives the devices, the application and an individual training session. Since 10.9% of patients withdrew in the trial of Mihevc et al. [19] because they could not use the equipment, training is treated as part of the intervention: a practical session, a written leaflet and a supervised first measurement. During the **follow-up period**, of at least five weeks, the patient measures at home according to the protocol above; readings are transmitted automatically, and adherence is checked weekly from the received data. **Clinician review** takes place continuously through the dashboard, each alert receiving a verdict. At the **end-of-study visit**, the devices are collected and the patient and the clinician give feedback on the system.

4.2.3 Collection instruments

The instruments are chosen from the device constraints established in the literature review (Section 2.3.4a).

4.2.3a What optical sensors can and cannot measure

Photoplethysmography (PPG) shines light into the skin and records the reflected fraction, which varies with the arterial blood volume at each heartbeat. The signal has a large quasi-static component (tissue, venous blood) and a small pulsatile component. Three parameters can in principle be derived from it, with very different reliability:

- **Heart rate** comes from the periodicity of the pulsatile component and needs no absolute calibration. It is the robust case, including with a smartphone camera.
- **Oxygen saturation** requires two controlled wavelengths (660 and 880 nm for the MAX30100) and an empirical calibration curve fitted on volunteers; skin pigmentation

is a known source of error. A camera under white light does not provide these conditions.

- **Blood pressure** is not measured but inferred from the pulse waveform through a model that needs calibration. Liu et al. [24] report promising agreement on 127 participants, but the WHO does not yet accept cuffless measurement for clinical decisions (Section 2.3.4b).

The admissible source of each parameter follows (Table 4.13). No device category reviewed by Liu and Wang [23] or Alhaddad et al. [20] supplies validated blood pressure without a cuff, or validated glucose without a finger-stick; the instrumented pipeline measures around the clinical problem rather than into it.

Table 4.13: Admissible source for each parameter of the proposed system.

Parameter	**Admissible source**	**Not admissible**
Blood pressure	validated automated upper-arm cuff; Bluetooth or manual entry	cuffless wristband; camera PPG as a decision input (exploratory only)
Blood glucose	glucometer; Bluetooth or manual entry	any optical method
Heart rate	cuff pulse reading, camera PPG, or optical sensor	—
Oxygen saturation	validated pulse oximeter	camera PPG
Temperature	digital thermometer or MAX30205 sensor	—

4.2.3b Consequences of the device constraint

The WHO position on measuring devices (Section 2.3.4a) has three consequences. Blood pressure must come from a validated automated upper-arm monitor; cuffless devices may provide context variables (activity, heart rate) but not the clinical measurement. Admissible devices must be restricted to those on a recognised validation register. And the higher cost of validated devices must enter the feasibility analysis, alongside the cost of treatment, which the WHO puts at 5 to 10 US dollars per patient per year for quality-assured generic medicines.

4.2.3c The acquisition decision

Two designs were on the table: an instrumented device built around an ESP32 microcontroller with MAX30100 and MAX30205 sensors, and a deliberately spare design using validated Bluetooth devices, manual entry and camera PPG. The decision adopted is the following. Patients are to use the spare configuration: a validated Bluetooth upper-arm cuff and glucometer, with manual entry as a fallback, and camera PPG for heart rate. The instrumented device is specified as a bench prototype only, to demonstrate the hardware design and the low-power transport, and is not given to patients (Section 4.4.5).

Four reasons support this choice. The sensors of the instrumented device do not measure the two parameters on which the clinical decision rests. Their validation would require a Bland–Altman study under movement that this work cannot conduct. Maintained devices in the field are the first casualty of the funding problems documented in Section 2.3.2d. And alerting on readings at sensor frequency is arithmetically unusable (Section 4.1.2c). Of the design of Efendi et al. [22], only offline-first storage is kept, because it addresses connectivity rather than sensing.

4.2.4 Data structure

Four principles govern the schema: one row per measurement, not per visit, so that time windows can be queried; provenance (stream, device, operator) on every value, because the streams have different errors; two timestamps, measurement and record, because offline synchronisation separates them; and no derived field stored next to its inputs, so that features cannot leak the label. The data model has three tables (Figure 4.7).

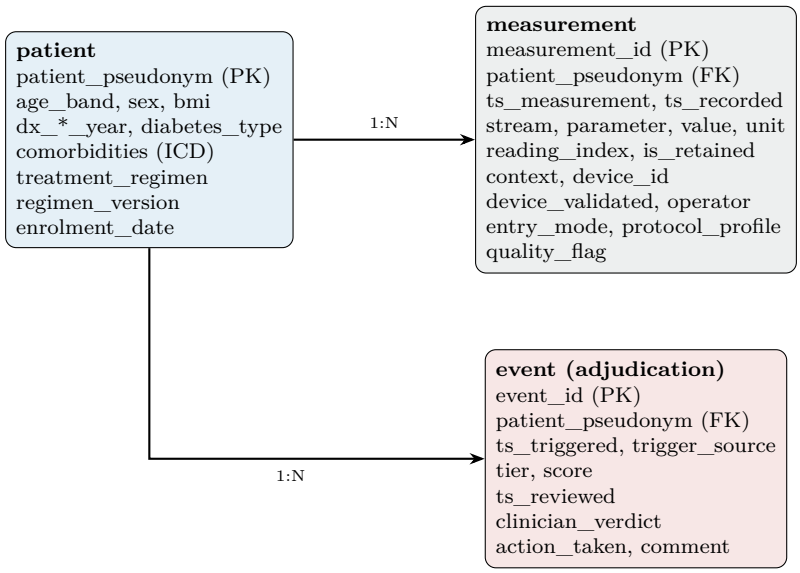


Figure 4.7: The three-table data model. The event table records clinician decisions, the only labels not derived from the thresholds the model sees.

The **patient record** holds the age in five-year bands, sex, body mass index, the year of each diagnosis, the diabetes type, ICD-coded comorbidities and the treatment regimen with a version number. Two of these fields deserve mention. The year of diagnosis gives the years since diagnosis, the strongest predictor of uncontrolled hypertension in Mendoza-Cano et al. [26]. The regimen version, incremented at every treatment change, distinguishes a patient drifting under an unchanged treatment from one whose treatment was just intensified, which is the clinical content of therapeutic inertia (Section 2.3.2b).

The **event record** stores each alert (trigger source, level, score) and the clinician's response: the verdict (confirmed, not clinically significant, uncertain) and the action taken

Table 4.14: Fields of the measurement record.

Field	Type	Notes
`measurement_id`	UUID	primary key; used to remove duplicates
`patient_pseudonym`	string	random, not derived from identity
`ts_measurement`	ISO 8601	when the reading was taken
`ts_recorded`	ISO 8601	when it entered the system
`stream`	enum	consultation, home, sensor
`parameter`, `value`, `unit`	enum, float, enum	sbp, dbp, glucose, hr, spo2, temp
`reading_index`, `is_retained`	int, bool	position in the reading pair; second reading retained
`context`	enum	fasting, post-prandial, morning, evening
`device_id`, `device_validated`	string, bool	validation required for sbp, dbp and glucose
`operator`, `entry_mode`	enum	patient, nurse, physician; Bluetooth, manual, transcribed
`protocol_profile`	enum	routine, 7-day BP profile, 6-point glucose profile
`quality_flag`	enum	ok, out of range, duplicate, carried forward, unit suspect

(none, advice, dose or drug change, consultation, referral). The verdict is the label that escapes the circularity of Section 3.6.4b, and the delay between triggering and review measures the bottleneck identified by Bediang et al. [5]. This table fills only once the system is in operation, but it must exist from the first deployment, since events that occur before it exists are lost.

4.3 Multi-Tier System Architecture

This section describes the target architecture; Table 4.22 states which parts are implemented.

4.3.1 The four layers

The decomposition follows Mucchi et al. [13] with two changes: the environment layer is omitted, since the system has no environmental sensors, and the application tier becomes a layer of its own, as in Efendi et al. [22] and Liu and Wang [23]. The resulting four layers are shown in Figure 4.8 and their functions in Table 4.15.

Three functions run at the edge, where most designs would place them in the cloud. The *rule-level evaluation* runs on the phone because the level decides how urgently a reading is transmitted; a phone that had to transmit in order to learn whether to transmit quickly would be caught in a circle. The *aggregation of high-frequency streams* runs there because alerting on each sensor reading is arithmetically unusable (Section 4.1.2c). And *local storage* makes the system usable under intermittent connectivity, a structural feature of the deployment context. In the cloud, the *registry* records which model version and

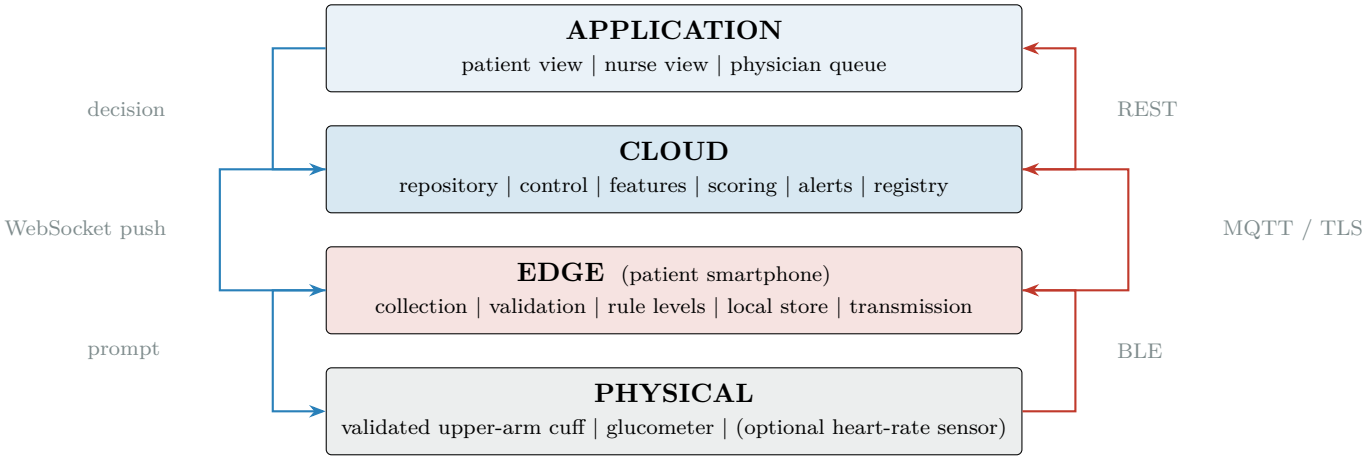


Figure 4.8: The four layers and the data flow of the target architecture. Red: upstream measurement path; blue: downstream decision path. In the built version the phone writes directly to the database over HTTPS (Table 4.22).

Table 4.15: Functions of each layer.

Layer	**Functions**	**Basis**
Physical	validated Bluetooth cuff (two readings, interval enforced, second retained); Bluetooth glucometer; optional heart-rate sensor; device register with validation reference	Sections 2.3.4a and 4.2.2c
Edge (smartphone)	Bluetooth reception and measurement timestamp; validation; rule-level evaluation; aggregation of any high-frequency stream; encrypted offline queue; MQTT transmission with level-dependent priority; patient interface	Sections 4.1.1b and 4.1.1d
Cloud	record keyed for window queries; identity, roles and audit log; features and normalisation; scoring, calibration and level mapping; alert queue and non-transmission detection; offline retraining; registry of model and threshold versions	Sections 3.6.1 and 3.6.5e
Application	patient, nurse and physician profiles (Table 4.16)	Mucchi et al. [13]

which threshold version produced every score, so that results remain reproducible after either changes.

Table 4.16: Application profiles.

Profile	Capabilities
Patient	enter or confirm readings; view own history and trend; receive prompts; view prescribed therapy and the outcome of clinician decisions
Nurse	the same for assigned patients; perform and record measurements; view flagged readings; cannot modify therapy
Physician	priority-ordered queue; patient record and trend; model score with its explanation; proposed protocol step; approve, override or request a consultation, each action written to the event record

4.3.2 Data flow and transmission priority

A reading travels from the device to the phone over Bluetooth, from the phone to the cloud over MQTT, through features, scoring and level mapping, and finally to the clinician's queue. Table 4.17 shows where each decision is taken and how long it takes.

Table 4.17: Where each decision is taken.

Decision	Layer	Time scale
Is the reading well formed? Which level under the rule? How urgently to send it?	edge	milliseconds
Which level under the model? Final level = max(rule, model)? Position in the queue?	cloud	seconds
Confirm, override or escalate?	physician	hours to days

Offline storage turns lost readings into delayed readings. For a routine reading this is a good trade; for a red reading it is not, since a crisis queued on a handset until the next connection is a failure caused by the architecture itself. Transmission priority therefore depends on the level (Table 4.18). The SMS fallback relies on the wide reach of mobile telephony in the region [7], although subscription counts overstate individual coverage. The local warning matters on its own: a patient with a critical reading is told so by the phone immediately, even when nothing can be sent. Advising a patient to seek care is not a therapeutic instruction, so it respects the boundary of Section 4.1.2d.

On the return path, any therapy change passes a clinician, and every decision is written to the event record *before* it is delivered, so that the label is kept even when delivery fails—which happens most often for the patients with the worst connectivity.

Table 4.18: Transmission behaviour by level, decided at the edge.

Level	Behaviour
Green	queued locally; sent with the next synchronisation
Yellow	sent immediately if possible; retried with increasing delays; synchronised within hours
Red	sent immediately; aggressive retry; SMS fallback after repeated failures; local warning to the patient whatever the connectivity

4.3.3 The latency budget

End-to-end delay can be decomposed into segments (Table 4.19). The engineering segments take seconds; scoring a shallow tree ensemble takes less than a millisecond, and a call to the risk service less than a second.

Table 4.19: Latency segments.

Segment	Symbol	Expected	Governed by
Device to phone	t_1	seconds	Bluetooth
Phone to cloud, online / offline	t_2 / t_2'	seconds / hours to days	connectivity
Ingestion, features, scoring	t_3	under a second	computation
Queue to clinician view	t_4	seconds	push notification
Clinician review	t_5	hours to days	staffing
Decision to patient	t_6	seconds to hours	return path

Bediang et al. [5] measured both ends of this chain in Cameroonian district hospitals. Tele-expertise reduced the time to perform an electrocardiogram from 7.0 to 1.3 days, about five times faster, but the specialist's opinion still took 10.3 days on average against 11.2 days in control centres—no improvement, and far beyond the 72 hours required by the protocol. The distributions were strongly skewed, so these means are indicative, but the conclusion is robust: technology accelerated acquisition, not the clinical decision. Clinician review exceeds all engineering segments by about five orders of magnitude (Figure 4.9).

Speeding up the engineering segments therefore changes nothing clinically: saving a second of computation against a total of 10.3 days is a gain of about one part in a million. Only clinician review can be meaningfully reduced, in two ways: by shortening the queue through triage (Section 3.6.5f), and by shortening the time per item through what the dashboard shows (Section 4.6.4). The dashboard is therefore a latency intervention, not a presentational detail. Offline delay t_2' is bounded by the level-dependent priority above, exactly where delay is harmful.

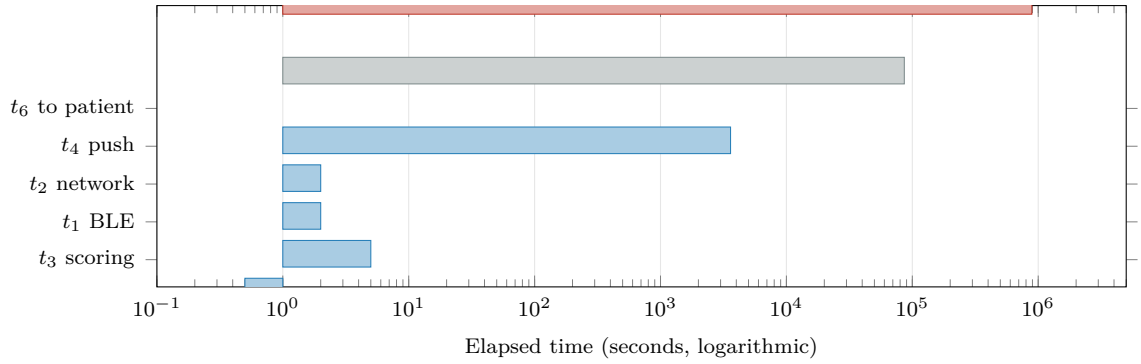


Figure 4.9: Latency segments on a logarithmic scale; t_5 is taken at the 10.3-day mean of Bediang et al. [5].

4.3.4 Failure modes

No reviewed source analyses what happens when a layer fails, although sustainability is the characteristic failure of telemedicine projects in the region. Table 4.20 lists the failure modes and the degraded behaviour designed for each.

Table 4.20: Failure modes and degraded behaviour.

Failure	**Detected by**	**Degraded behaviour**	**Residual risk**
Device lost, broken or flat	absence of readings	non-transmission alarm	unmonitored until noticed
Phone unavailable	absence of readings	none at the edge	no local alerting
Connectivity lost	edge	local queue; red by SMS	yellow delayed
Cloud unavailable	edge retries	local queue; edge rules still assign levels	no model score, no queue
Model fails	cloud	final level = rule level	loss of personalisation only
Calibration drifts	monitoring	thresholds revert to the rule	over- or under-alerting
Clinician unavailable	queue age	escalation to a named alternate	delayed review
Guideline revised	manual	new threshold version installed; registry keeps provenance	old scores not comparable

Three rows carry the design. When the model fails, the system falls back to the guideline rule, so its worst case equals the current standard of care. The phone is a single point of failure: a patient without a working handset is outside the system, and the only remedy is detection of the silence by the cloud, since a failed device cannot report its own failure. And clinician unavailability requires an escalation rule; without it, the long delays of a skewed distribution become a regular occurrence for some patients.

4.3.5 Security

Security is set by the weakest segment [13], so the design enumerates the segments and specifies each one (Table 4.21).

Table 4.21: Security segments and their protection.

#	Segment	Protection
1	Device to phone	Bluetooth LE Secure Connections; pairing bound to the patient account
2	Phone at rest	operating-system encryption required; sandboxed store; no identifiers in logs
3	Phone to cloud	TLS 1.2 or later; certificate pinning; MQTT over TLS
4	Cloud at rest	encryption at rest; server-side rules enforcing roles
5	Cloud to clinician	TLS; authenticated session with short expiry; re-authentication before a therapy change
6	Clinician device	outside system control: short sessions, no local caching, access logged
7	Research export	the de-identified analysis file of Section 3.7.1

Segments 6 and 7 are the weakest and are not addressed by any reviewed source. The clinician often works on a personal phone or a shared workstation, whose encryption and locking the system cannot enforce; it can only limit sessions, avoid caching data and log access. The research export is the one copy of the data outside all other protections; the de-identification measures of Section 3.7.1 are its security control. The many advanced techniques surveyed by Liu and Wang [23]—differential privacy, blockchain, steganography—are not proportionate to a thirty-patient pilot. The proportionate set is encryption in transit and at rest, role-based access, an append-only access log, and de-identification. What is implemented is stated separately from what a production deployment would add (Table 4.22).

4.3.6 Implementation status

The reviewed architectures rarely distinguish what was built from what was only designed. Table 4.22 makes that distinction for the present system.

4.4 IoT Hardware Prototype and Communication Protocol

The instrumented subsystem measures heart rate, oxygen saturation and temperature; it does *not* measure blood pressure or glucose, which come only from the validated cuff and glucometer (Table 4.13). It is therefore an optional covariate channel, specified as a bench prototype to demonstrate the hardware design and the low-power transport (Section 4.2.3c). If it is not built, or fails, the system loses feature columns and nothing clinically (Table 4.23). Note also that an automated cuff already reports the pulse rate at

Table 4.22: Implementation status of the architecture.

Status	Components
Built	patient application with manual entry, history and trend chart; rule levels computed on the phone, offline; Firebase authentication with patient and clinician roles, enrolment by clinician code; Firestore record of readings and messages with offline cache and security rules; risk service (FastAPI, local) running the research pipeline, with the rule floor and a check of the threshold version; clinician view of assigned patients; patient–clinician messaging; health assistant; nearby-care search
Specified, not built	Bluetooth acquisition; sensor prototype and MQTT transport; priority-ordered queue and verdict capture; event record; non-transmission alarm; SMS fallback; queue-age escalation; model registry; ICD and SNOMED CT coding; online hosting of the risk service
Deferred	environment layer; nurse profile; multi-site deployment; production security beyond the proportionate set

each measurement, so the prototype and the camera add temporal density to heart rate rather than a new parameter.

Table 4.23: Origin and clinical role of each parameter.

Parameter	Acquired by	Clinical decision?
Systolic, diastolic pressure	validated Bluetooth cuff	yes
Blood glucose	Bluetooth glucometer	yes
Heart rate	cuff pulse, camera PPG or MAX30100	no, covariate
Oxygen saturation	MAX30100	no, covariate
Temperature	MAX30205	no, covariate

4.4.1 Components and interconnection

The components are those of Efendi et al. [22] (Table 4.24), who give their characteristics and pin lists but no wiring, pin assignment or configuration.

Table 4.24: Components of the prototype, after Efendi et al. [22].

	ESP32	MAX30100	MAX30205
Function	microcontroller with Wi-Fi and Bluetooth	heart rate and SpO_2 (reflective PPG, 660/880 nm)	body temperature, accuracy 0.1 °C
Supply	3.3 V logic	3.3–5 V	2.7–3.3 V
Interface	36 pins; WPA/WPA2	I^2C; interrupt pin INT	I^2C; alarm pin OS; address pins A0–A2
Memory	520 kB SRAM; 8 kB retained in deep sleep	—	—

Three design decisions follow from these characteristics. First, the supply ranges of the two sensors meet only at 3.3 V, so a single regulated 3.3 V rail is the only configuration satisfying both, and it matches the logic level of the microcontroller. Second, both sensors

share one I²C bus, which requires distinct addresses: the oximeter's address is fixed, the temperature sensor's is set by its A0–A2 pins, which must be wired to ground or supply and never left floating. A floating address pin causes intermittent bus failures that would appear in the data as missing readings, indistinguishable from a patient who did not measure. Third, the sensors' INT and OS outputs are wired as interrupts, so the processor can sleep until data are ready instead of polling. Table 4.25 gives the proposed assignment, based on the default I²C pins of the microcontroller; bus addresses, pull-up resistors and currents are taken from the manufacturers' datasheets.

Table 4.25: Proposed pin assignment (design of this work).

ESP32	**MAX30100**	**MAX30205**	**Function**
3.3 V / GND	VIN / GND	VCC / GND	supply and reference
GPIO21	SDA	SDA	I²C data
GPIO22	SCL	SCL	I²C clock
GPIO19 (input)	INT	—	data-ready interrupt
GPIO18 (input)	—	OS	over-temperature alarm
—	—	A0, A1, A2	tied to GND or 3.3 V

4.4.2 Power and duty cycle

Efendi et al. [22] identify battery dependence as their main limitation: their device sampled every second with the radio permanently on. With a duty cycle d, active current I_{act}, sleep current I_{slp} and battery capacity C,

$$d = \frac{t_{\mathrm{act}}}{T}, \qquad I_{\mathrm{avg}} = d\, I_{\mathrm{act}} + (1 - d)\, I_{\mathrm{slp}}, \qquad \mathrm{life} = \frac{C}{I_{\mathrm{avg}}}. \tag{4.5}$$

Under the protocol of two measurement occasions a week ($T = 3.5$ days $= 302\,400$ s) with a 60-second active window, $d = 1.98 \times 10^{-4}$. With illustrative currents of 120 mA active and 10 µA in sleep, the average current falls from 120 mA to 0.034 mA, and the life of a 2 000 mAh battery rises from 17 hours to several years (Table 4.26).

Table 4.26: Power regimes (illustrative currents, to be replaced by datasheet values).

Regime	**Average current**	**Life on 2 000 mAh**
Continuous, as Efendi et al. [22]	120 mA	17 h
Duty-cycled, $d = 1.98 \times 10^{-4}$	0.034 mA	≈ 6.7 years

A less intuitive result follows. At this duty cycle the active term still contributes 70% of the average current. The two terms are equal for an active window of $t^{\star} = I_{\mathrm{slp}} T / I_{\mathrm{act}} \approx 25$ s; above it, shortening the active window is the effective lever, below it, the sleep current dominates. Since 60 s is above $t^{\star}$, the first optimisation is a shorter acquisition (faster

sensor settling, smaller payload), not a lower sleep current. The 8 kB of memory retained in deep sleep holds the state between cycles (last transmission, unacknowledged readings, last level), and the device wakes on the cuff's Bluetooth advertisement, so the covariates are sampled at the same moment as the clinical measurement.

Matching sampling to the clinical protocol thus solves three problems at once: battery life here, alert volume (Section 4.1.2c) and data cost (Section 4.4.3).

4.4.3 MQTT client design

Messages are published on the topic `telemon/v1/<site>/<pseudonym>/<channel>/<measurement_id>`. The version prefix allows the schema to evolve; the pseudonym keeps identities out of broker logs; and the channel segment lets the scoring service subscribe to clinical channels only.

MQTT offers three delivery guarantees (Table 4.27). Exactly-once delivery (QoS 2) seems natural for red readings but is the wrong choice: it needs two more round trips, precisely on the degraded links where red delays are harmful, and the duplicate suppression it provides is already ensured by the unique `measurement_id`. Red readings therefore use at-least-once delivery, with urgency expressed through retries and the SMS fallback.

Table 4.27: Delivery guarantee by level.

Message	QoS	Guarantee	Reason
Green, routine covariate	0	at most once	one lost sample is immaterial
Yellow and red readings	1	at least once	must arrive; duplicates removed by identifier

At connection, each client registers a "Last Will" message that the broker publishes if the connection drops without a clean disconnect. This provides, at no cost, the non-transmission signal that a failed device cannot send itself, and distinguishes a device sleeping as designed from one that has failed.

Each record is about 300 bytes, and each measurement occasion sends three (blood pressure, glucose, covariates). The duty-cycled design transmits about 7.7 kB per patient per month, against 778 MB for continuous 1 Hz streaming—five orders of magnitude less (Table 4.28). On Cameroonian mobile tariffs this is the difference between a recurring cost for the patient and a negligible one, and it makes the system sustainable as it grows. Transport uses MQTT over TLS, a credential per device, and broker rules restricting each device to its own topics.

Table 4.28: Transmitted volume, at 300 bytes per record.

Regime	Per patient per month	30 patients per month
Continuous 1 Hz	778 MB	23 GB
Duty-cycled, 2 occasions per week	7.7 kB	231 kB

4.4.4 Camera photoplethysmography

The smartphone camera is used for heart rate, which it measures robustly; it is not used for oxygen saturation, which requires controlled illumination at two wavelengths. Blood pressure estimated by camera [24] may be recorded alongside cuff readings for research, but never enters a clinical decision (Section 4.2.3a). The proposed signal chain is: fingertip on the rear camera with the flash on and automatic exposure disabled; mean red intensity of a central region at 30 frames per second; removal of the slow baseline; band-pass filtering between 0.7 and 3.5 Hz (42 to 210 beats per minute); rate estimation by spectral peak, cross-checked against peak-to-peak intervals; and the median over 30 seconds. Three quality gates—finger contact, motion, and periodicity—reject unusable acquisitions, which then count as missing rather than as values.

This channel comes with a free validation study, to be run in a prospective collection. The cuff reports the pulse at every measurement, so pairing a camera acquisition with each cuff reading would yield about 300 paired observations from thirty patients over five weeks. Because each patient contributes about ten pairs, agreement is assessed with the repeated-measures variant of Bland–Altman analysis, which separates between-patient from within-patient differences. The study reports the bias and limits of agreement, the proportion within ±5 and ±10 beats per minute, and the rejection rate of each gate. It must be planned from the start, because the two acquisitions must be simultaneous.

4.4.5 The three options compared

Table 4.29 compares the three acquisition options. Option C is adopted: patients would receive exactly the spare configuration of option A, and the instrumented device exists only as a bench prototype. This keeps the hardware design without requiring thirty devices to be built and maintained by one researcher; the low-cost acquisition itself is not evaluated in this work (Section 3.3.2).

4.5 Backend

The backend has two parts: a managed document database, Cloud Firestore, which holds the record and synchronises the applications, and a risk service that scores a patient's recent readings with the model selected by the benchmark. The database is a proprietary

Table 4.29: Acquisition options compared.

	A: spare	**B: instrumented**	**C: staged (adopted)**
BP and glucose	validated Bluetooth devices	validated Bluetooth devices	validated Bluetooth devices
Heart rate	cuff pulse, camera	MAX30100	cuff pulse, camera
SpO_2, temperature	—	MAX30100, MAX30205	bench prototype only
Devices to build	none	30 units	one prototype
Validation burden	camera against cuff pulse	both sensors, under movement	camera only
Clinical risk if it fails	none	none	none

service and the risk service is plain Python, so the domain logic stays outside the platform.

4.5.1 Data model

The logical model of Section 4.2.4 is implemented as collections of documents (Table 4.30). Readings and messages are subcollections of the patient, so that a security rule on the patient governs them, and each patient's readings are read in time order with a single ordered query. The published database of Efendi et al. [22], by contrast, holds one current value per parameter; it stores the present and discards the past, which makes a seven-day mean impossible.

Table 4.30: Firestore collections of the built system.

Path	**Content**
`users/{uid}`	profile: role (patient or clinician), age, sex, conditions, medications, assigned clinician, last-seen time
`users/{uid}/readings/{id}`	one reading: time, systolic and diastolic pressure, pulse, glucose in mmol/L, measurement context
`users/{uid}/messages/{id}`	patient–clinician thread: sender, role, text, time, read time
`clinician_codes/{code}`	enrolment code mapped to a clinician

Three parts of the logical model are not yet implemented: the event record with clinician verdicts, the device register, and the separation of measurement time from record time, since a manual entry is stamped when it is saved. The threshold table is versioned in a configuration file shared with the research code, and its version is checked at every call to the risk service (Section 4.5.4).

4.5.2 Offline operation

The database client keeps a local cache on the phone. A reading entered without connectivity is written to the cache, shown at once in the history, and sent when the connection returns; a lost reading becomes a delayed reading. The rule level of the latest readings is

computed on the phone (Section 4.6.5), so the patient sees a level even offline; only the model score needs the network.

4.5.3 Authentication and authorisation

Identity is established by Firebase Authentication; authorisation is enforced by security rules evaluated on the server for every read and write, which are the entire access-control layer since clients address the database directly. A patient joins a clinician with an enrolment code that the clinician gives them; the code is resolved before the account is created, and the role cannot be chosen by the client. Table 4.31 summarises the rules.

Table 4.31: Access matrix of the built system.

Path	Patient	Assigned clinician
`users/{own}`	read; create; update except protected fields	read
`users/{own}/readings`	read, write	read
`users/{own}/messages`	read; send	read; send
`clinician_codes`	look up one code	list

The specification of Section 4.2.4 makes the record append-only, a correction being a new reading that supersedes the old one. The built rules still let a patient modify or delete their own readings, which keeps the record editable by the person it describes; enforcing append-only writes is the first change needed before clinical use. Server-side validation of each write (plausible range, unit, device) is likewise specified but, in the built version, applied by the application only.

4.5.4 The risk service

The model is served by a local web service (FastAPI). The application sends the readings of the last four days, with age and sex and without any identifier; the service applies the quality controls, computes the features or sequences with the same code as the research pipeline, and returns, for each condition, the probabilities of the four levels and the alert probability. It also applies the guideline rule to the last reading of each channel and returns the final tier, the more severe of the rule tier and the model tier. Because the service and the benchmark share one implementation, the features seen in deployment cannot drift from those seen in training; and because the service refuses a model labelled with a threshold version different from its own, and the application refuses a service whose version differs from its own, a revision of the guidelines cannot silently produce inconsistent levels.

The service runs on the development machine only; hosting it online is outside the scope of this thesis. The application works without it, with the rule floor alone, and says so on the risk card. The server functions of the production design—escalation of yellow

and red events to a queue, notification, non-transmission sweep and queue-age sweep—are specified in Table 4.32 but not built.

Table 4.32: Server functions specified for the production version.

Function	Fires on	Action
Measurement check	new reading	re-validate; set the quality flag; recompute the rule level
Scoring	a completed window	call the risk service; apply max(rule, model)
Escalation	yellow or red level	create the event; add it to the clinician's queue
Red alert	red level	push notification; SMS after repeated failures
Verdict	clinician verdict	close the event; store the label; inform the patient
Non-transmission sweep	schedule	raise an event when a patient exceeds their checkup period [14]
Queue-age sweep	schedule	escalate an unreviewed event to a named alternate

4.5.5 Cost and residency

Firestore is billed on stored volume and on the number of documents read and written. A reading is one document of a few hundred bytes, and the history screen reads the patient's readings once and then only the changes, so thirty patients following the protocol of Section 4.2.2c generate a few hundred writes a month, far below the free quota of the service. The cost per patient and month at a thousand-patient scale is to be computed from the provider's current prices, a figure that none of the reviewed sources provides.

Two choices are legal as much as technical. The database region is fixed at creation and cannot be changed without migrating the data; a region outside Cameroon is a transfer abroad, which Law No. 2024/017 subjects to the authorisation of the supervisory authority (Section 3.7.2). And Firebase is a proprietary service whose rules do not port to other providers; the domain logic—rule engine, features, scoring—is therefore kept in plain, provider-independent code, and a scheduled export to a neutral format serves as backup and as portability insurance.

4.6 Application Interfaces

Three interfaces are described: the patient application, the health assistant it contains, and the clinician dashboard.

4.6.1 Design basis

The literature offers half of what is needed. The application of Efendi et al. [22] has a single user, the monitored person, who sees the prediction directly; there is no clinician. Yazdani et al. [14] design a clinician view ordered by priority but did not implement it.

The priority queue specified below therefore has no implemented precedent; the built version is a simpler roster.

Three findings guide the patient application. Commercial applications contain a median of nine behaviour change techniques [15], just below the eleven associated with the larger effect in Zhou et al. [17]. The clinical effect of Mihevc et al. [19], however, came from treatment adjustment by the clinician, not from behaviour change. And a tenth of their patients dropped out because they could not use the equipment (Section 4.2.2d). The primary function of the patient application is therefore reliable capture and prompt escalation, with as little burden as possible; behaviour change techniques are delivered by the structure of the system rather than by advice content.

4.6.2 Patient mobile application

Platform. The patient application is built with Flutter, a cross-platform toolkit, for one code base on Android and iOS and consistent rendering on low-end phones; the same application serves the clinician. This is an engineering choice, made for development effort rather than by benchmark.

What is built. The patient signs up with the code given by their clinician, enters readings, sees their history and trend, sees the risk tier of their latest readings, exchanges messages with their clinician, asks the health assistant (Section 4.6.3), and can open a search for nearby pharmacies, health centres, hospitals and laboratories. The search is handed to Google Maps with a "near me" query, so the application never requests the phone's location and never stores it.

Manual entry. Manual entry is the only acquisition path of the built version, and three safety requirements apply to it. The *unit* of glucose is fixed to mmol/L for the whole application and displayed beside the field: a glucose of 7 is a normal value in mmol/L and a dangerous one in mg/dL, and no range check can tell them apart; a value outside the plausible range of Table 3.5 is refused with a message that names the unit. The *coherence* of a pressure pair is checked (systolic above diastolic by at least 10 mmHg), as in the quality controls of Section 3.4.2b. The measurement *context* (fasting, after a meal, after medication, after exercise, random) is recorded, because the applicable glucose threshold depends on it. The specification adds two points not yet built: entering the two readings of the protocol, of which the second is retained, and dating a reading taken earlier.

Trends and what the patient sees. The chart shows the fourteen-day window with the target band drawn behind the series, so that the patient sees their position relative to the target; it is rendered from local storage, so it does not go blank when the connection drops. The patient sees their values, their position relative to the band, the tier and, at

yellow or red, the permitted instruction. The specification adds that the patient never sees a model score or a probability; the built risk card still displays the alert probability of each condition, which is to be removed from the patient view and kept for the clinician.

Behaviour change techniques. Most of the useful techniques are properties of a telemonitoring system rather than advice content (Table 4.33). This mapping is a design intention of this work; it is to be confirmed by formal coding against the taxonomy of behaviour change techniques before any count is reported as a result.

Table 4.33: Behaviour change techniques delivered by the structure of the system.

Technique	**Delivered by**
Self-monitoring of outcome	the measurement protocol
Feedback on outcome	the trend chart with target band
Prompts and cues	reminders based on the checkup period
Goal setting; discrepancy between state and goal	the target band and the reading plotted against it
Review of outcome goals	the clinician's decision relayed to the patient
Action planning; habit formation	a fixed measurement schedule
Practical social support	nurse-assisted measurement
Credible source	advice attributed to the clinician and the guideline
Information about health consequences	explanation of the target band
Behavioural contract	the enrolment agreement

Notifications. Notifications are to follow the level (Table 4.34), and their content respects the boundary of Section 4.1.2d: a patient may be told that a reading is high and that they should contact their health centre, never to change a dose.

Table 4.34: Notification by level and recipient (specified).

Level	**Patient**	**Clinician**
Green	silent; visible in the application	appears in the queue
Yellow	prompt for the verification profile	queued at normal priority
Red	immediate notification and local warning, whatever the connectivity	immediate notification; SMS fallback

When a patient stops sending data, the reminder is to be sent by SMS rather than by push notification, since the push channel is precisely the one that has gone silent [14]; neither is built yet.

Minimum device. The device requirements of Efendi et al. [22] are adopted (Table 4.35), with one exception: connectivity. A patient sends about 7.7 kB a month, which a second-

generation mobile connection transfers in seconds, so requiring a 4G connection would exclude the settings the system targets for no reason.

Table 4.35: Minimum device specification.

Requirement	Value
Free storage	1 GB
RAM	2 GB
Processor	1.2 GHz quad-core
Connectivity	intermittent, any generation (not 4G)

4.6.3 The health assistant

The patient application contains a conversational assistant that answers questions in the patient's language, French or English. It relies on a general-purpose language model (Gemini, in its fast variant, chosen for answers in a few seconds on a slow connection) reached through Firebase AI Logic. At the start of each conversation it receives an instruction that describes its role and the patient's context: age, sex, recorded conditions, prescribed medications, and the seven most recent readings.

Role. The assistant explains what a reading means, what hypertension and diabetes are and why follow-up matters; it gives general information on diet, salt, physical activity and adherence to the treatment already prescribed; it helps the patient judge whether a situation can wait for the next appointment, needs a visit soon or needs care today; and it encourages regular measurement. These functions belong to the automated side of the boundary of Table 4.8: return of a reading and its meaning, generic lifestyle guidance, and no instruction beyond "contact your health centre".

Guardrails. Four rules, stated as overriding every other instruction and every request of the user, and restated at the end of the instruction, keep the assistant within that boundary (Table 4.36). The provider's safety filters add a second layer: a blocked answer is replaced by a message directing the patient to their clinician. And the assistant has no access to the record beyond what it is told: it cannot write a reading, change a profile, message the clinician or alter a risk level.

Limits. These guardrails are instructions, not guarantees: a language model can be led outside its instructions by an unusual question, and the provider's filters are not designed for clinical safety. Their effect has not been measured; before use with patients, the assistant must be tested on a fixed set of adversarial questions (requests for a dose change, symptoms of an emergency, readings in each level, questions in French and in

Table 4.36: Guardrails of the health assistant and the risk each one addresses.

Rule given to the model	Risk addressed
Never recommend, name, adjust, increase, reduce or stop a medication or dose; refer the question to the doctor	antihypertensive and antidiabetic drugs have narrow margins: a wrong dose causes hypotension or hypoglycaemia (Section 4.1.2d)
Never give a diagnosis	a diagnosis requires repeated measurement and a clinician (Section 4.2.2a)
Never reassure in a way that could discourage seeking care	false reassurance delays care, the costliest error of triage
Emergency symptoms (chest pain, breathlessness, one-sided weakness, speech difficulty, fainting, confusion): tell the patient first and plainly to go to hospital or call emergency services	stroke and myocardial infarction, the complications the follow-up aims to prevent

English), with the proportion of answers that breach a rule reported. The assistant also sends personal data to a provider outside Cameroon: the patient's name, conditions, medications and readings leave the country at every conversation. This conflicts with the pseudonymisation applied elsewhere (Section 3.7.1) and is a transfer abroad under Law No. 2024/017 (Section 3.7.2); it requires the patient's informed consent and the authorisation of the supervisory authority, and the name, which the assistant does not need, is to be removed from its context. Finally, the instruction tells the model that the risk level is computed from the last seven readings, whereas the built risk card uses the readings of the last four days; the two must be aligned.

4.6.4 Clinician dashboard

What is built. The built clinician view lists the patients assigned to the clinician, each with their latest reading, updated live, and gives access to each patient's history, risk card and message thread; it also displays the clinician's enrolment code. The rest of this subsection specifies the dashboard the system needs, which this roster prefigures.

A queue, not a patient list. The unit of clinical work is an unreviewed event, not a patient. The dashboard is to read the queue ordered by priority and then by age. Since clinician review dominates the end-to-end delay (Section 4.3.3), the dashboard is the system's main lever on that delay, and the queue-age sweep prevents long waits from becoming routine.

Three levels, not two. The initial specification of the subject proposed two levels, normal and alert. With two levels, the intermediate readings must go somewhere: folded into normal, borderline patients are never re-examined; folded into alert, the queue fills

with false alerts (Section 4.1.2c). The three levels of the only validated protocol solve this.

What a queue row shows. Each row must allow a triage decision without opening the patient record (Table 4.37).

Table 4.37: Content of a queue row.

Field	Purpose
Level	the triage decision
Trigger source: rule, model or non-transmission	the strength of the evidence
Triggering value, with unit and context	the clinical fact
Direction of the fourteen-day trend	isolated excursion or deterioration
Days since last review	escalation
Pseudonym and age band	identification without identity
Model and threshold versions	traceability

Two display rules. First, a raw score from a weighted model is never displayed: an output of 0.5 may correspond to a true probability of 0.02 (Section 3.6.5e); only the corrected, calibrated value is shown, and the level with its reason often communicates more honestly than any decimal. Second, the trigger source is always shown. A rule trigger is a guideline threshold crossing; a model trigger, on this sample size, is weaker evidence. Showing both the same way would make the system distrusted after its first false alarm.

The verdict is the label pipeline. The physician's verdict on each event is the source of the clinically adjudicated labels of Section 3.6.4e. If recording it is tedious, the label store stays empty. The verdict must therefore take two or three interactions—true alert, false alert or indeterminate, plus the action taken—with an optional comment.

Roles and adoption. The nurse sees a separate worklist of measurements and assistance, not a greyed-out physician screen. And the dashboard must *replace* a task rather than add one, since health professionals in the region see telemedicine as additional work [7]: it must become the place where home readings are reviewed instead of the paper register, show the same window as the patient's application, and let the verdict both close the alert and serve as the clinical note.

4.6.5 Edge inference

Running the trained model on the phone is a natural extension. The model itself is small: a shallow tree ensemble takes tens of kilobytes, and even the LSTM of Section 3.6.5a, with $4[16(10+16)+16] = 1\,728$ parameters, takes under 7 kB. The difficulty is fidelity:

a local score is only valid if the window, the features, the normalisation constants, the masking, the calibration and the thresholds are reproduced exactly as in the cloud. A feature pipeline rewritten for the phone will diverge in small ways, and a phone that silently produces different scores from the cloud is worse than one that produces none. Local scoring therefore requires one shared implementation, or one exported pipeline, with agreement tests on a fixed dataset before each release.

The rule engine, by contrast, is a short list of threshold comparisons with no numerical drift. It delivers most of the offline value at a fraction of the risk, and degrades exactly to guideline conformity. The built version therefore ships the rule engine only, as a mirror of the threshold table of the research code whose levels are checked against those of the Python reference by an automated test; local model scoring follows once the agreement tests pass. Every locally computed level carries its threshold and model versions, and the cloud re-scores the window on reconnection and records any discrepancy.

A local red level may do exactly two things: warn the patient to seek care, and queue the event for transmission. It may not advise a dose or replace review. If connectivity is lost, the event is to reach the clinician by two independent routes: the queued reading when the connection returns, and the non-transmission alarm in the meantime.

4.7 Summary and Limitations

Table 4.38 shows where each element of the design comes from. The elements marked "this work" are original contributions of the design.

Table 4.38: Origin of the main design elements.

Design element	**Origin**
Four-layer decomposition with an edge tier	Mucchi et al. [13]
Offline-first storage	Efendi et al. [22]
Three-level alerting with a verification profile	Mihevc et al. [19]
Priority-ordered queue; configurable checkup period	Yazdani et al. [14]
Offline training separated from online scoring	Liu and Wang [23]
Validated-device restriction	World Health Organization [1]
Level-dependent transmission priority	this work (Section 4.3.2)
Alert-volume arithmetic and aggregation at the edge	this work (Section 4.1.2c)
Rule floor, final level = max(rule, model)	this work (Section 3.6.5f)
Latency decomposition and the dominance of clinician review	this work (Section 4.3.3)
Failure-mode table and model/threshold version registry	this work (Sections 4.3.1 and 4.3.4)
Duty cycling matched to the protocol (battery, data, alerts)	this work (Section 4.4.2)
Shared implementation of research and risk service, with threshold version check	this work (Section 4.5.4)
Guardrails of the health assistant	this work (Section 4.6.3)

Six limits remain. No architecture of this kind has yet been validated in routine care, in the literature or here. Part of the design is specified but not built (Table 4.22), in

particular the priority queue, the event record and append-only writes. The guardrails of the health assistant are untested, and its use transfers personal data abroad. Camera heart rate is unvalidated until the sub-study of Section 4.4.4 is run. The efficiency of MQTT over HTTP is assumed rather than measured; a bench comparison of bytes, round trips and energy per record on the prototype would turn it into a result. And the usability of the interfaces, in particular the two-to-three-interaction verdict, has not been measured; it requires a prospective study with patients and clinicians.

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Appendix A

TRIPOD+AI Checklist

The modelling study is reported according to the TRIPOD+AI statement [85], and its risk of bias can be assessed with PROBAST+AI [86]. Table A.1 gives, for each item of the checklist, where it is addressed. Items 20 to 27 concern results and their discussion, which are reported in the experimental part.

Table A.1: TRIPOD+AI items and where they are addressed.

Item	**Topic**	**Where**
1	Title	title page
3a–b	Background, intended use	Chapter 1; Section 2.5
3c	Health inequalities	Section 2.4 (Central Africa absent from the evidence)
4	Objectives	Sections 1.3 and 2.5
5a	Data sources	Sections 3.3.2 and 3.5
5b	Dates	Section 3.3.2 (Douala, 2026); eICU admissions 2014–2015 [27]
6a–b	Setting, eligibility	Sections 3.3, 3.3.1 and 3.5
6c	Treatments	not recorded in either dataset; Section 3.8
7	Data preparation	Sections 3.4.2b, 3.6.1a and 3.6.1b
8a	Outcome	Sections 3.6.2 and 3.6.4c
8b–c	Outcome assessment, blinding	rule-based label; blinded relabelling by the partner physician (Section 3.6.2)
9a–b	Predictors	Section 3.6.3 and Tables 3.12 and 3.13; instruments in Section 3.4.2a
9c	Subjective predictors	none; pulse read from the chart curve (Section 3.4.2a)
10	Sample size	Section 3.5 and Table 3.7
11	Missing data	Section 3.6.1d
12a	Data partitioning	Section 3.6.5a
12b	Predictor handling	Table 3.14
12c	Model type, tuning	Section 3.6.4a; hyperparameter search
12d	Heterogeneity across clusters	outer folds grouped by hospital (Sections 3.6.5a and 3.6.5c)
12e	Performance measures	Section 3.6.5c and Table 3.18
12f	Model updating	intercept update in external validation (Section 3.6.5e)

Item	Topic	Where
12g	How predictions are computed	published code (Section 3.6.6)
13	Class imbalance	Sections 3.6.5b and 3.6.5e
14	Fairness	performance by sex and age band (Section 3.6.5c)
15	Model output, thresholds	Sections 3.6.5e and 3.6.5f
16	Development versus evaluation data	Section 3.5 (intensive care in the United States against Douala wards)
17	Ethical approval	Section 3.7
18c–d	Protocol, registration	this chapter fixes the protocol before the results (Section 3.6.5h); not registered
18e–f	Data and code sharing	Section 3.6.6
19	Patient and public involvement	none
20–27	Results, discussion, usability	experimental part